

X+2
SYSTEM 5.0

D

C

B

A

IMAGERS

AP_PMIC

DISPLAY

AUDIO_FE

XMCS

SLICE_IF

USER_INPUT

POWER

BLK_WTR3925

RFFE_CA+DSDA

CONNECTIVITY

ANTENNAS

REVISIONS				
ZONE	LTR	DESCRIPTION	DATE	APPROVED

D

C

B

A

APPROVALS		BLOCK TITLE:	
ENGINEER: MOTOROLA MOBILITY		84016440001_a	
LAST MODIFIED BY: NATE CONNELL		PROJECT NAME	
Wed Apr 08 11:47:13 2015		84016440001_a.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED		REV	
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CAD FILE NAME:		PAGE 1 OF 78	

X+2 ME/SHIELDS

BUNKER HILL REFERENCE DESIGNATOR RANGES BY BLOCK

000 - 499: ANTENNA

500 - 599: USER INPUT

600 - 799: POWER

1000-2999: MSM8994+PM8994+PMI8994

3000-3199: WTR39X5

4000-4499: CONNECTIVITY

4500-4999: AUDIO FE

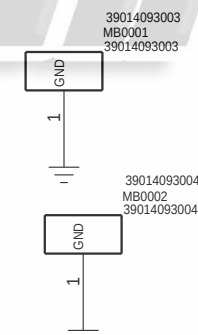
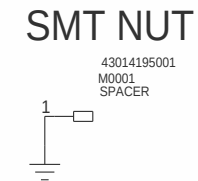
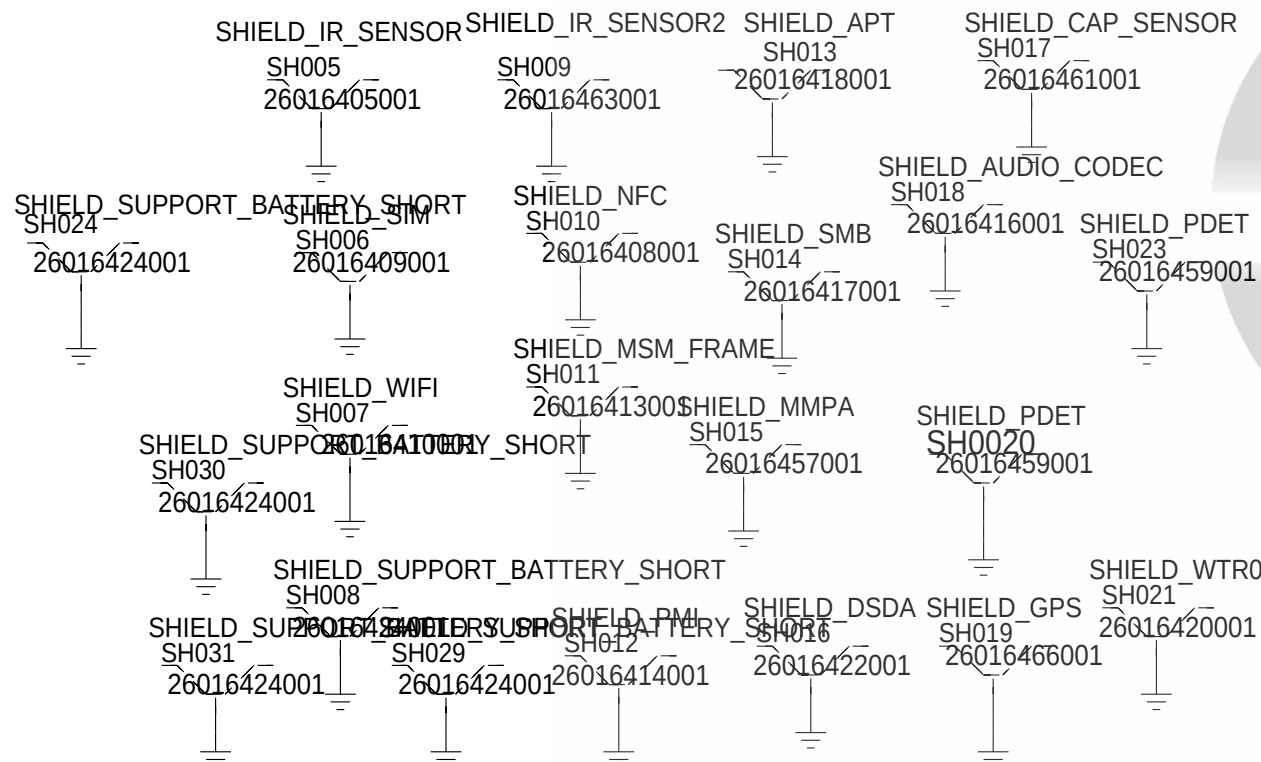
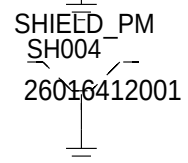
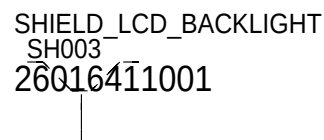
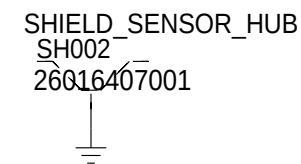
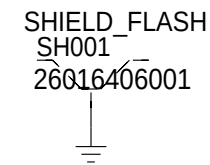
5000-5499: DISPLAY

5500-5999: CAMERA

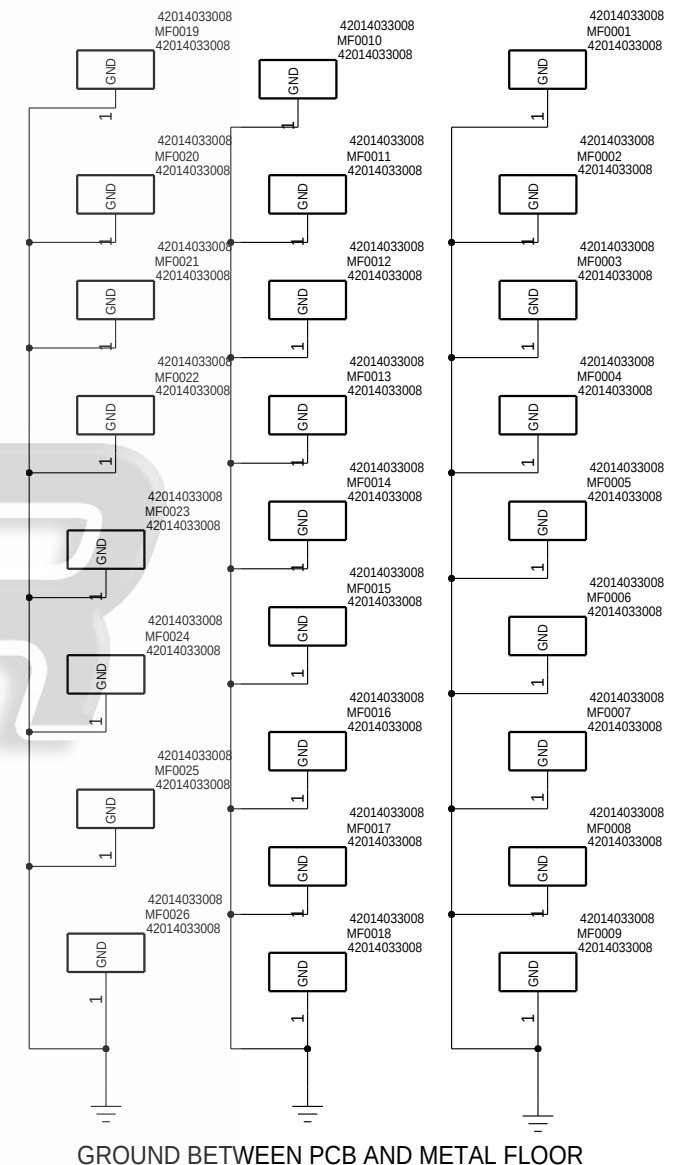
6000-6999: XMCS

7000-8999: RFFE

9000-9999: PRODUCT-SPECIFIC (FILTERING, 100PF CAPS)



GROUND BETWEEN PCB AND ENDO



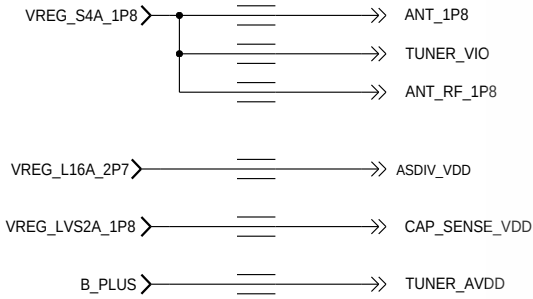
GROUND BETWEEN PCB AND METAL FLOOR

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED				
APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER:	BLOCK TITLE: 84016440001_a			
LAST MODIFIED BY: Tue May 12 10:24:12 2015				
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		CAD FILE NAME:		PAGE 2 OF 78

ANT: PORTS

REFDES: 000 - 499

SUPPLIES



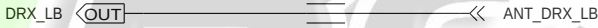
TRX ULTRA LOW/MID BAND ANTENNA



TRX LOW/HIGH BAND ANTENNA



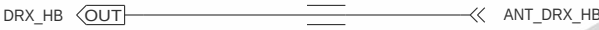
DRX LOW BAND ANTENNA



DRX MID BAND ANTENNA



DRX HIGH BAND ANTENNA



GPS ANTENNA



BT/WLAN ANTENNA



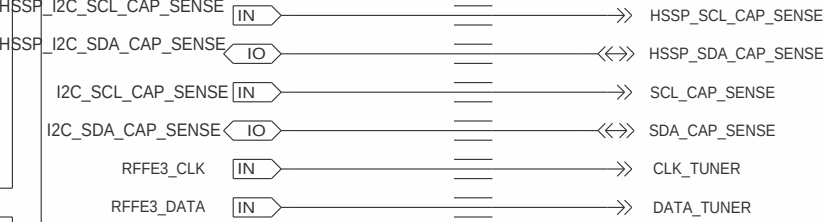
WLAN ANTENNA 2 (FOR 2X2 MIMO SUPPORT)



COUPLER PORTS



INTERFACES



GPIO - INPUT



GPIO - OUTPUT



APPROVALS

ENGINEER: NATE CONNELL

LAST MODIFIED BY: NATE CONNELL
Tue May 12 10:24:05 2015

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BLOCK TITLE:

antennas

PROJECT NAME
84016440001_a.cpm
REV
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CAD FILE NAME:
PAGE 3 OF 78

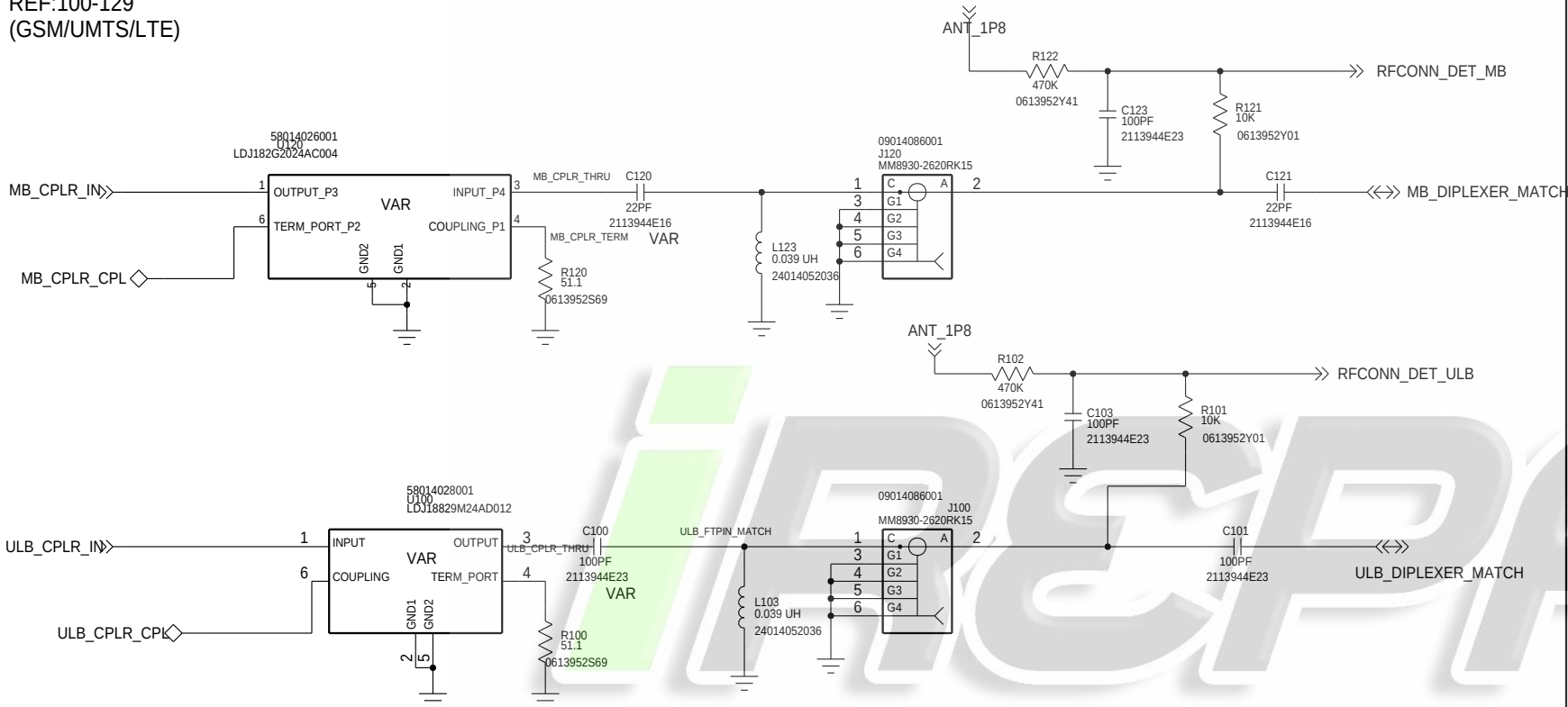
ANT: ANTENNAS

REFDES: 100 - 199

TRX ULTRA LOW/ MID BAND ANTENNA - ASM2

REF:100-129

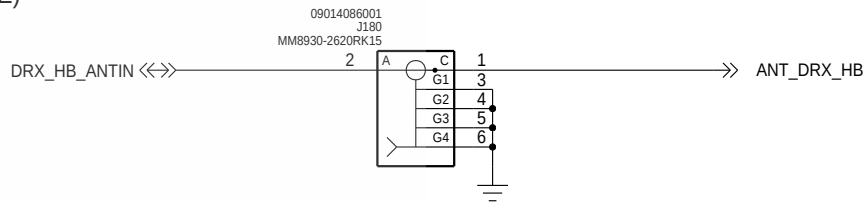
(GSM/UMTS/LTE)



DRX HIGH BAND ANTENNA

REF:180-189

(GSM/UMTS/LTE)

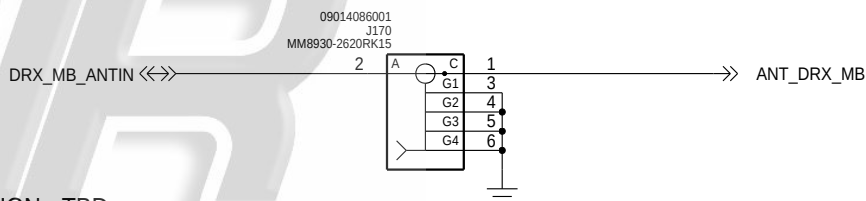


DEVICE LOCATION - TBD

DRX MID BAND ANTENNA

REF:170-179

(GSM/UMTS/LTE)

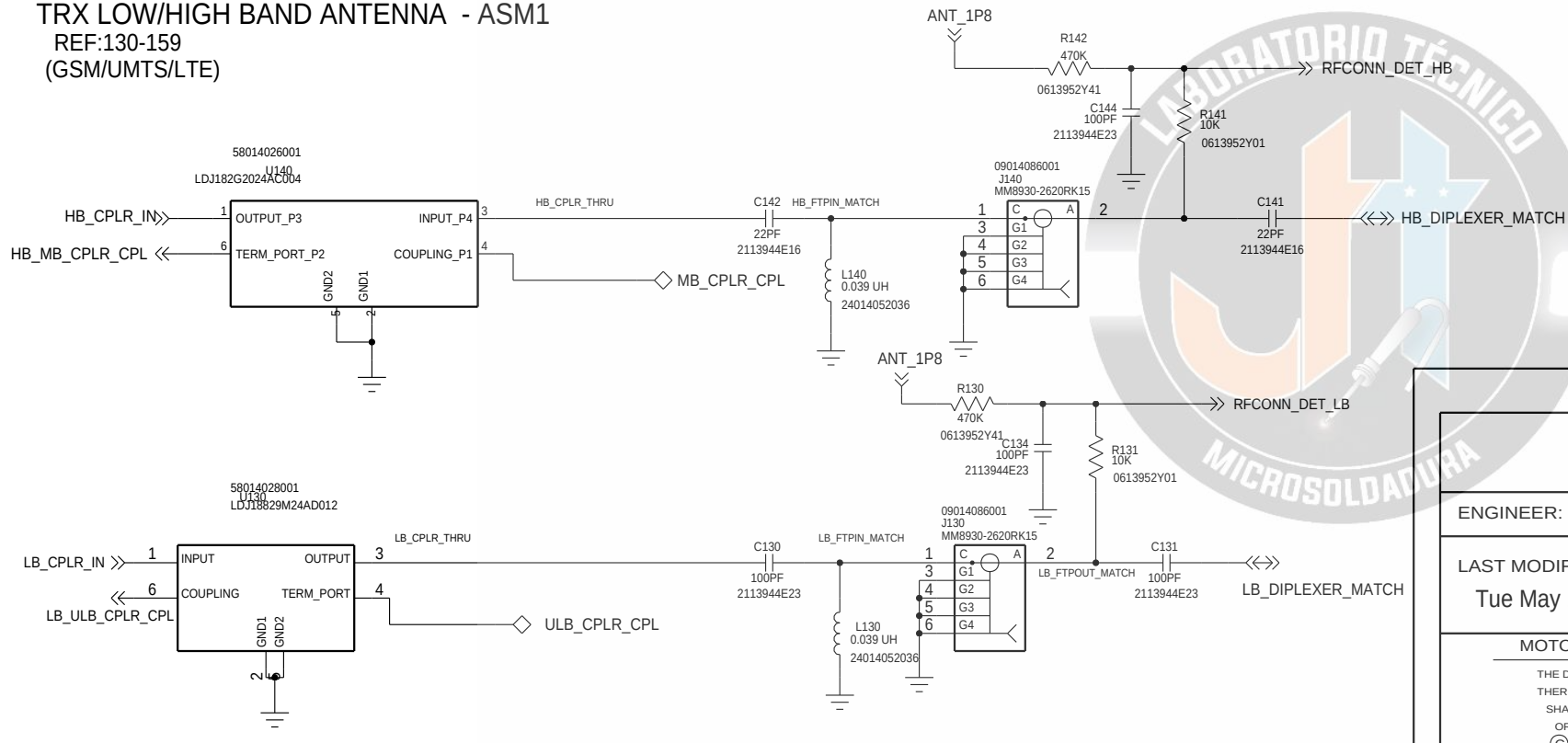


DEVICE LOCATION - TBD

TRX LOW/HIGH BAND ANTENNA - ASM1

REF:130-159

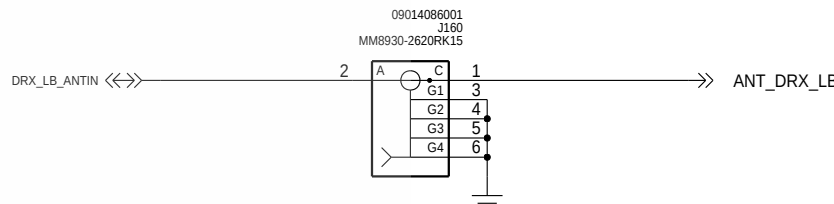
(GSM/UMTS/LTE)



DRX LOW BAND ANTENNA

REF:160-169

(GSM/UMTS/LTE)



DEVICE LOCATION - TBD

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
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MOTOROLA

MOBILE DEVICES

BLOCK TITLE:

antennas

PROJECT NAME

B 84016440001_a.cpm

CAD FILE NAME:

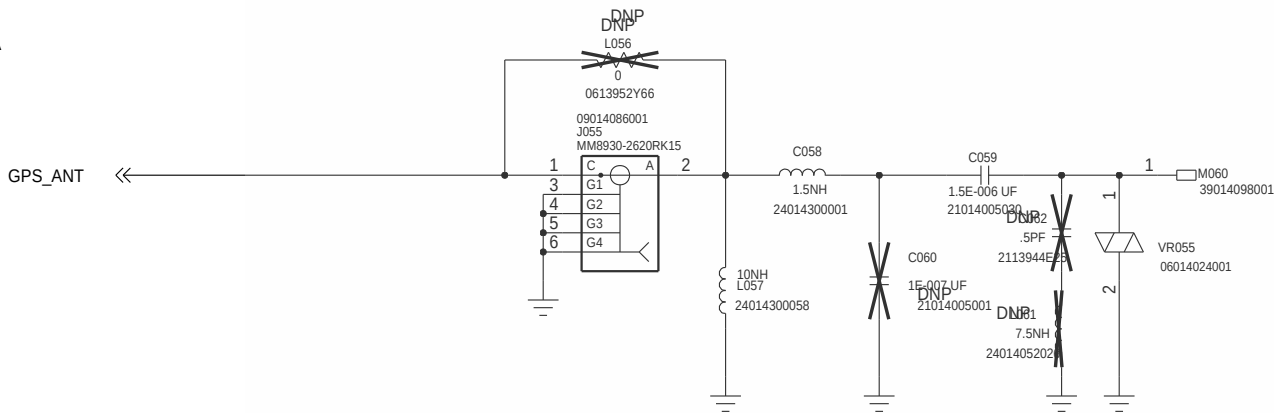
REV

01

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ANT: GPS/WLAN ANTENNAS

GPS ANTENNA
REF: 055-069



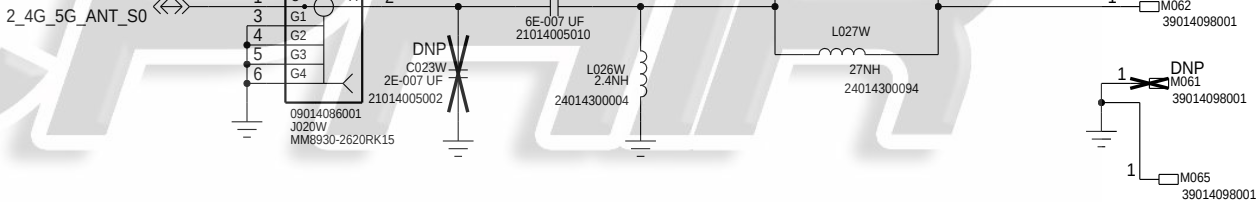
LOCATION - TBD

WLAN CHAIN1 ANTENNA

REF: 060-85

MIMO AND MAIN PATHS ARE SWAPPED BEFORE THE RF CONNECTORS J010 AND J020W

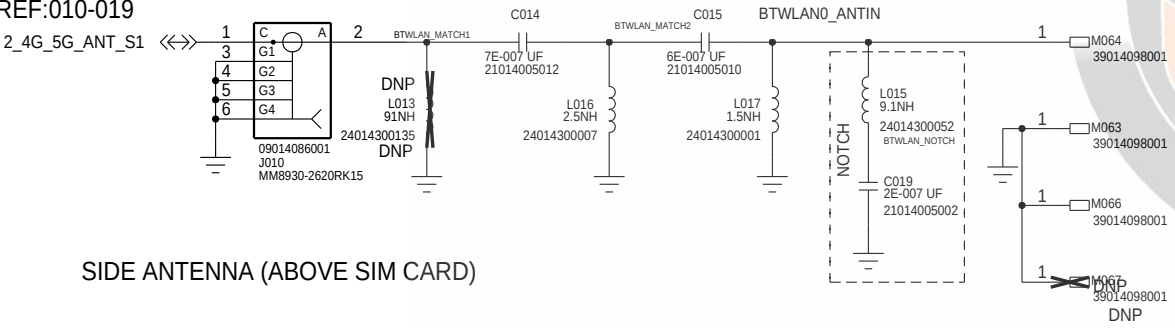
REF:020W-029W



DEVICE LOCATION - TBD, APPLIES ONLY IF QCA6174 IS USED
ALL CHAIN1 ANTENNA COMPONENTS HAVE W SUFFIX FOR NOW.

BT/WLAN CHAIN0 ANTENNA
REF:010-019

MIMO AND MAIN PATHS ARE SWAPPED BEFORE THE RF CONNECTORS J010 AND J020W



SIDE ANTENNA (ABOVE SIM CARD)

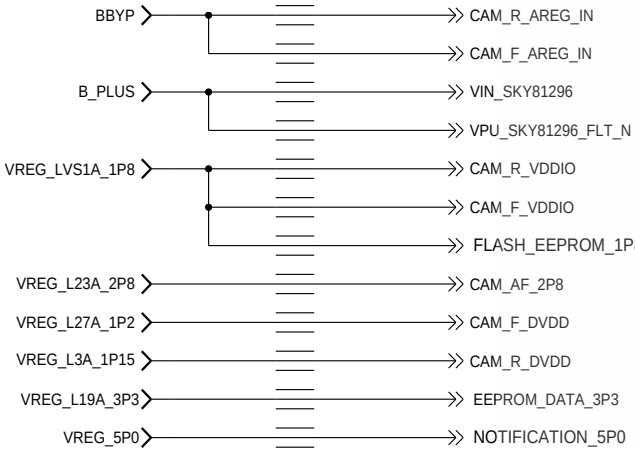
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: antennas	
LAST MODIFIED BY: MOTOROLA Tue May 12 10:24:07 2015		PROJECT NAME 84016440001_a.cpm	
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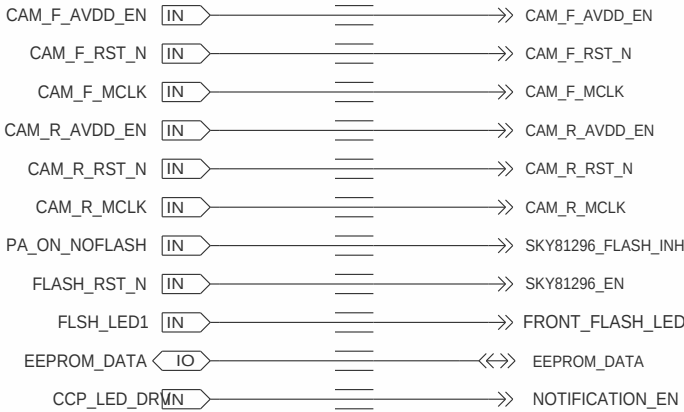
CAM: PORTS

REF 5500-5999

SUPPLIES



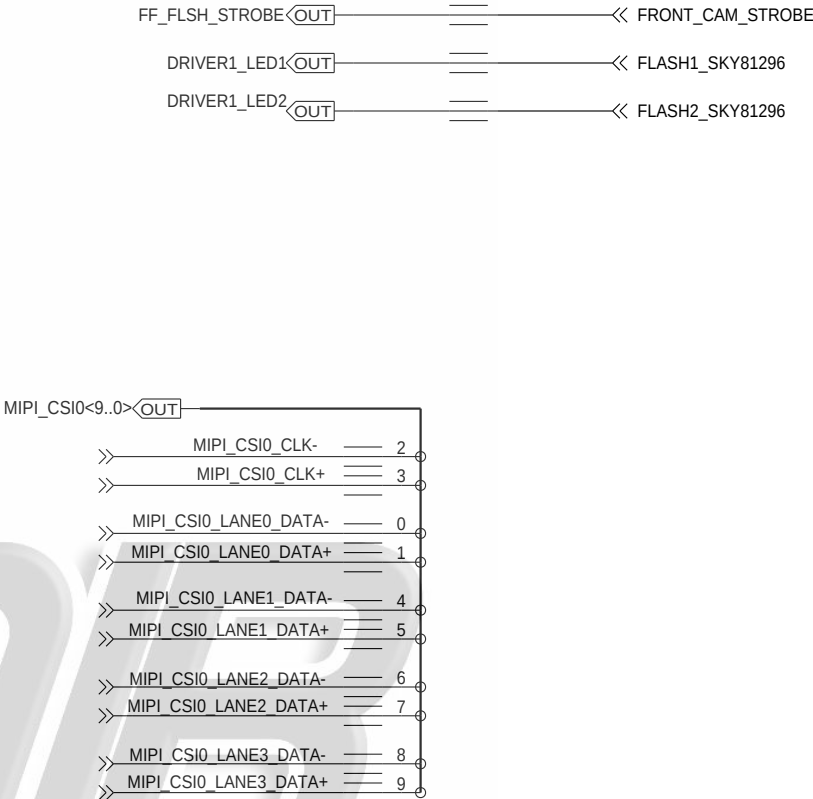
INPUTS



INTERFACES



OUTPUTS



APPROVALS

ENGINEER: NATE CONNELL

LAST MODIFIED BY: NATE CONNELL
Tue May 12 10:24:02 2015

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

imagers

B

PROJECT NAME
84016440001_a.cpm

REV
01

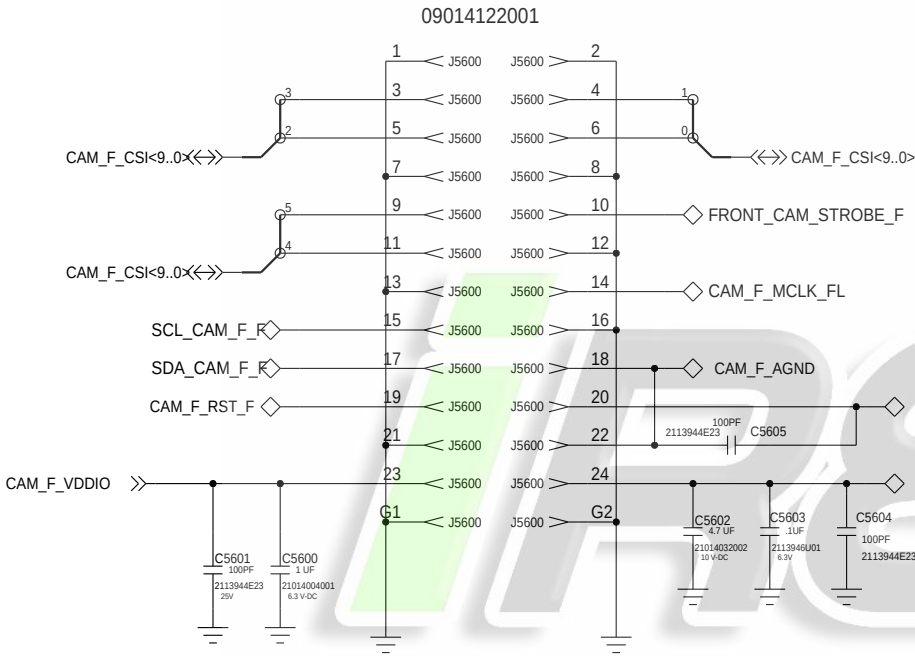
CAD FILE NAME:

PAGE 10 OF 78

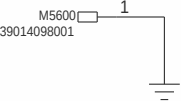
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REF 5600-5699

*** IMAGE SENSOR I2C ADD. 0X6C

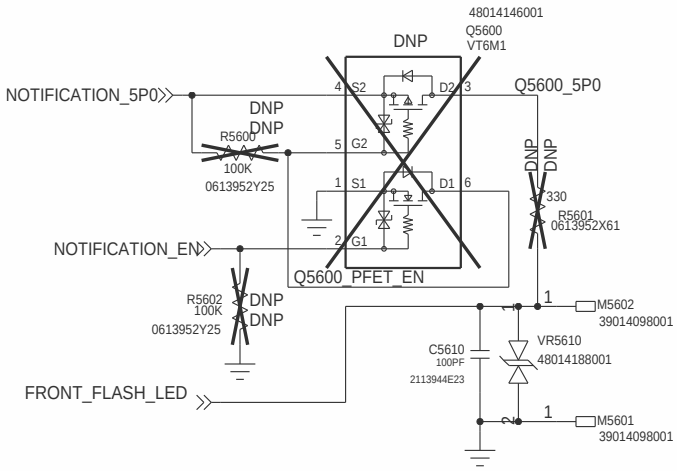
MATES WITH 2871491E05



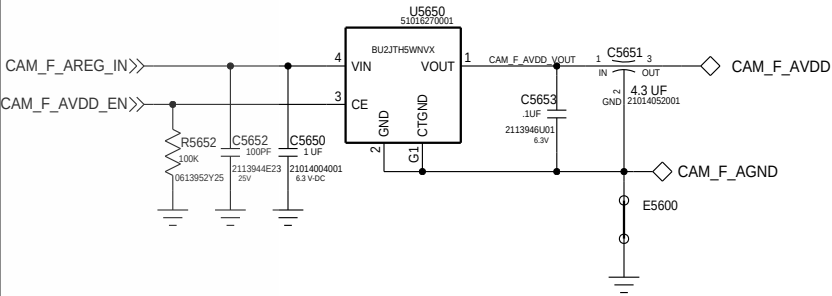
MECH UNIV GROUND CONTACT



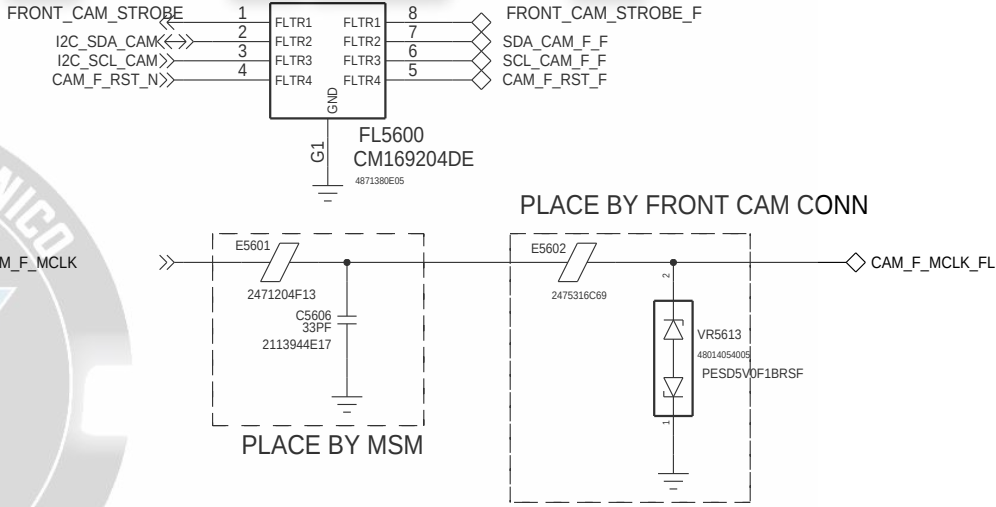
FRONT CAMERA FLASH



ANALOG REGULATOR



EMI/ESD FILTER



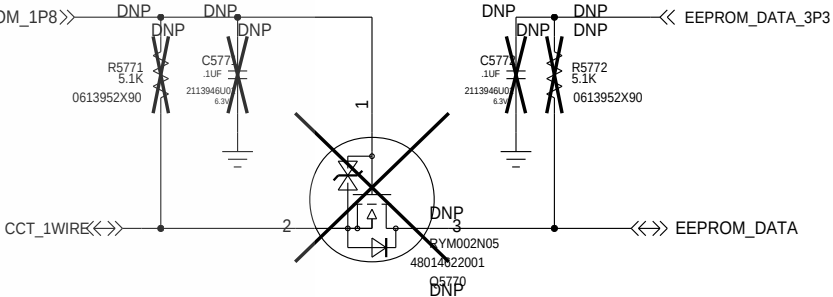
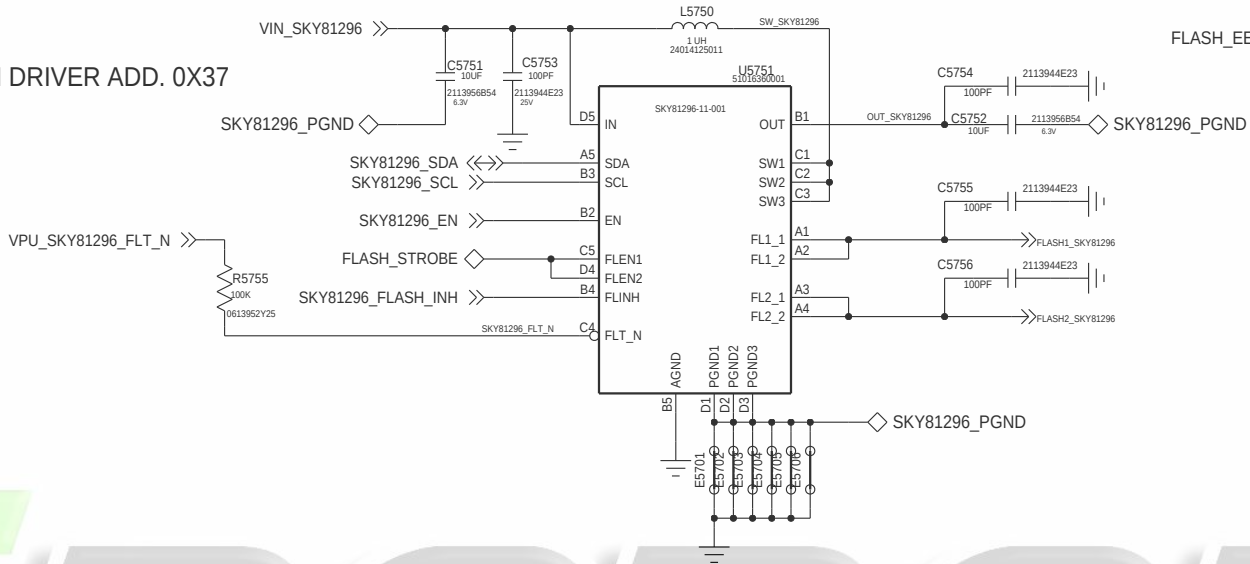
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN KRAMER		BLOCK TITLE: imagers	
LAST MODIFIED BY: UID ELRAN Tue May 12 10:24:03 2015		PROJECT NAME 84016440001_a.cpm	
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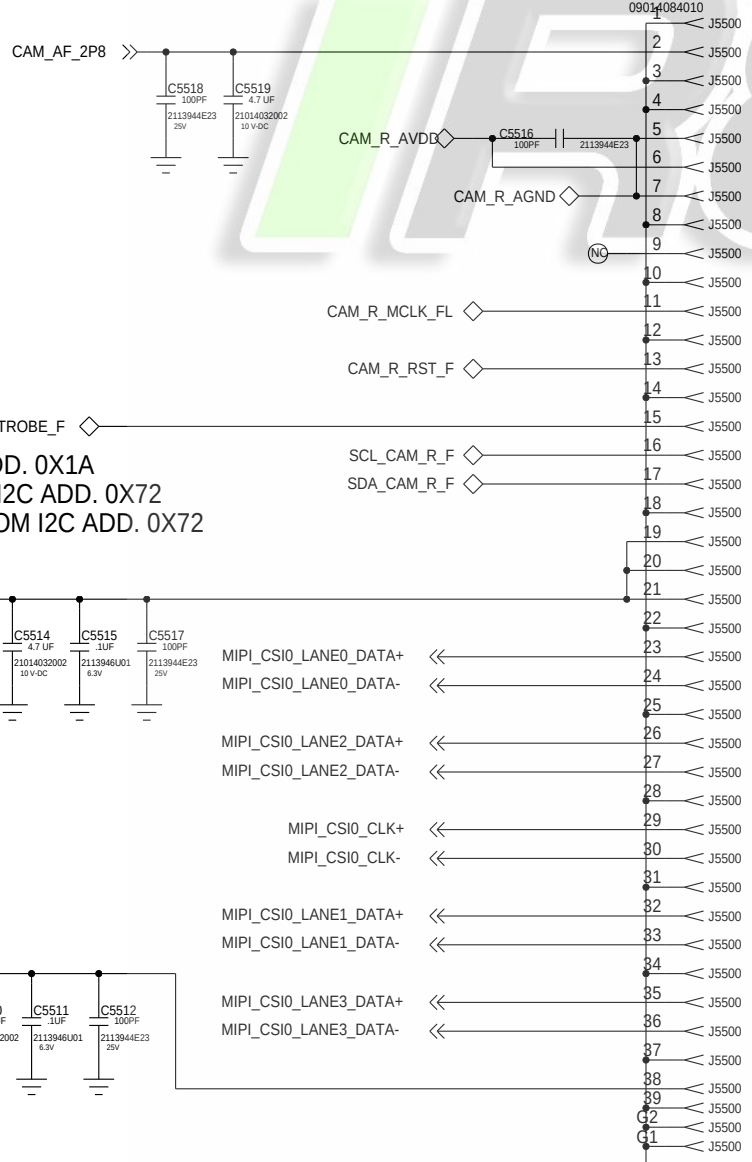
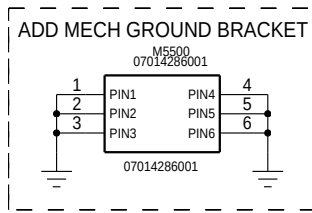
CAM: REAR
REF:5500-5599

LED FLASH DRIVER
REF:5750-5799

*** FLASH DRIVER ADD. 0X37

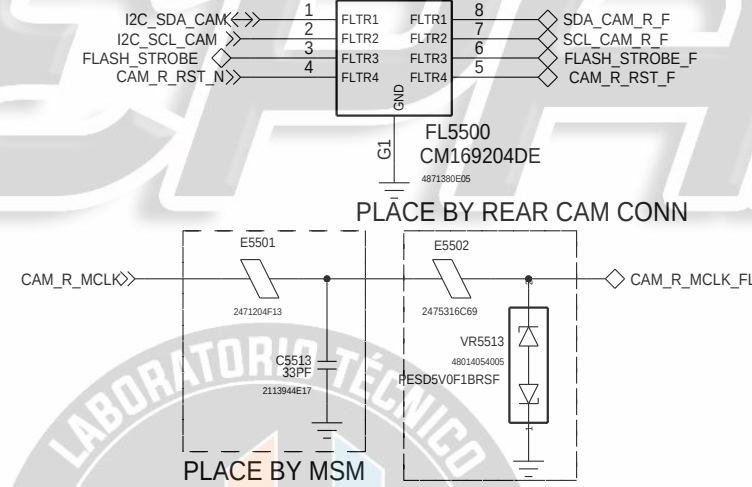


IMAGER FLEX CONNECTOR

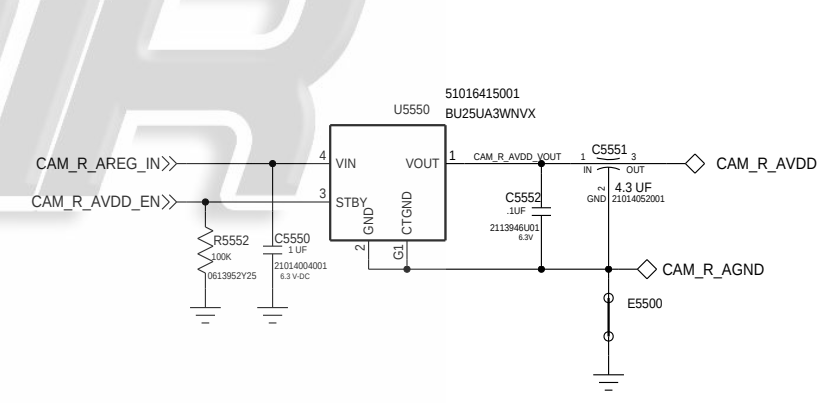


*** IMAGE SENSOR I2C ADD. 0X1A
*** AUTO FOCUS DRIVER I2C ADD. 0X72
*** IMAGE SENSOR EEPROM I2C ADD. 0X72

EMI/ESD FILTER



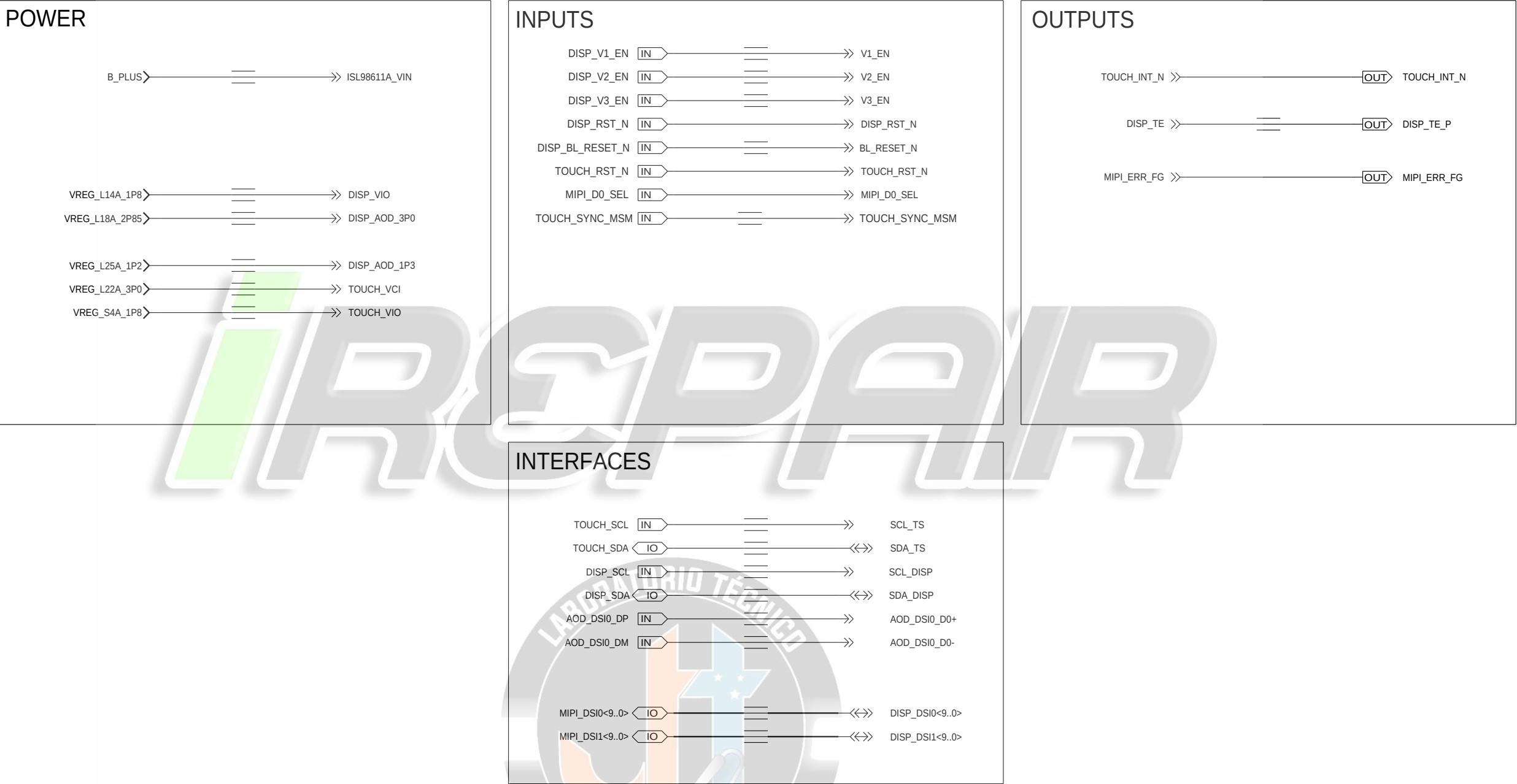
ANALOG REGULATOR



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue May 12 10:24:04 2015		imagers	
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CAD FILE NAME:		PAGE 12 OF 78	

DISP: PORTS

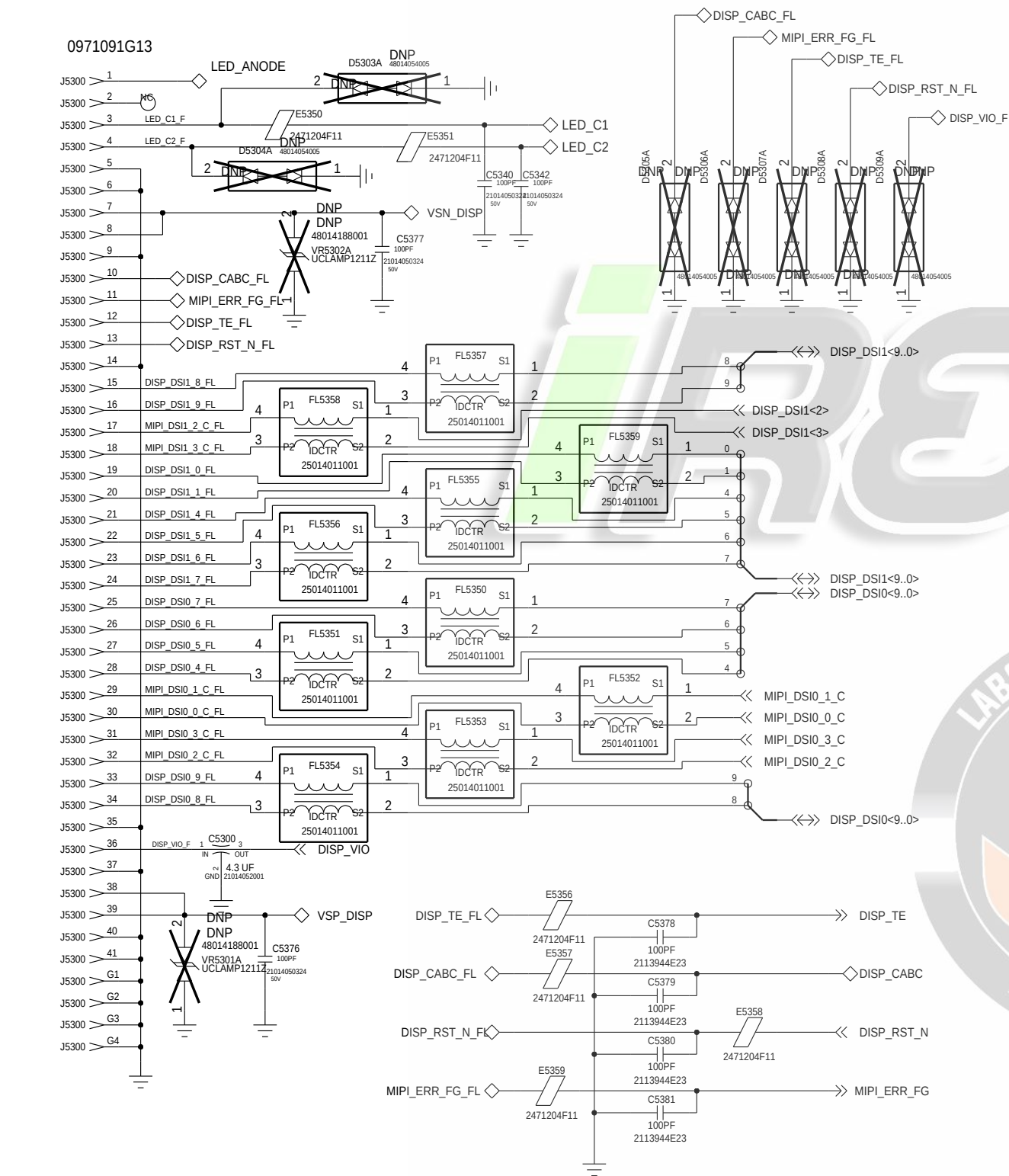
REF 5000-5499



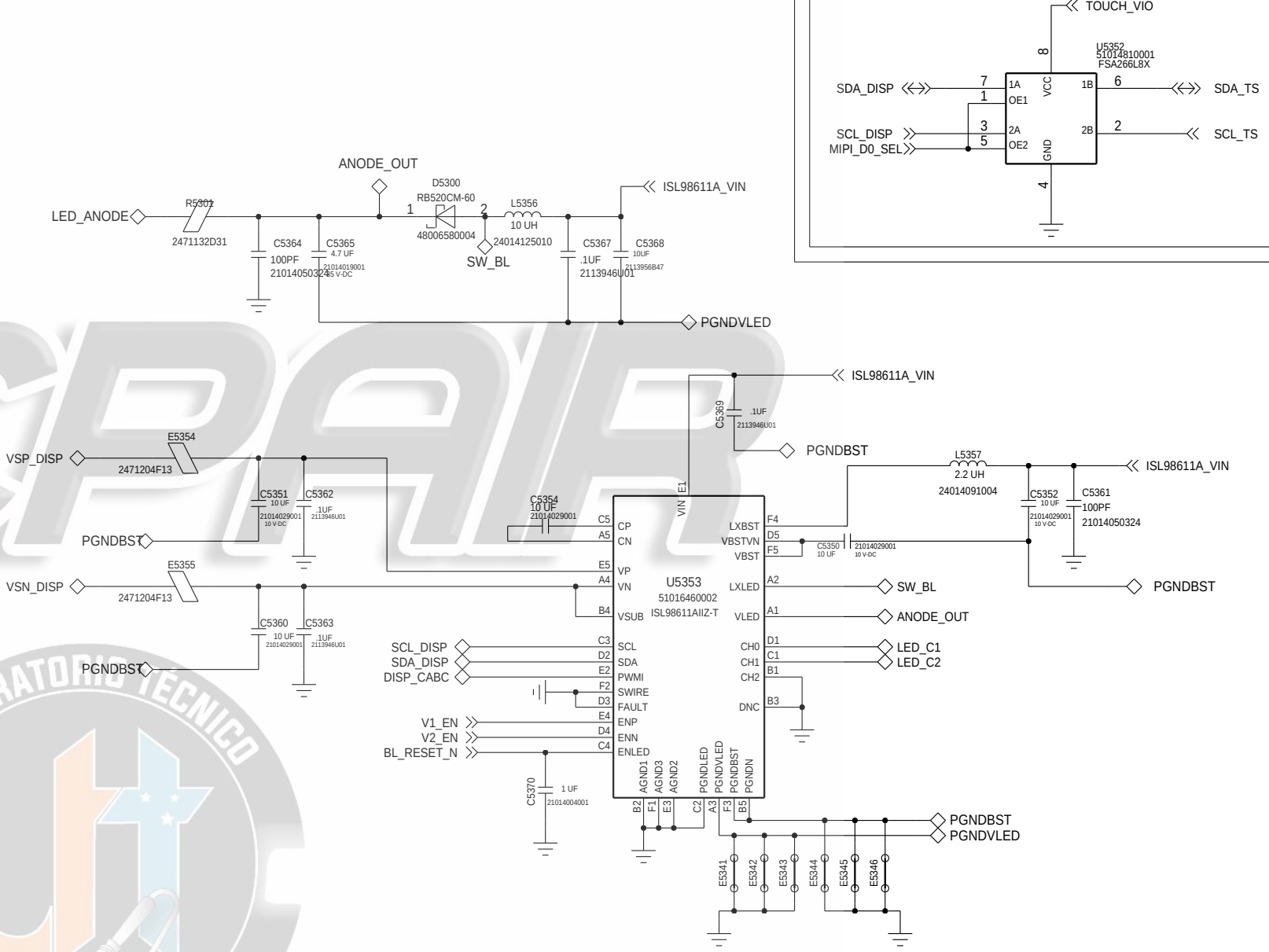
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Tue May 12 10:23:58 2015		display	
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		B 84016440001_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 13 OF 78	

DISP: CONNECTOR
REF 5300-5349

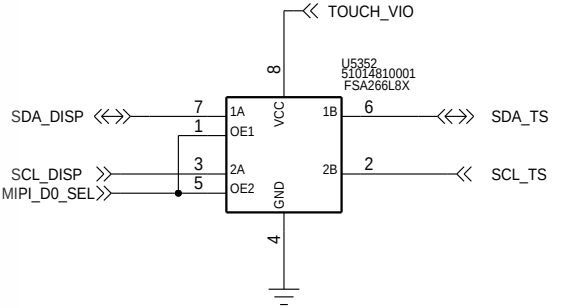
DISPLAY CONNECTOR



DISPLAY DUAL OUTPUT POWER SUPPLY
*** I2C ADD. 0X29



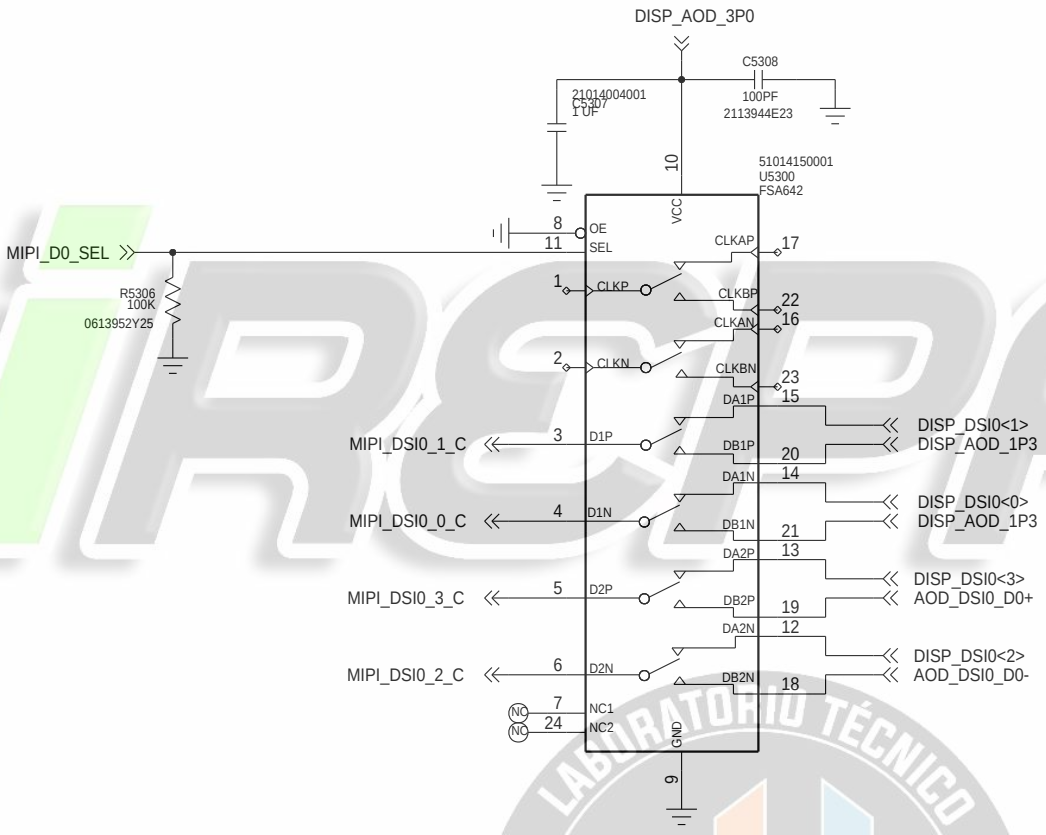
DISPLAY I2C PATH SWITCH



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		BLOCK TITLE:	
ENGINEER: NATE CONNELL		display	
LAST MODIFIED BY: NATE CONNELL Tue May 12 10:23:59 2015		PROJECT NAME 84016440001_a.cpm	
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CAD FILE NAME:		PAGE 14 OF 78	

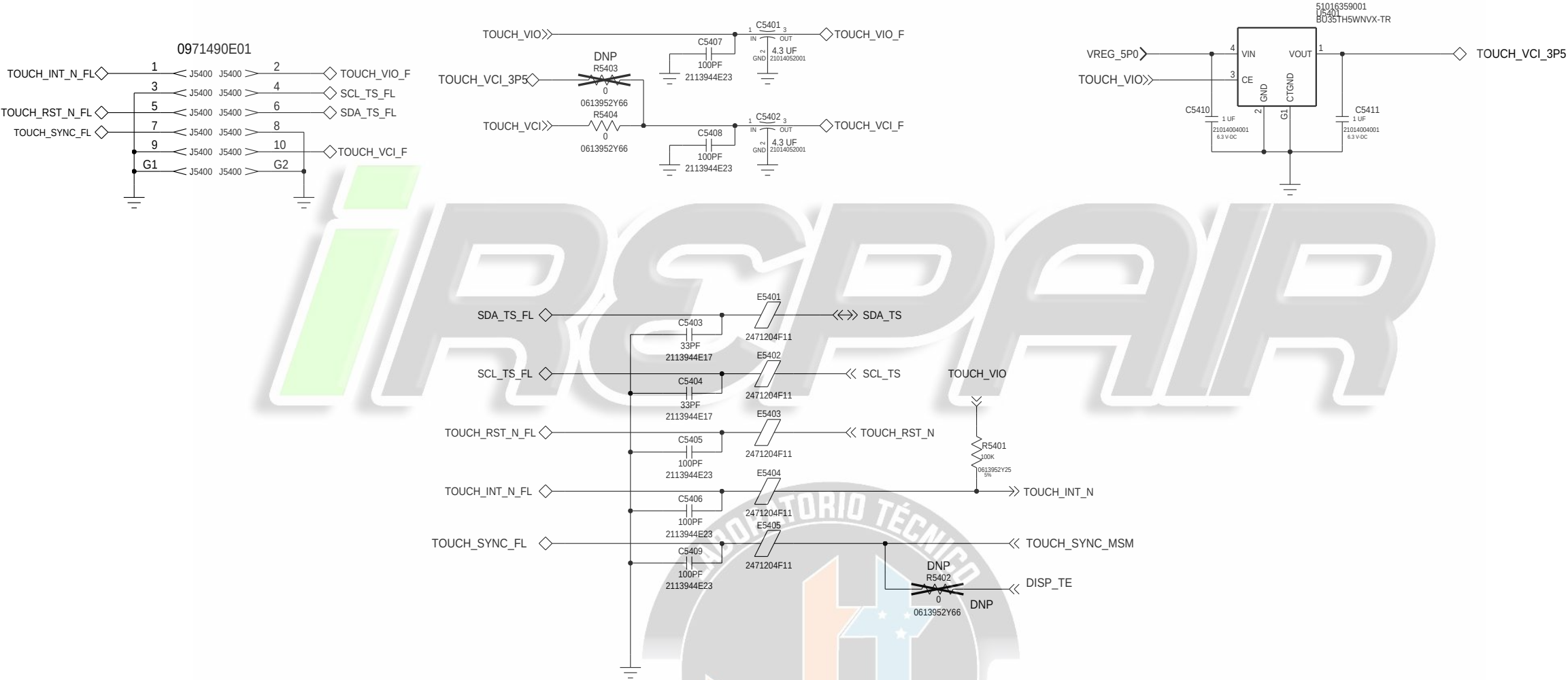
DISP: AOD



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Tue May 12 10:24:00 2015		display	
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		REV 01	
		CAD FILE NAME:	
		PAGE 15 OF 78	

TOUCH: CONNECTOR
REF 5400-5439

*** TOUCH SENSOR I2C ADD. 0X20

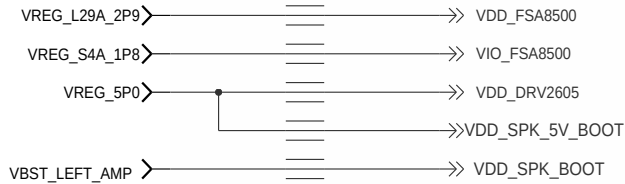


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

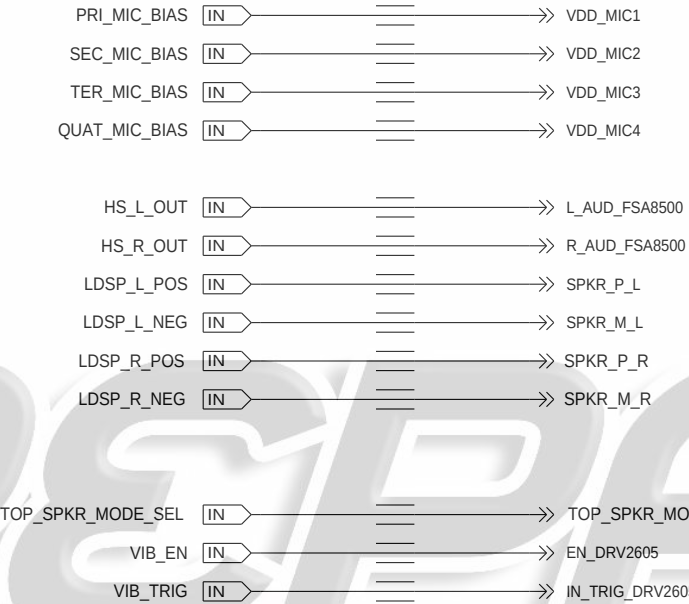
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ENGINEER: NATE CONNELL		BLOCK TITLE: display	
LAST MODIFIED BY: NATE CONNELL Tue May 12 10:24:01 2015			
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		CAD FILE NAME: PAGE 16 OF 78	

AFE: PORTS
REF 4500-4999

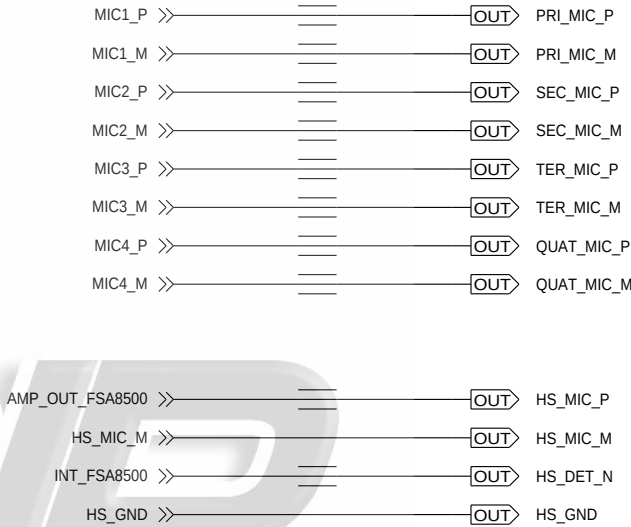
POWER



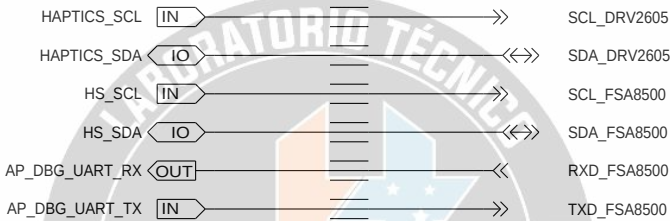
INPUTS



OUTPUTS

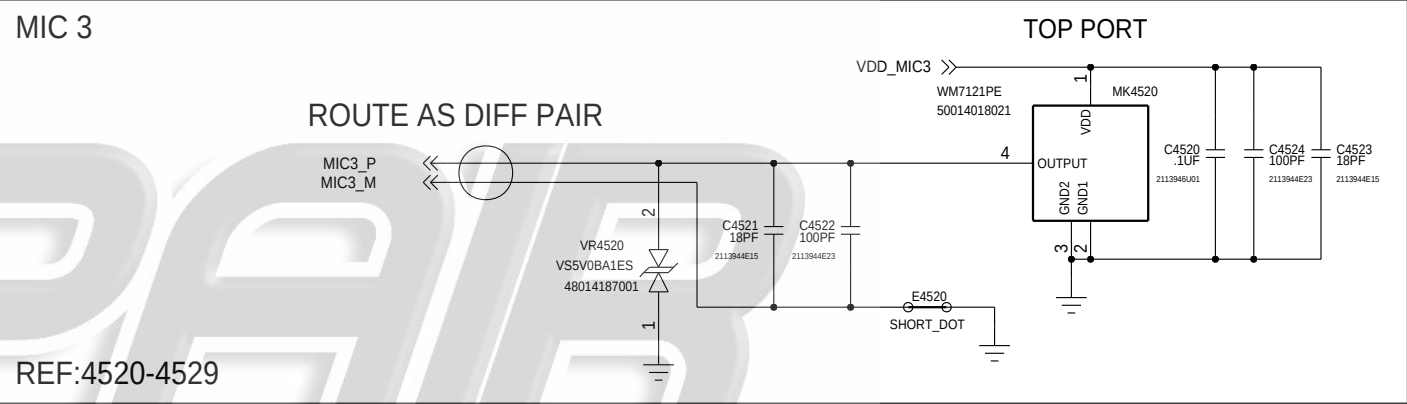
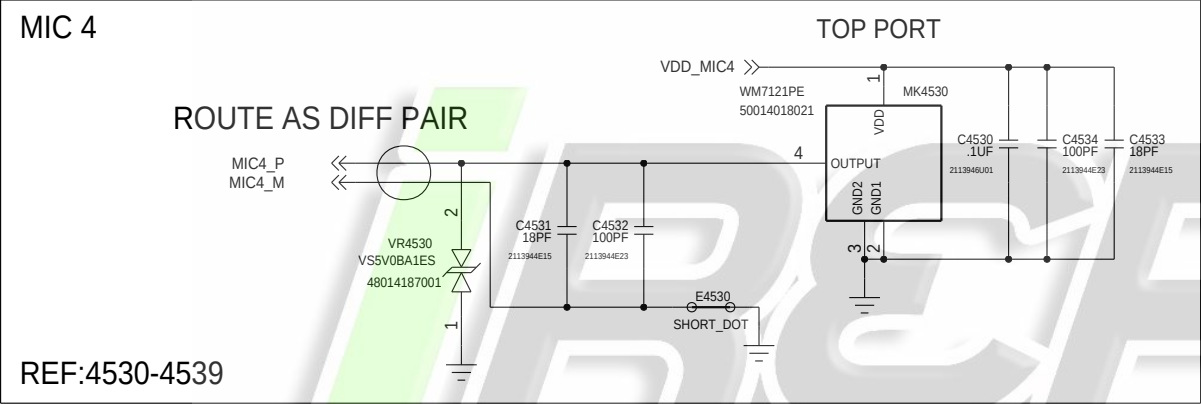
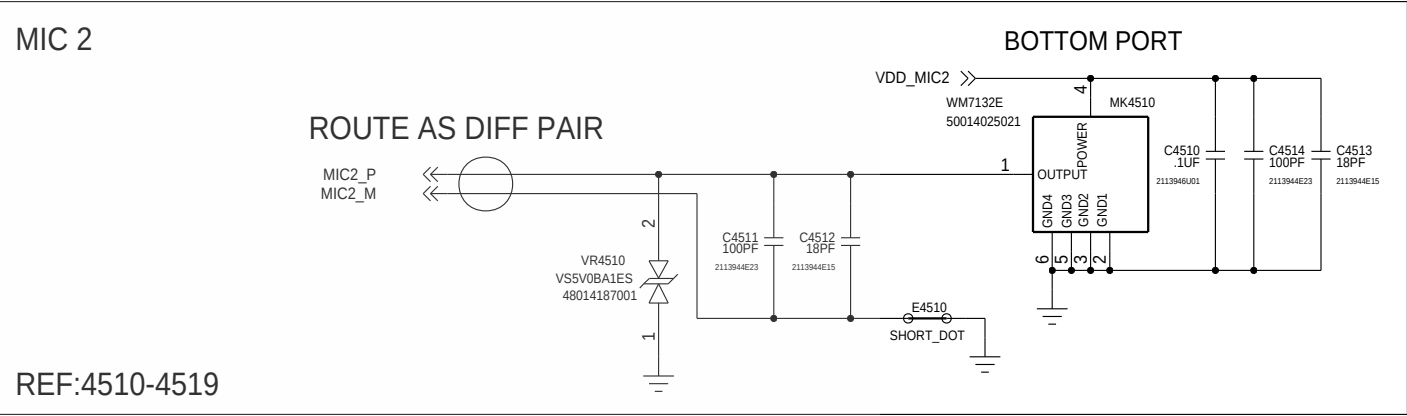
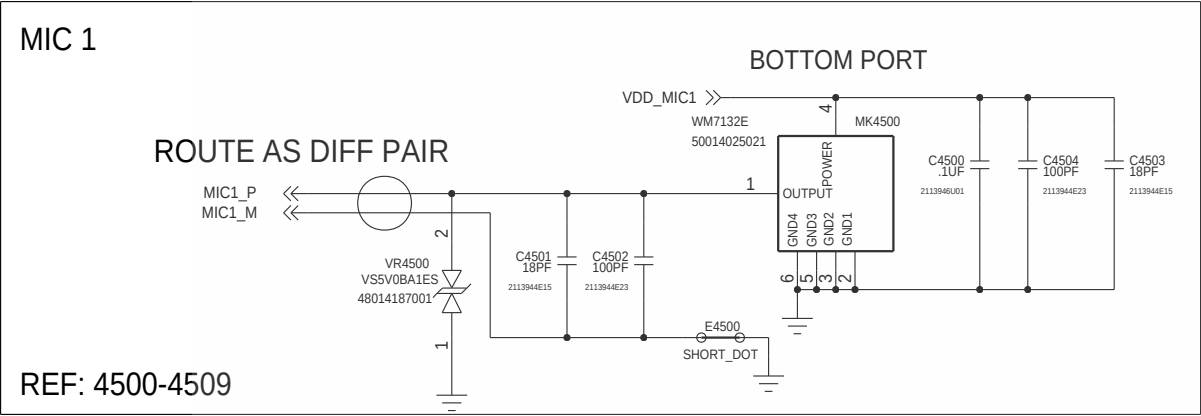


INTERFACES

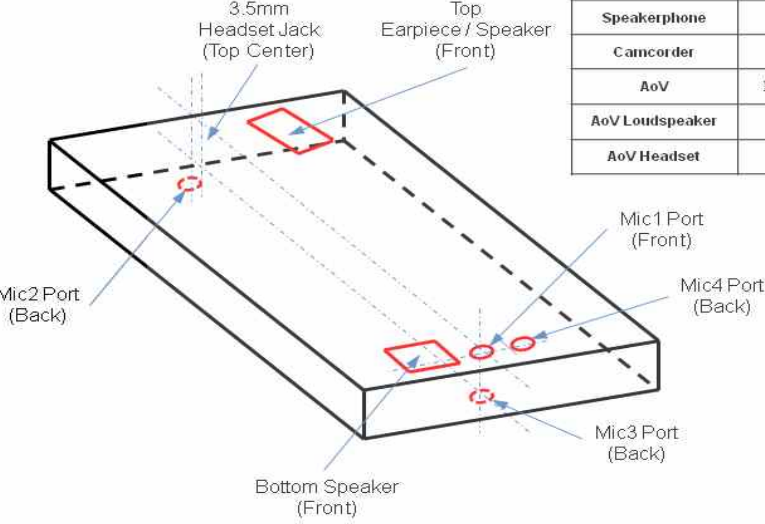


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Tue May 12 10:23:54 2015		audio_fe	
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		REV 01 PAGE 17 OF 78	

AFE: MICS
REF: 4500-4599



Clark (Front View)

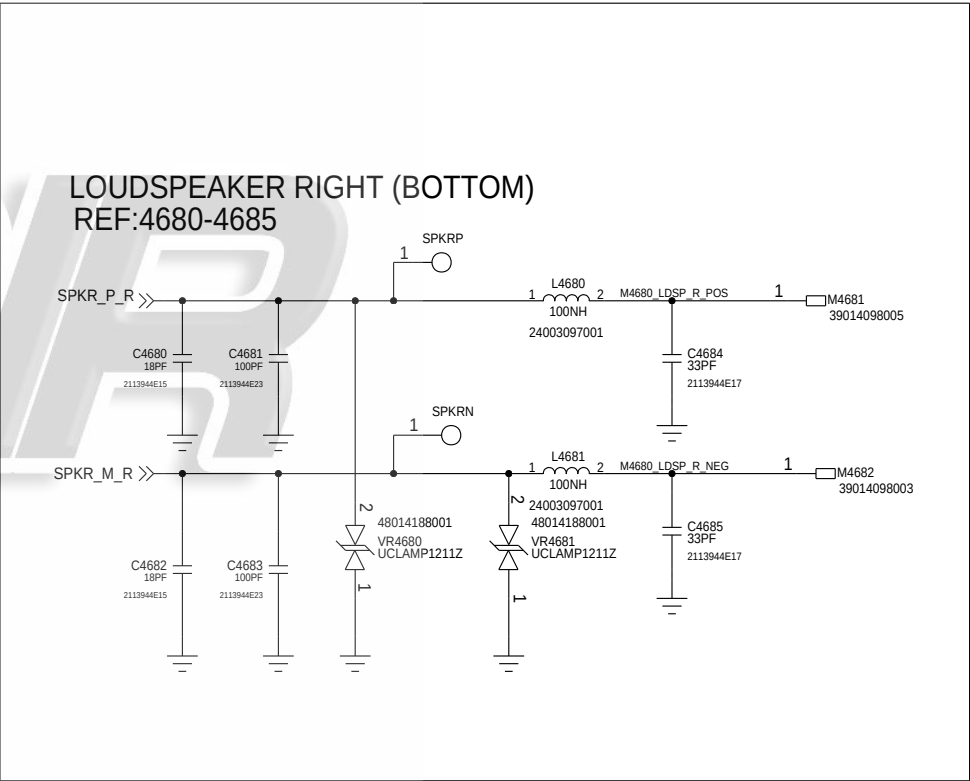
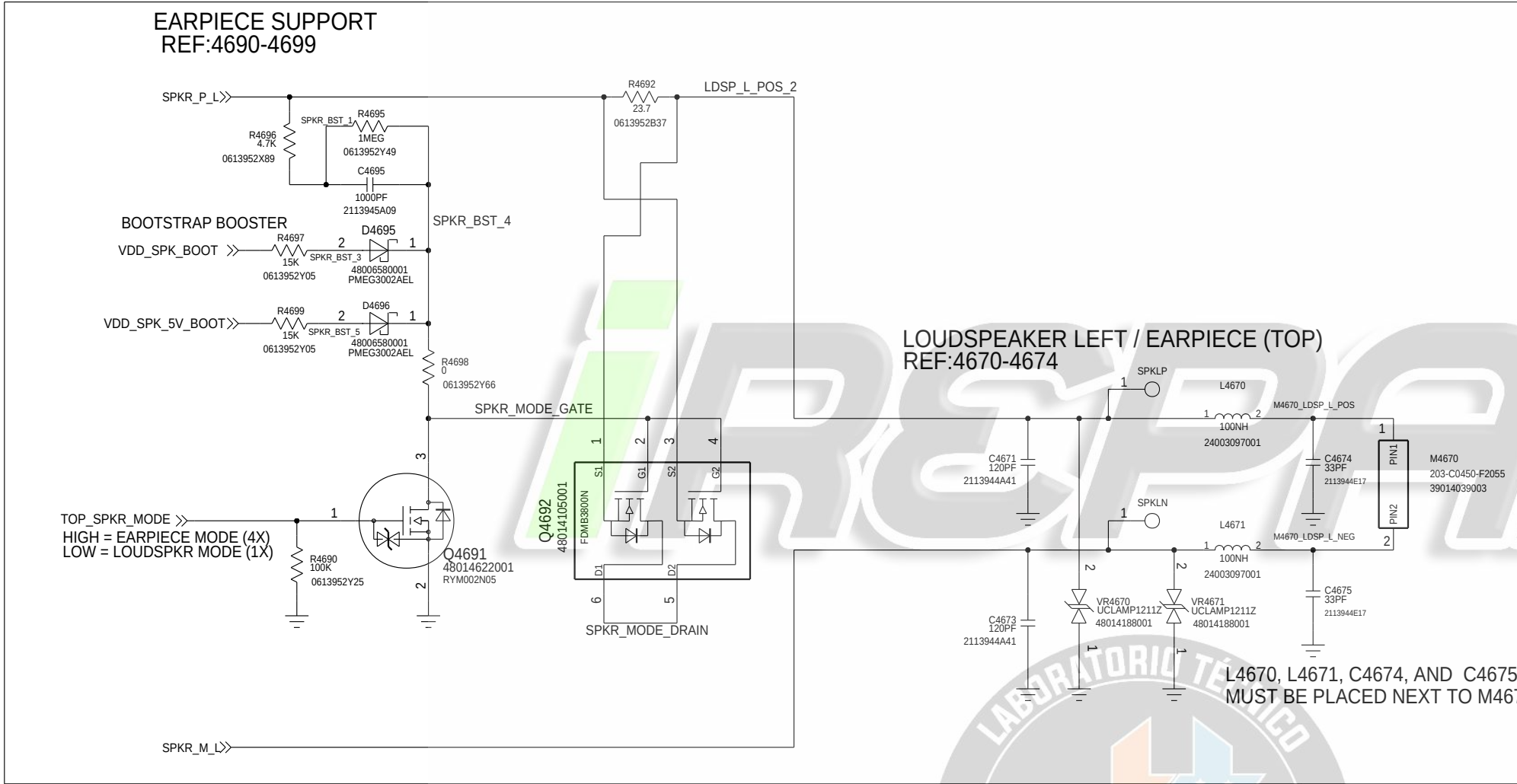


	Mic1	Mic2	Mic3	Mic4	HS Mic
Voice Handset	X	X			
Speakerphone		X			
Camcorder		X	X		
AoV	X		X	X	
AoV Loudspeaker		X			
AoV Headset					X

APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MOTOROLA		BLOCK TITLE: audio_fe		
LAST MODIFIED BY: MOTOROLA Tue May 12 10:23:55 2015				
MOTOROLA CONFIDENTIAL RESTRICTED		B	PROJECT NAME 84016440001_a.cpm	REV 01
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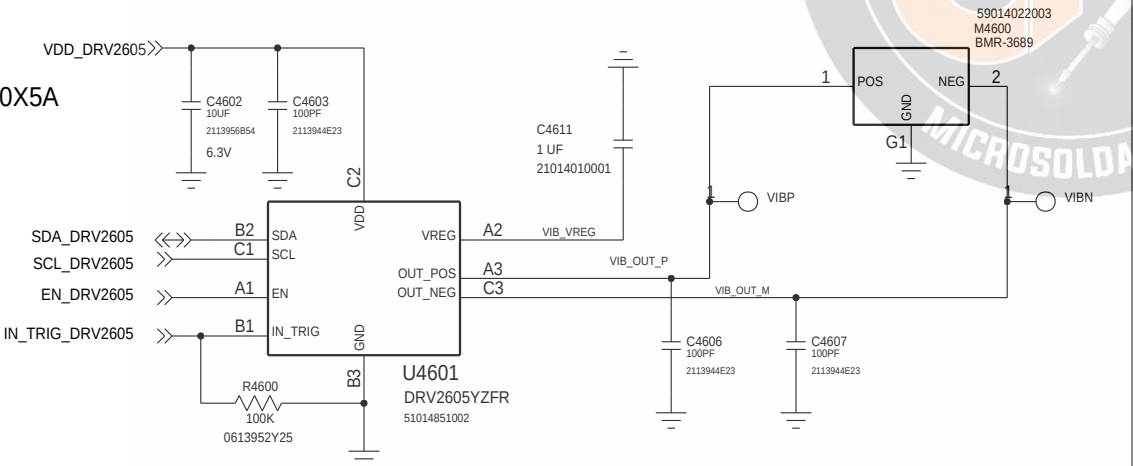
AFE: SPEAKERS

REF: 4600-4699



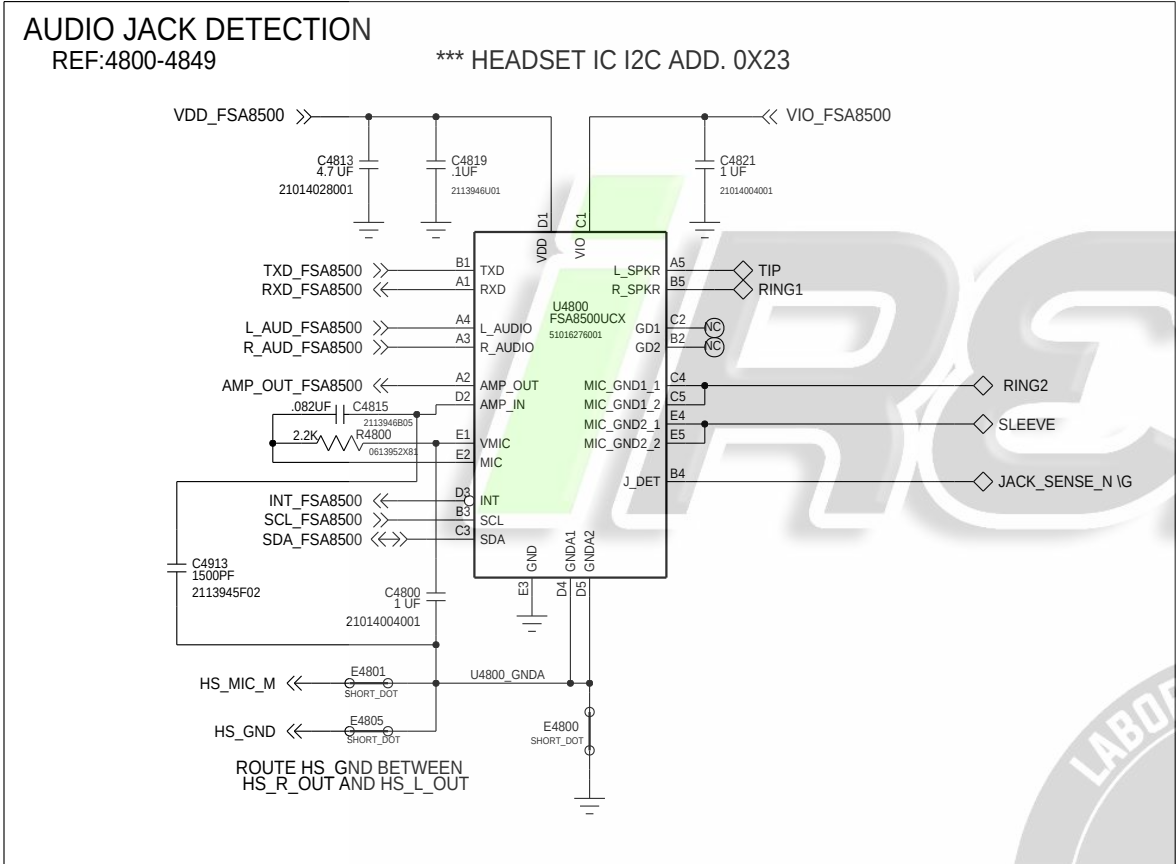
ROTARY VIBRATOR
REF DES 4600-4619

*** VIBRATOR DRIVER I2C ADD. 0X5A

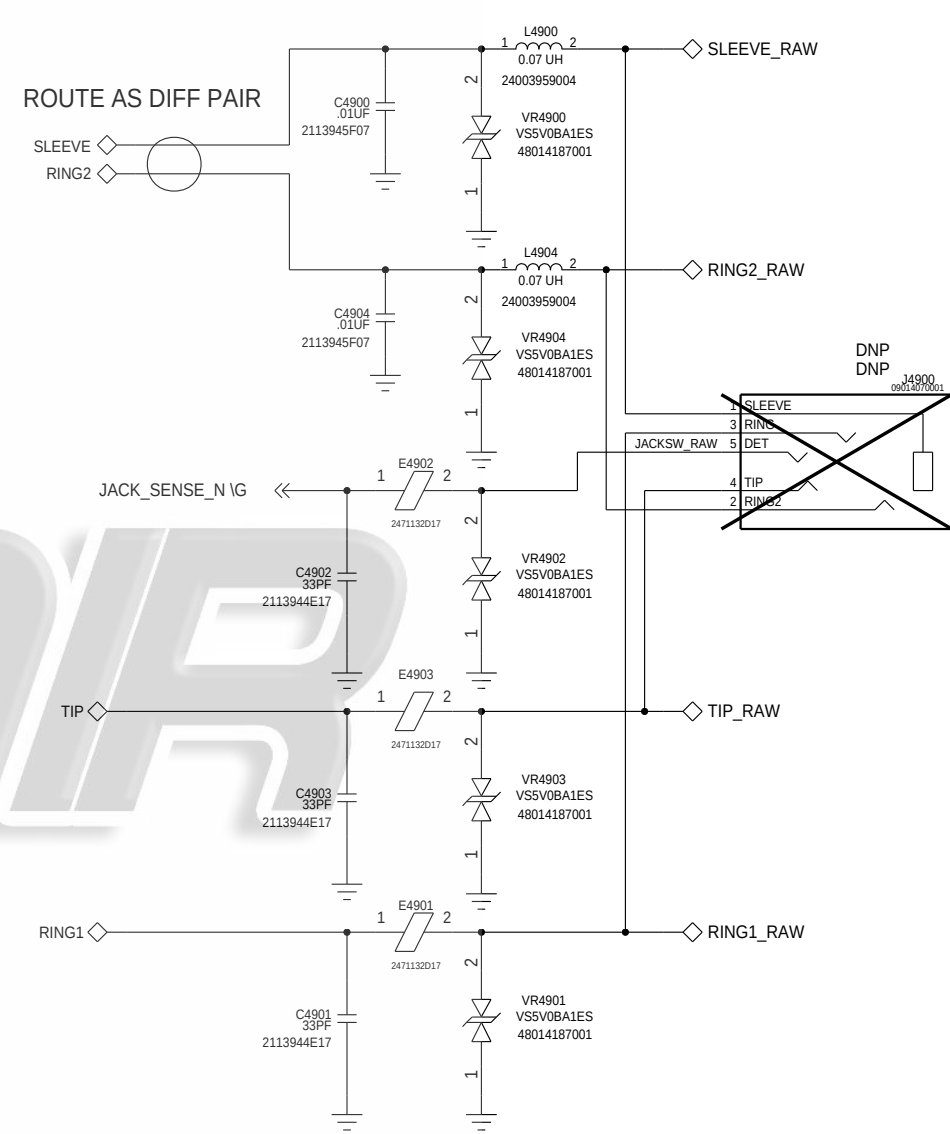


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: MOTOROLA Tue May 12 10:23:56 2015		PROJECT NAME 84016440001_a.cpm	
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		PAGE 19 OF 78	

AFE: 3.5MM HEADSET



3.5MM HSJ
REF:4900-4949



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

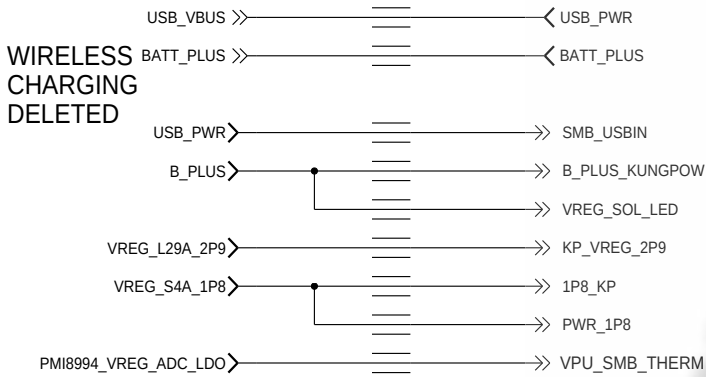
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MATT MOTTIER		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: NATE CONNELL Tue May 12 10:23:57 2015		PROJECT NAME 84016440001_a.cpm	
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REV 01			

PWR: PORTS

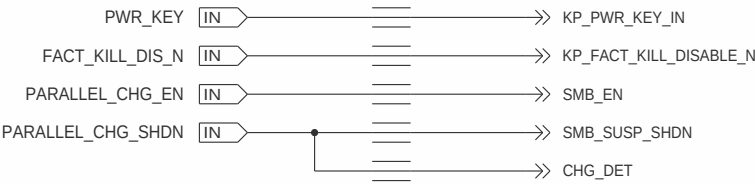
REF 500-799

POWER

WIRELESS
CHARGING
DELETED



INPUTS



OUTPUTS



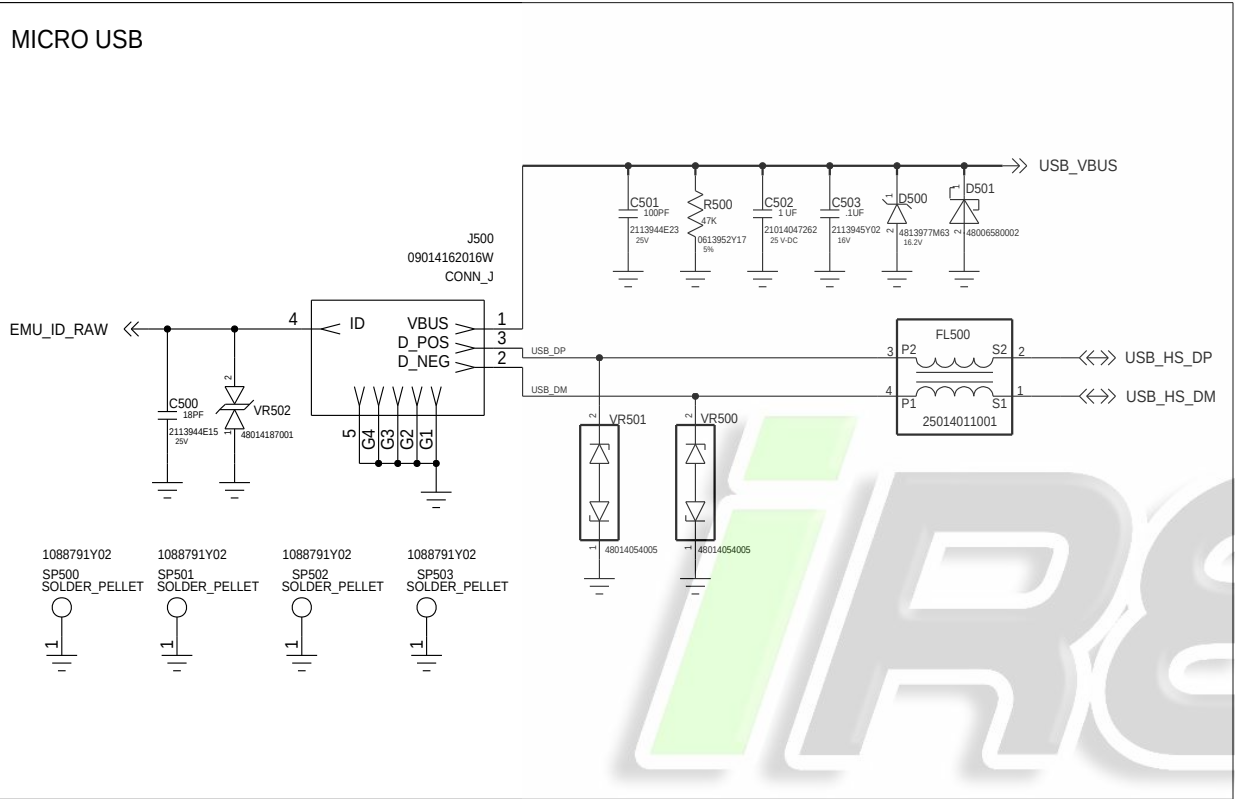
INTERFACES



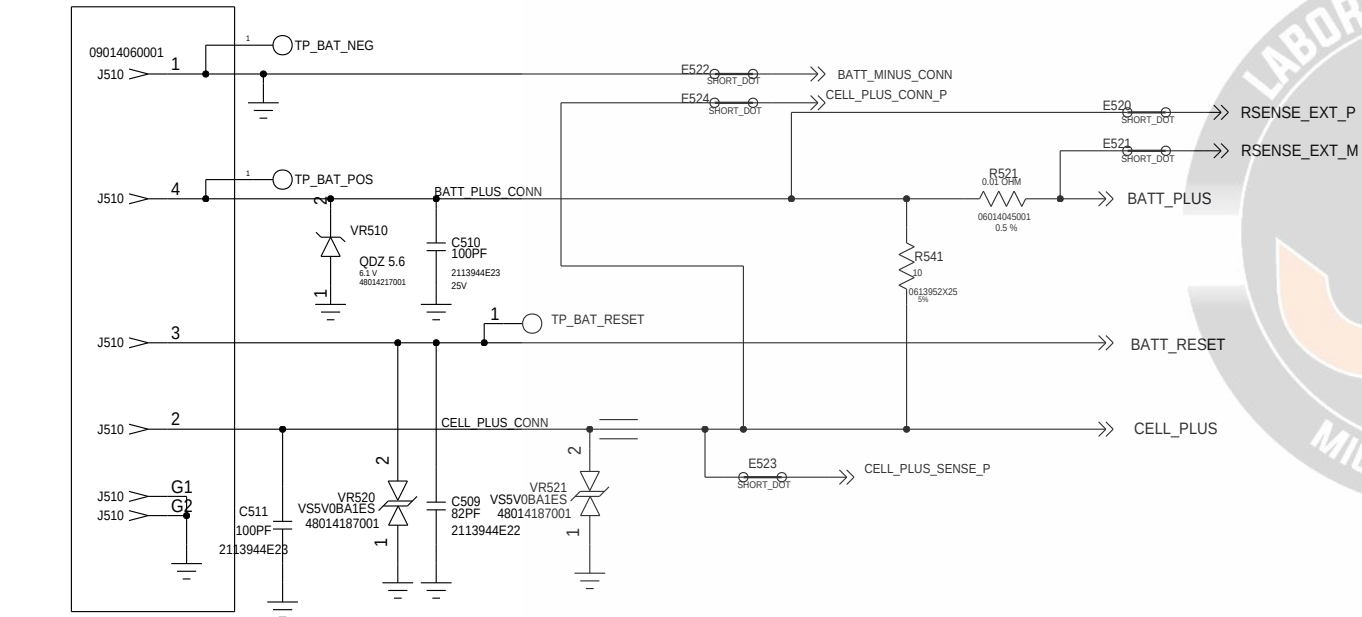
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Tue May 12 10:23:51 2015		power	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B PROJECT NAME 84016440001_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 21 OF 78	

PWR: USB2 / BATT

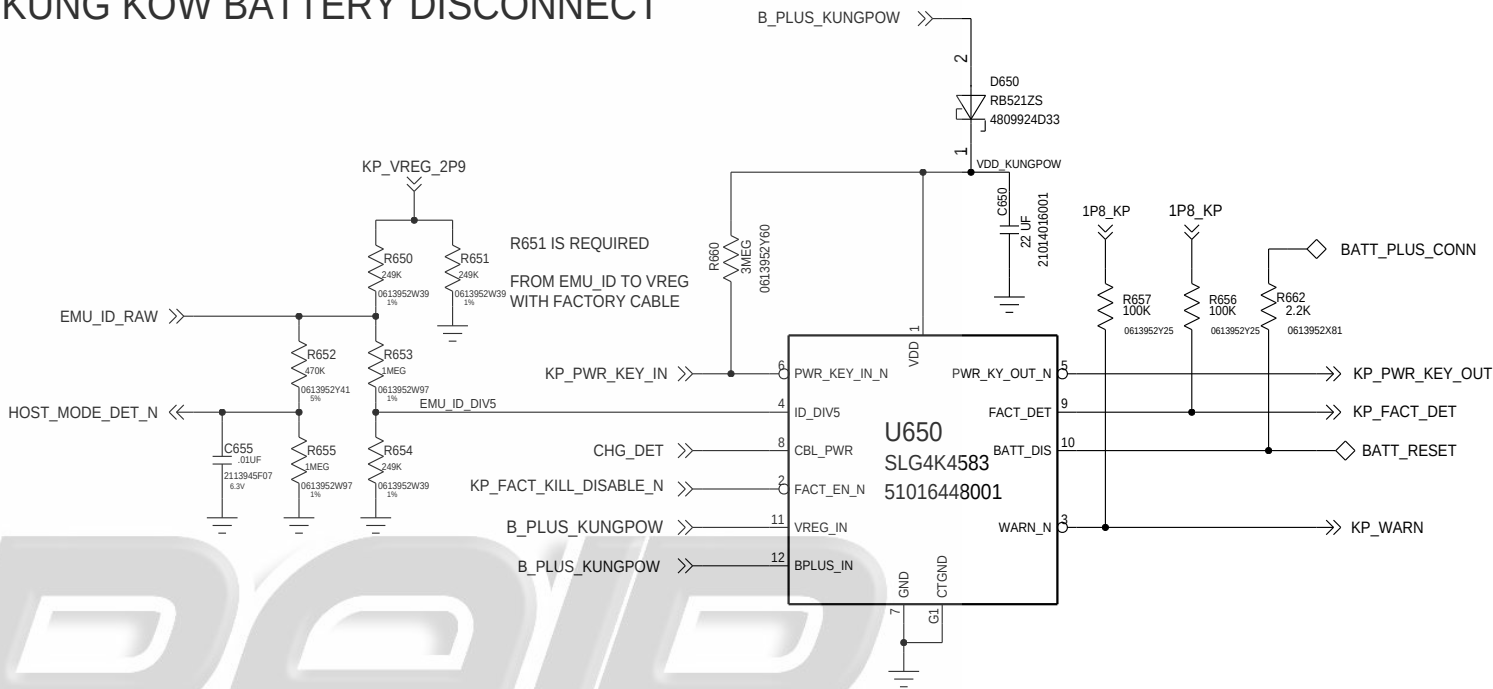
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BATTERY BLOCK



KUNG KOW BATTERY DISCONNECT

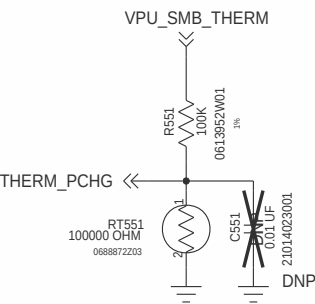
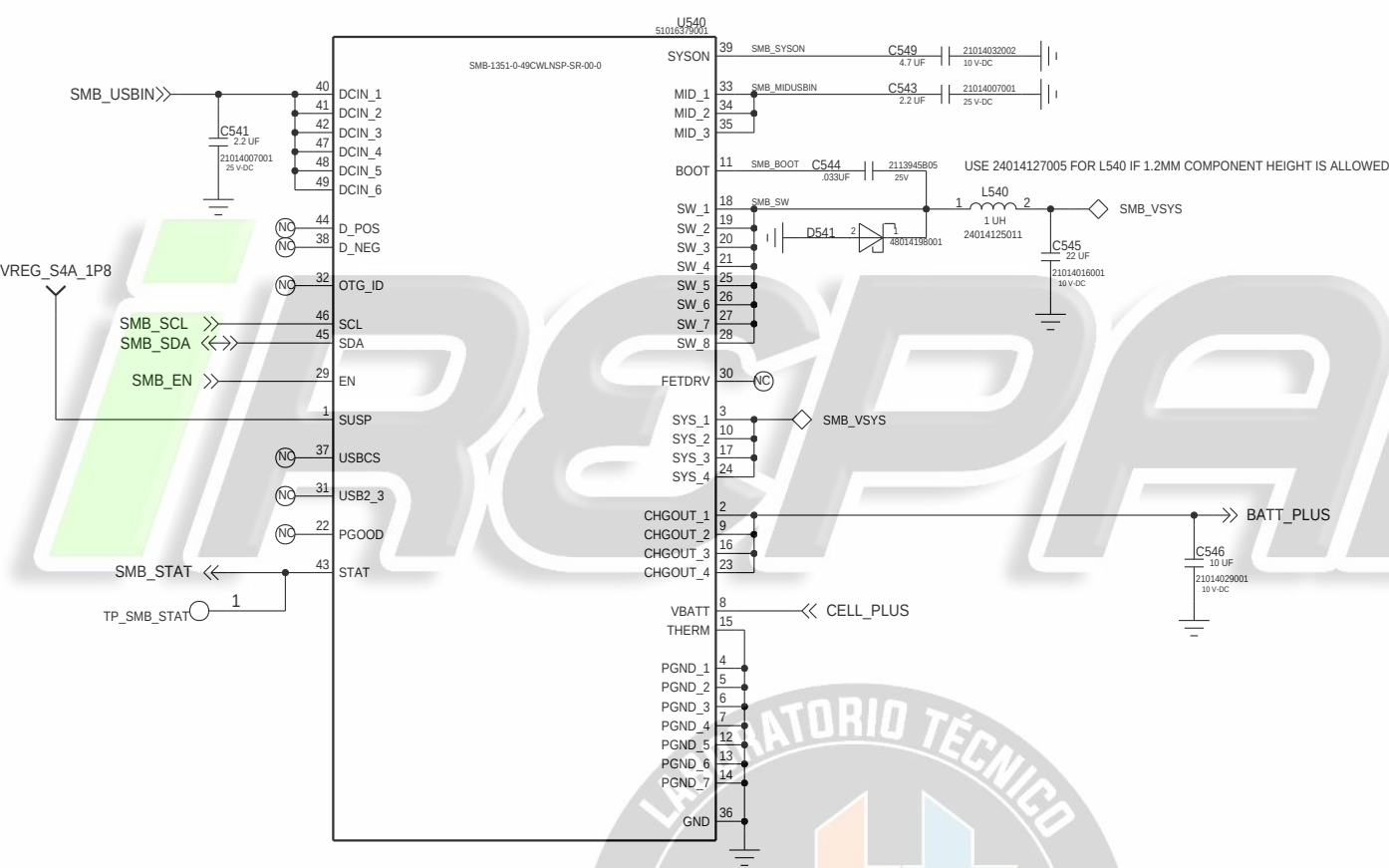


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG		BLOCK TITLE: power	
LAST MODIFIED BY: UDI ELRAN Tue May 12 10:23:52 2015		PROJECT NAME 84016440001_a.cpm	
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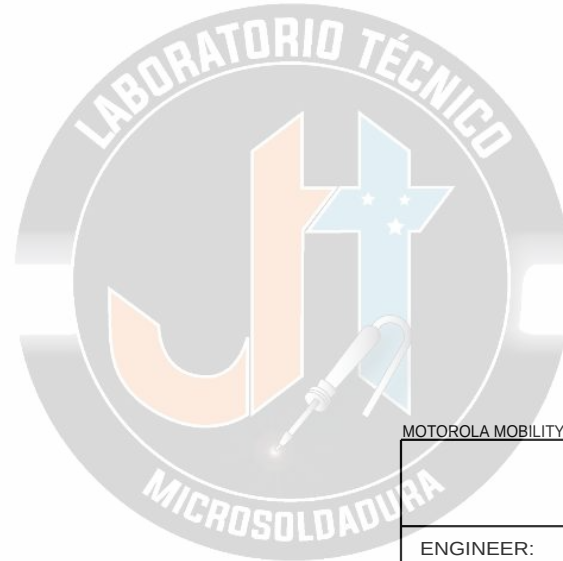
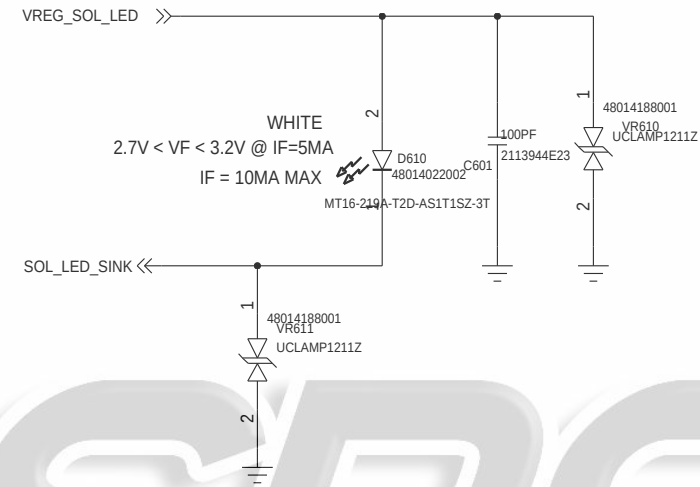
PWR: EXT CHG

REF: 540-559



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: R. NG Tue May 12 10:23:53 2015		power	
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		CAD FILE NAME:	PAGE 23 OF 78

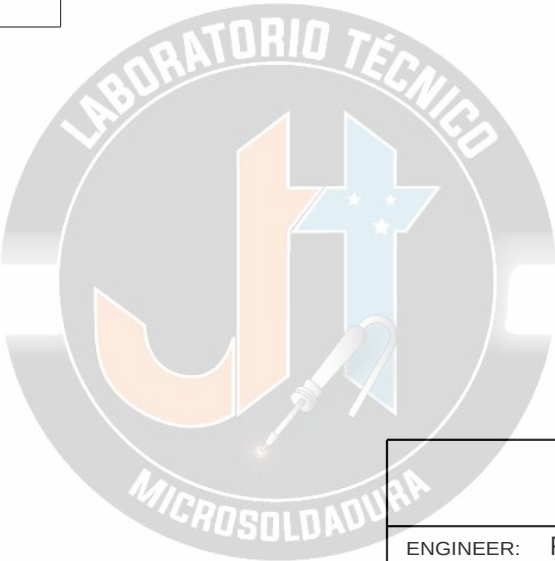
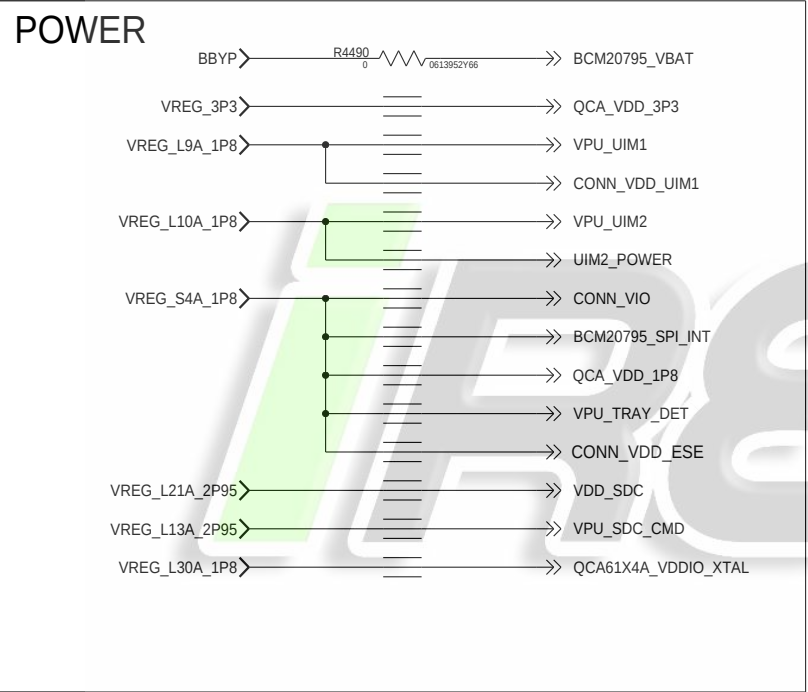
SOL LED
REF: 610-619



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: power		
LAST MODIFIED BY: TIM MILLER Tue May 12 10:23:53 2015			
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		CAD FILE NAME:	PAGE 24 OF 78

CONN: POWER
REF 4490-4499



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE: connectivity	
LAST MODIFIED BY: FROEHLING Tue May 12 10:23:44 2015			
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		CAD FILE NAME: PAGE 25 OF 78	

CONN: PORTS

REF 4000-4499

BT/WLAN SYSTEM

BT_WLAN_32KHZ_CLK_IN IN QCA61X4_32KHZ_CLK_IN

WLAN

WLAN_ENABLE IN QCA61X4_WL_EN

PCIE

PCIE_CLK_REQ_OUT_N OUT QCA61X4_PCIE_CLKREQ_N
PCIE_CLKIN_DP IN QCA61X4_PCIE_REFCLKP
PCIE_CLKIN_DM IN QCA61X4_PCIE_REFCLKN
PCIE_RX_DP IN QCA61X4_PCIE_RXP
PCIE_RX_DM IN QCA61X4_PCIE_RXN
PCIE_TX_DP OUT QCA61X4_PCIE_TXP
PCIE_TX_DM OUT QCA61X4_PCIE_TXN
PCIE_RST_IN_N IN QCA61X4_PCIE_RST_N
PCIE_HOST_WAKE_OUT_N OUT QCA61X4_PCIE_WAKE_N

BT CONTROLS AND INTERFACES

BT_ENABLE IN QCA61X4_BT_EN

BT_PCM_IN IN QCA61X4_PCM_IN
BT_PCM_OUT OUT QCA61X4_PCM_OUT
BT_PCM_CLK IN QCA61X4_PCM_CLK
BT_PCM_SYNC IO QCA61X4_PCM_SYNC

BT_UART_TX OUT QCA61X4_UART_TXD
BT_UART_RX IN QCA61X4_UART_RXD
BT_UART_CTS_N IN QCA61X4_UART_CTS_N
BT_UART_RTS_N OUT QCA61X4_UART_RTS_N

CELLULAR MODEM/GPS COEXISTENCE

MODEM_LTE_SYNC_UART_RXD IN QCA61X4_LTE_SYNC
WLAN_LTE_PRI_UART_TXD OUT QCA61X4_LTE_PRI
WLAN_GPS_COEX OUT QCA61X4_GPS_COEX

NFC

NFC_CLK_IN IN BCM20795_EXT_CLK_IN
NFC_CLK_REQ_OUT OUT BCM20795_CLK_REQ_OUT
NFC_ENABLE_IN IN BCM20795_ENABLE_IN
NFC_IRQ_OUT OUT BCM20795_IRQ_OUT
NFC_SCL IN BCM20795_SCL
NFC_SDA IO BCM20795_SDA
NFC_POWER_ON IN BCM20795_REG_PU

SDCARD

SDC_CLK IN CLK_SDC
SDC_CMD IN CMD_SDC
SDC_DATA<3..0> IO DATA_SDC<3..0>

UIM1

UIM1_DATA IO UIM1_DATA
UIM1_CLK IN UIM1_CLK
UIM1_RST IN UIM1_RST

UIM2

UIM2_DATA IO UIM2_DATA
UIM2_CLK IN UIM2_CLK
UIM2_RST IN UIM2_RST

ANTENNA

STREAM0_BT IO BT_WLAN_ANT0
STREAM1 IO WLAN_ANT1

TRAY_DET OUT TRAY_DET

APPROVALS

ENGINEER: FROEHLING

LAST MODIFIED BY: FROEHLING
Mon Mar 30 17:43:19 2015

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

connectivity

PROJECT NAME
84016440001_a.cpm
REV
01

CAD FILE NAME:
PAGE 26 OF 78

QCA_VDD_3P3

QCA61X4_VDD33

C4013
10UF
2113956B54
6.3V

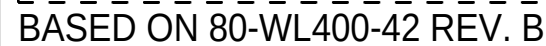
QCA_VDD_1P8

QCA61X4_DVDD_GPIO0

QCA61X4_DVDD_GPIO1

QCA61X4_DVDD_GPIO2

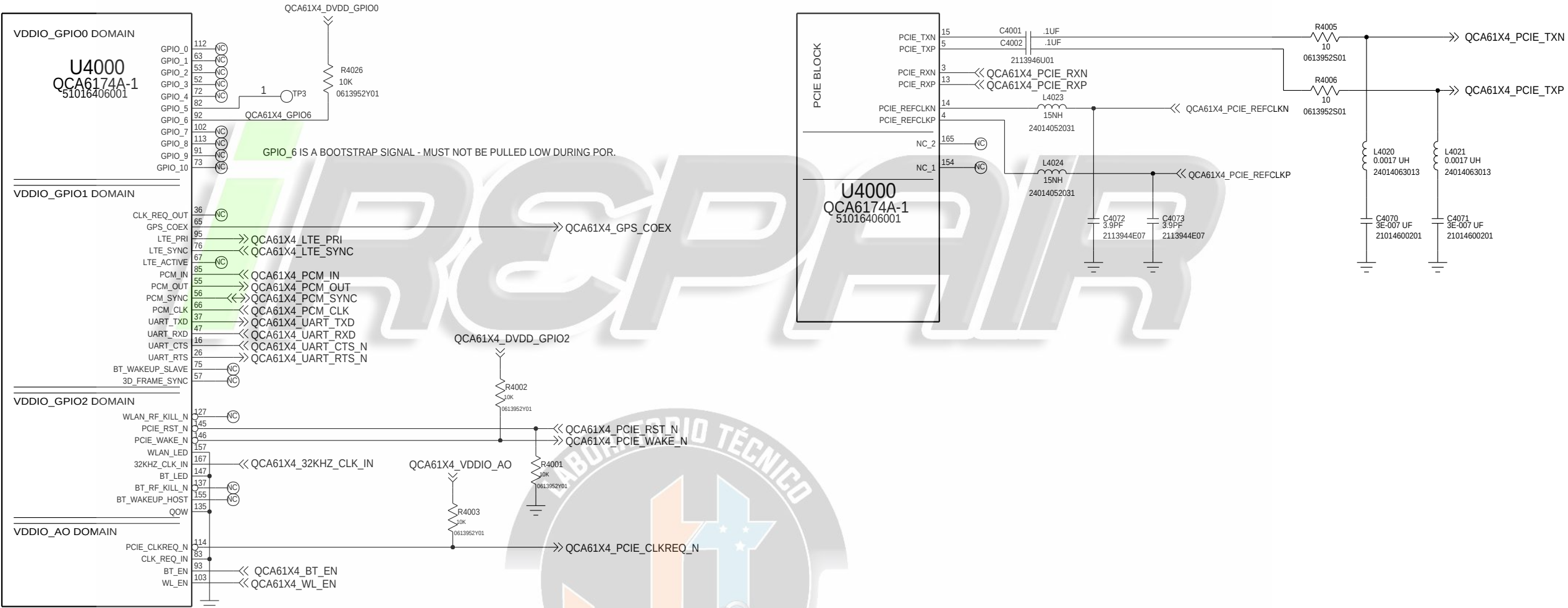
QCA61X4_VDDIO_AO



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

CONN: QCA61X4A HOST I/F

REF: 4000-4099

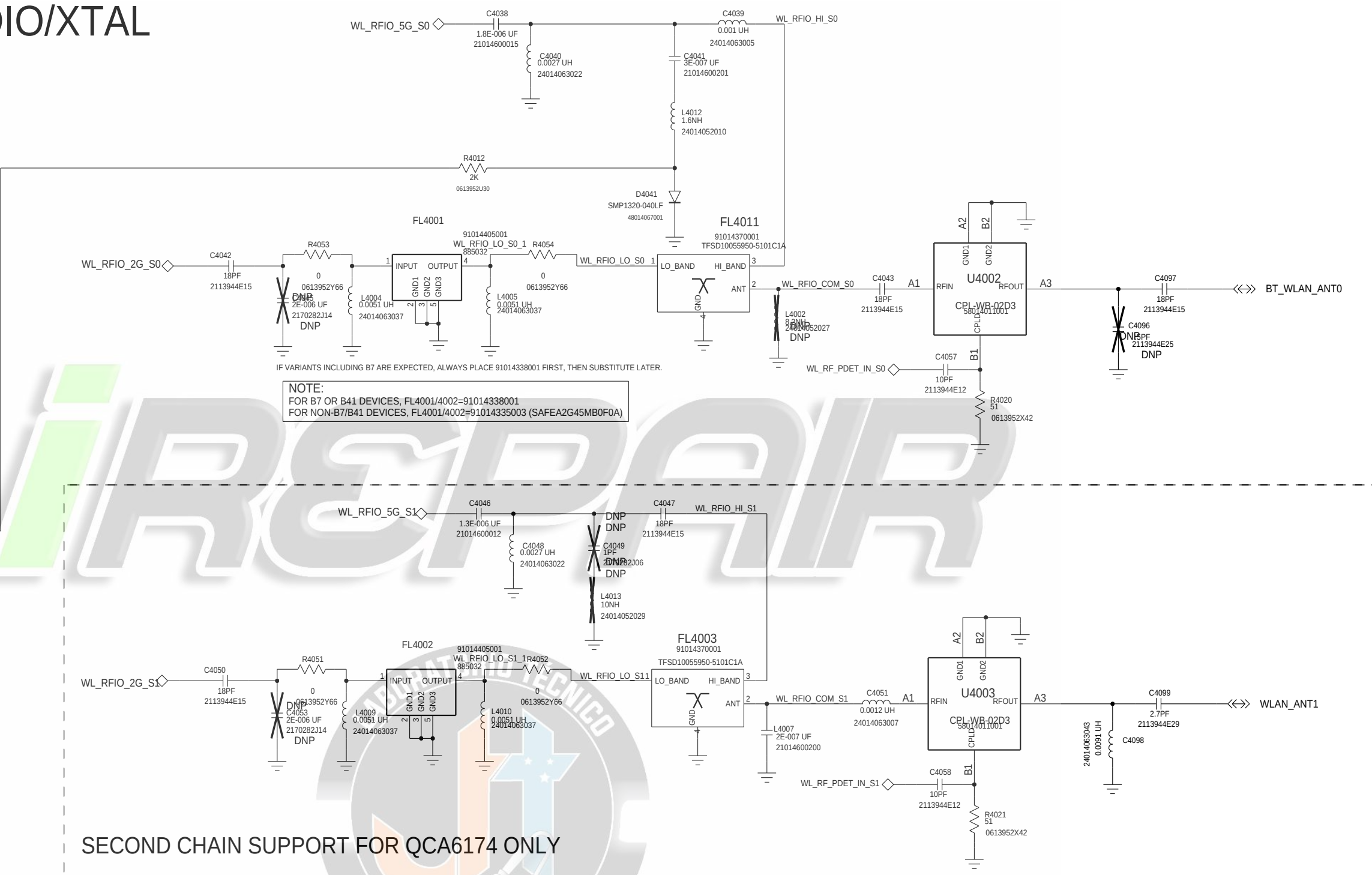
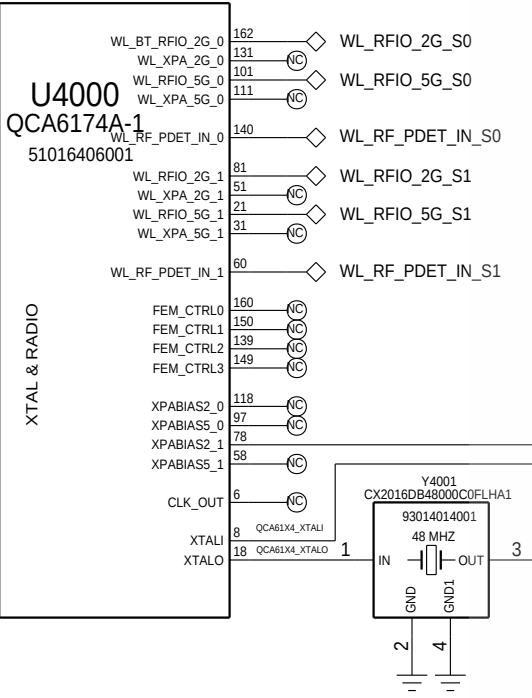


BASED ON 80-WL400/405-42 REV. B

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE: connectivity	
LAST MODIFIED BY: FROEHLING Tue May 12 10:23:46 2015		PROJECT NAME 84016440001_a.cpm	
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		CAD FILE NAME:	
		PAGE 28 OF 78	

CONN: QCA61X4A RADIO/XTAL
REF 4000-4099



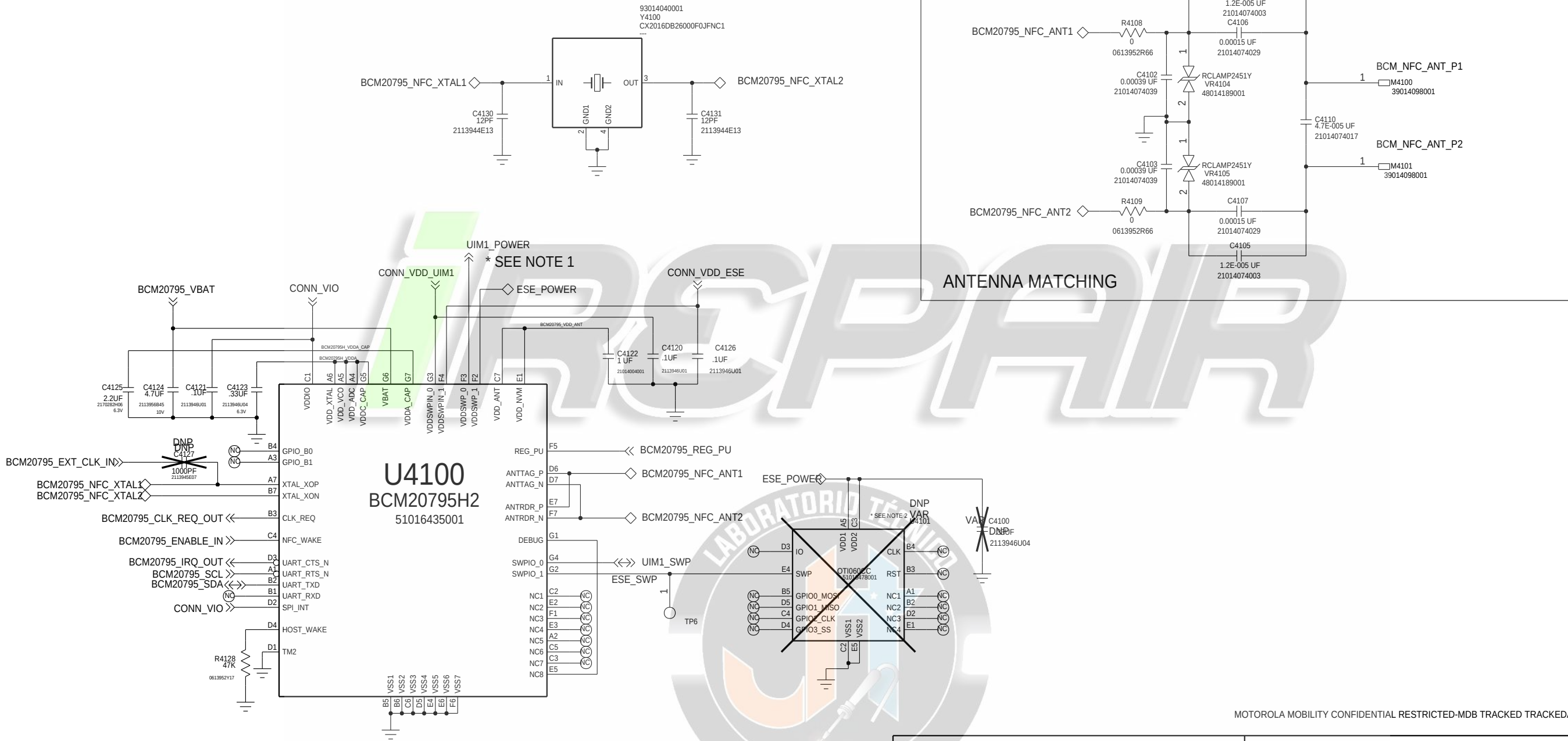
BASED ON 80-WL400-42 REV. B

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Tue May 12 10:23:47 2015		connectivity	
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CAD FILE NAME:		PAGE 29 OF 78	

CONN: BCM20795 NFC

REF: 4100-4199
I2C ADDR: 0X77 (BOOT DEFAULT WHEN PIN 3 IS PULLED TO GROUND)
OR
I2C ADDR: 0X76 (BOOT DEFAULT WHEN PIN 3 IS FLOATED/PULLED HIGH)



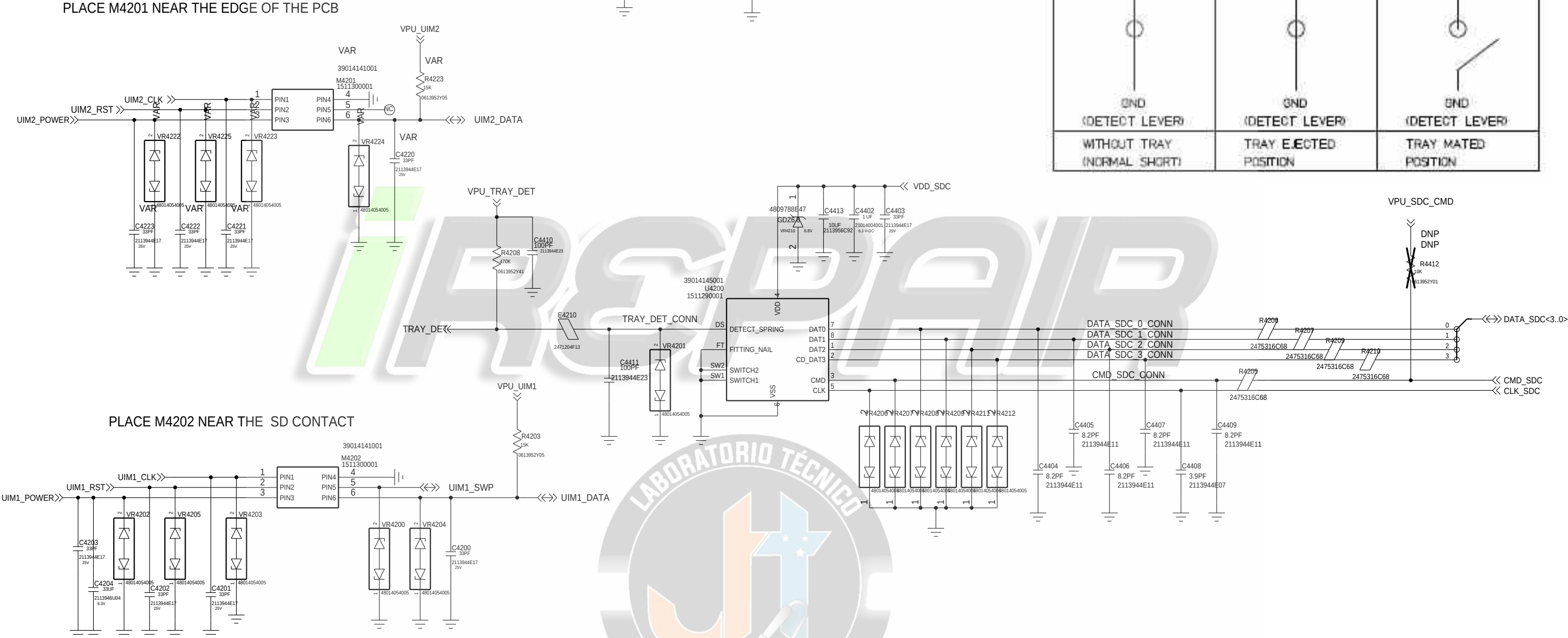
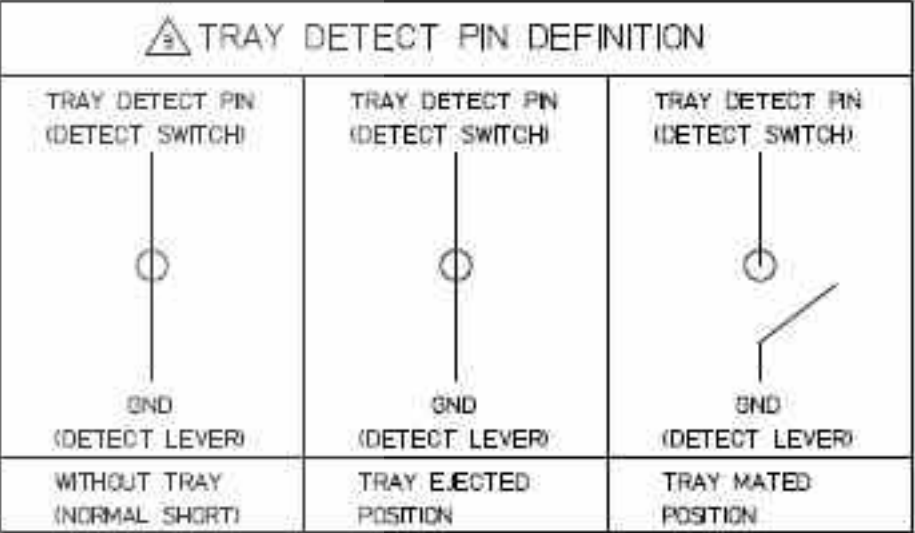
NOTES:
1. PER BROADCOM RECOMMENDATION, CAPACITANCE ON UIM1_POWER NET SHOULD BE 0.33UF. TOO LITTLE COMPROMISES REGULATOR STABILITY, TOO MUCH CAN CAUSE INADVERTENT POWER-ON RESETS AT VDD.
2. SECURE ELEMENT U4101 ONLY PLACED ON DSDA SKU (CHINA) REFER TO BELOW CHART:

	U4100	C4100
ATT/ROW	DNI	DNI
VZW/ SPRINT/ USC	DNI	DNI
TMO	DNI	DNI
CHINA	PLACE	PLACE

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Tue May 12 10:23:48 2015		connectivity	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		CAD FILE NAME:	PAGE 30 OF 78

CONN: UIM1/UIM2/SDCARD

REF:4200-4219



NOTE 1: USE 24.9K PULL-UP TO SUPPORT HOT SWAP WITH FREEZE-IO FEATURE.

NOTE 2: UIM_PWR BYPASS OF 0.33UF RECOMMENDED BY NFC IC SUPPLIER, PLACE NEAR CONNECTOR.

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Tue May 12 10:23:49 2015		connectivity	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		CAD FILE NAME:	PAGE 31 OF 78

USR: PORTS

REF: 800-999

CAMERA_FLASH

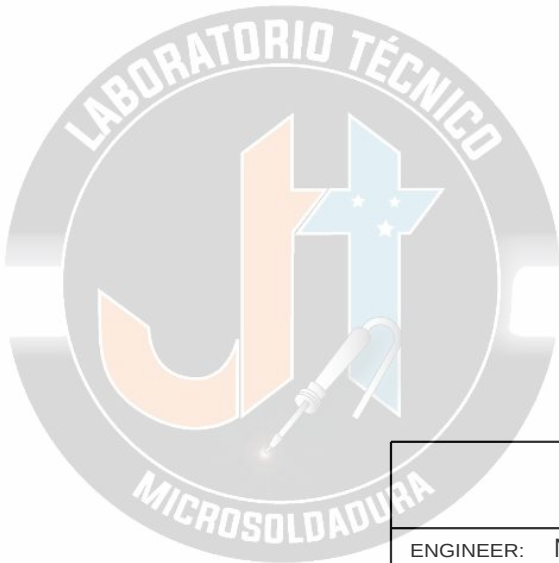
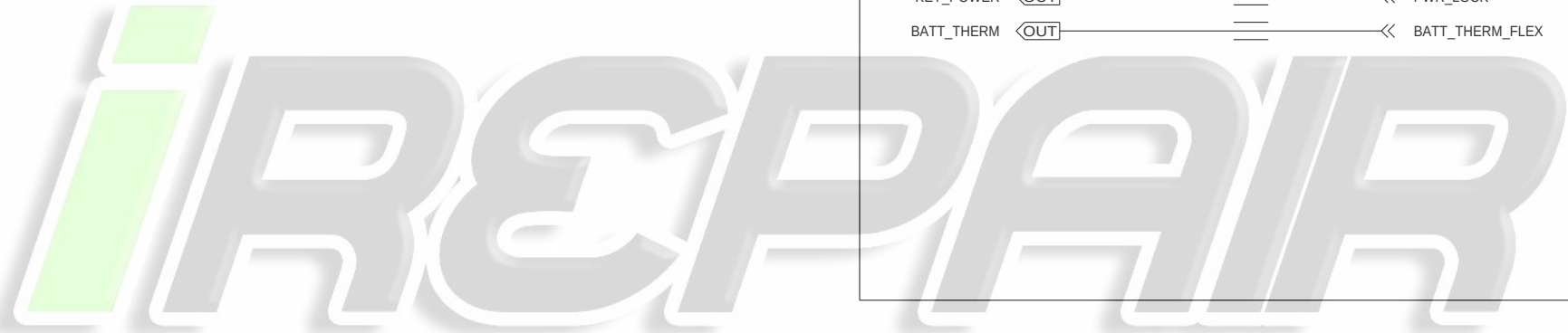


SLICE INTERFACES



GPIO - INPUT

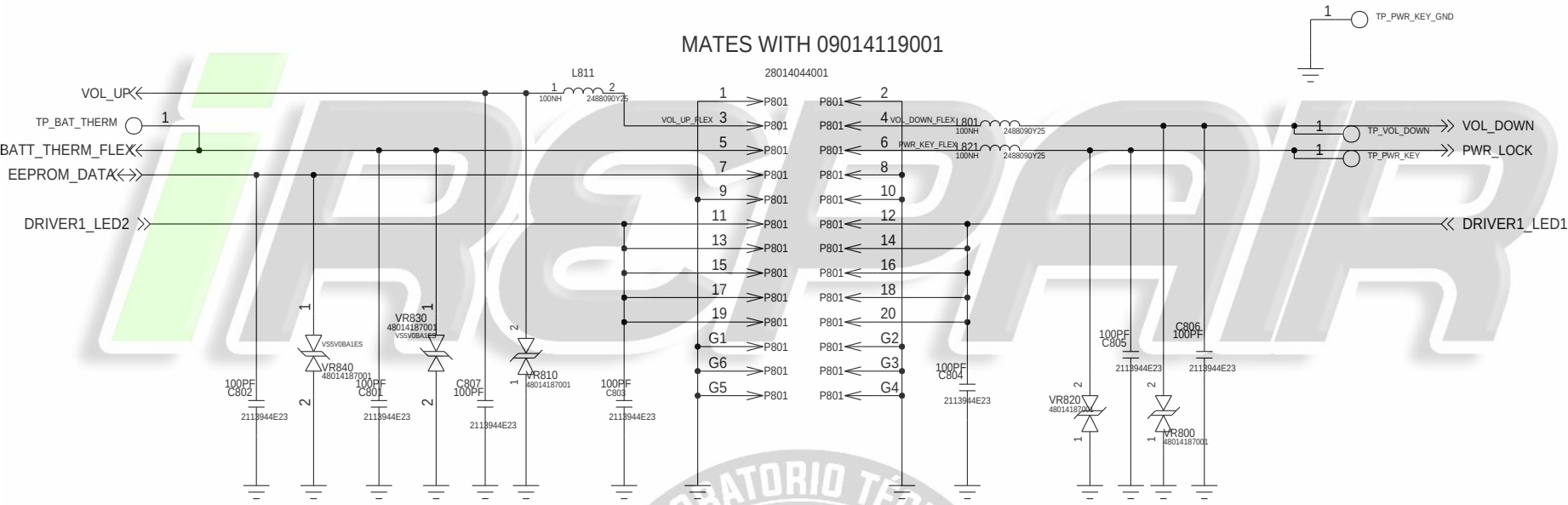
GPIO - OUTPUT



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: NATE CONNELL		BLOCK TITLE: user_input		
LAST MODIFIED BY: NATE CONNELL Wed Nov 26 14:04:42 2014				
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		B 84016440001_a.cpm		01
		CAD FILE NAME:		PAGE 40 OF 78

USR: INPUT

REF: 800-899

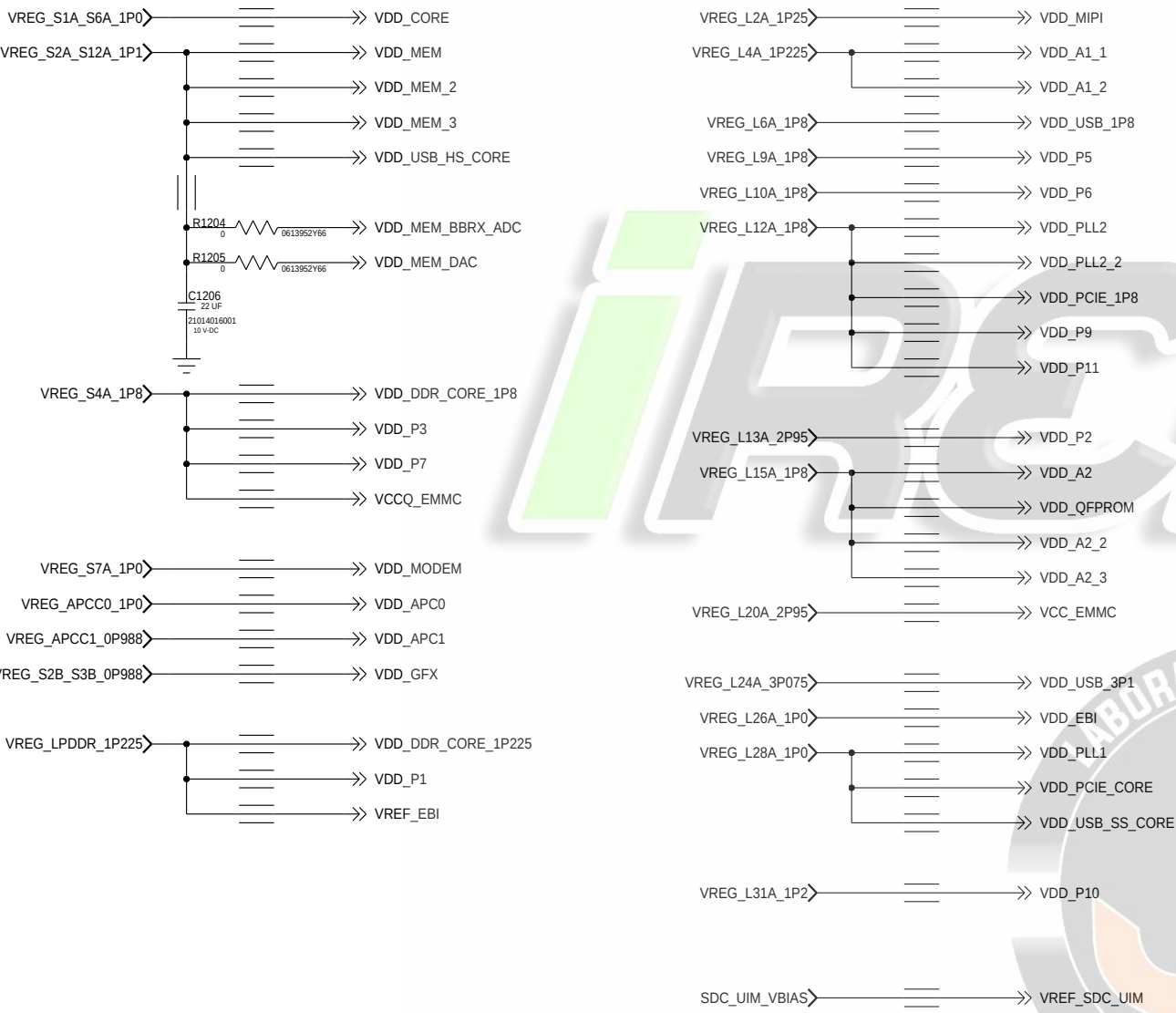


APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: NATE CONNELL Tue May 12 10:23:36 2015		user_input	
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		CAD FILE NAME:	
		REV 01	
		PAGE 41 OF 78	

MSM8992: POWER

REFDES: 1200-1299

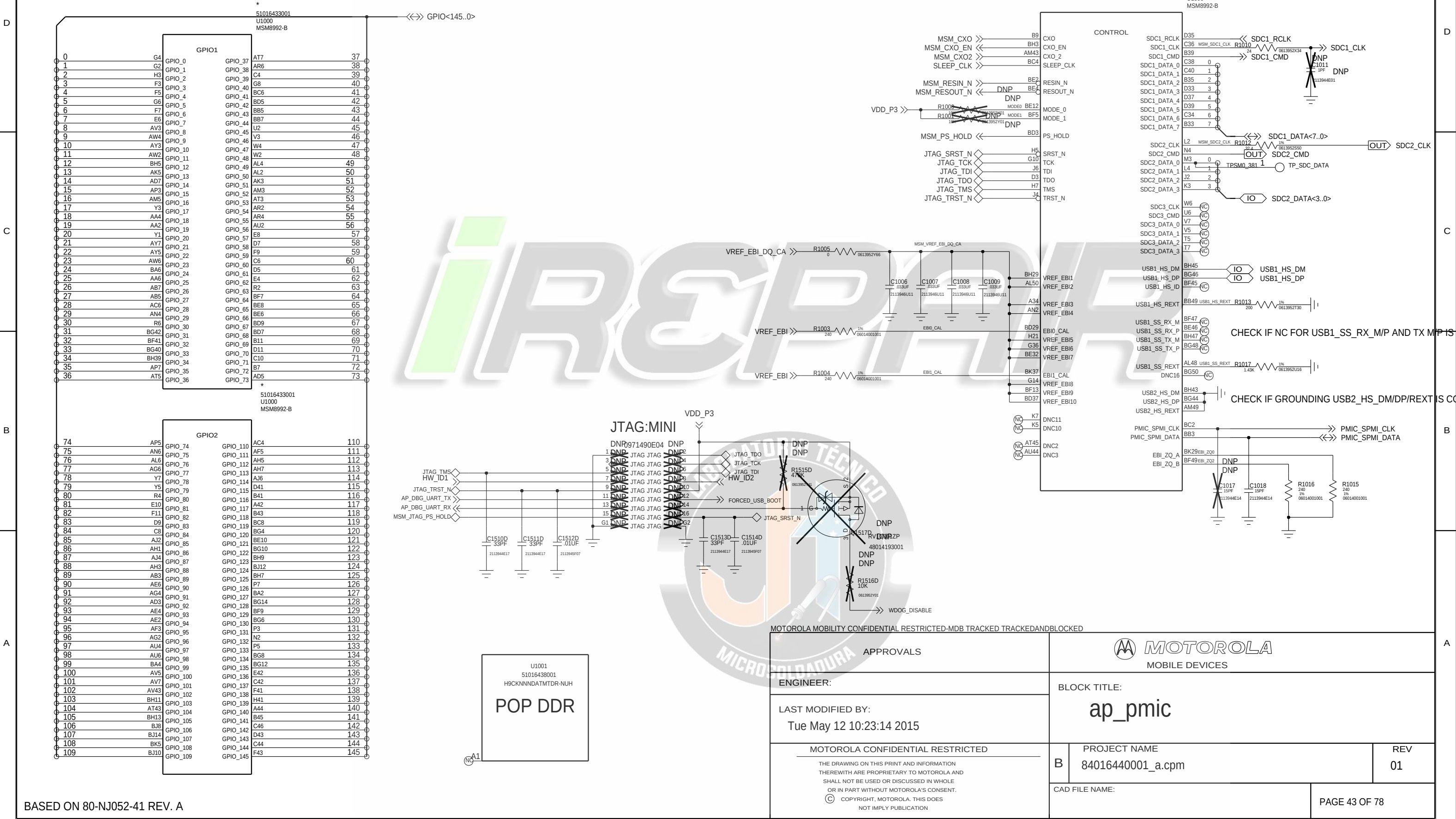
SUPPLIES



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Tue May 12 10:23:13 2015		ap_pmic	
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		REV 01	
		PAGE 42 OF 78	

MSM8992: CTRL/USB/GPIO

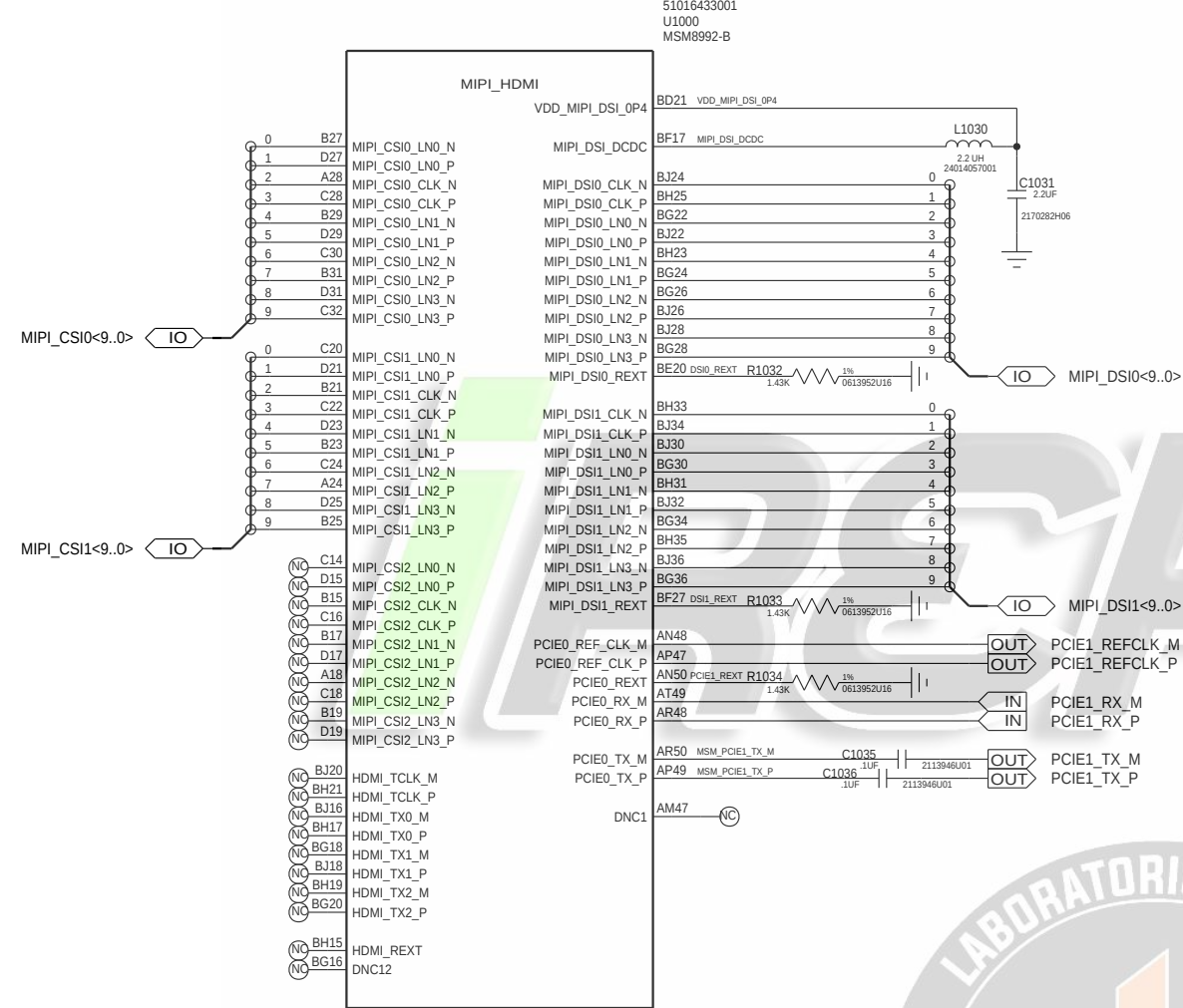
REFDES: 1000-1099



MSM8992: MIPI/HDMI/PCIE

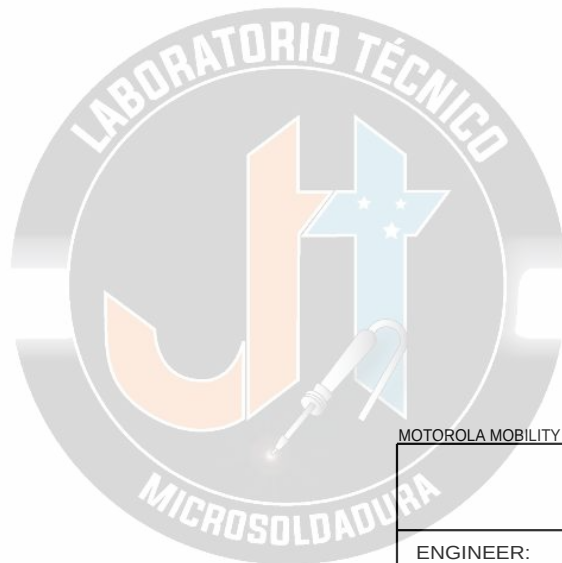
REFDES: 1030-1039

QCOM REFERENCE DESIGN HAS REAR CAMERA ON CSI0 AND FRONT CAMERA ON CSI2.



QCOM REFERENCE DESIGN HAS 1.43K 1% PULL-DOWN ON THE HDMI_REXT PIN.

BASED ON 80-NJ052-41 REV. A



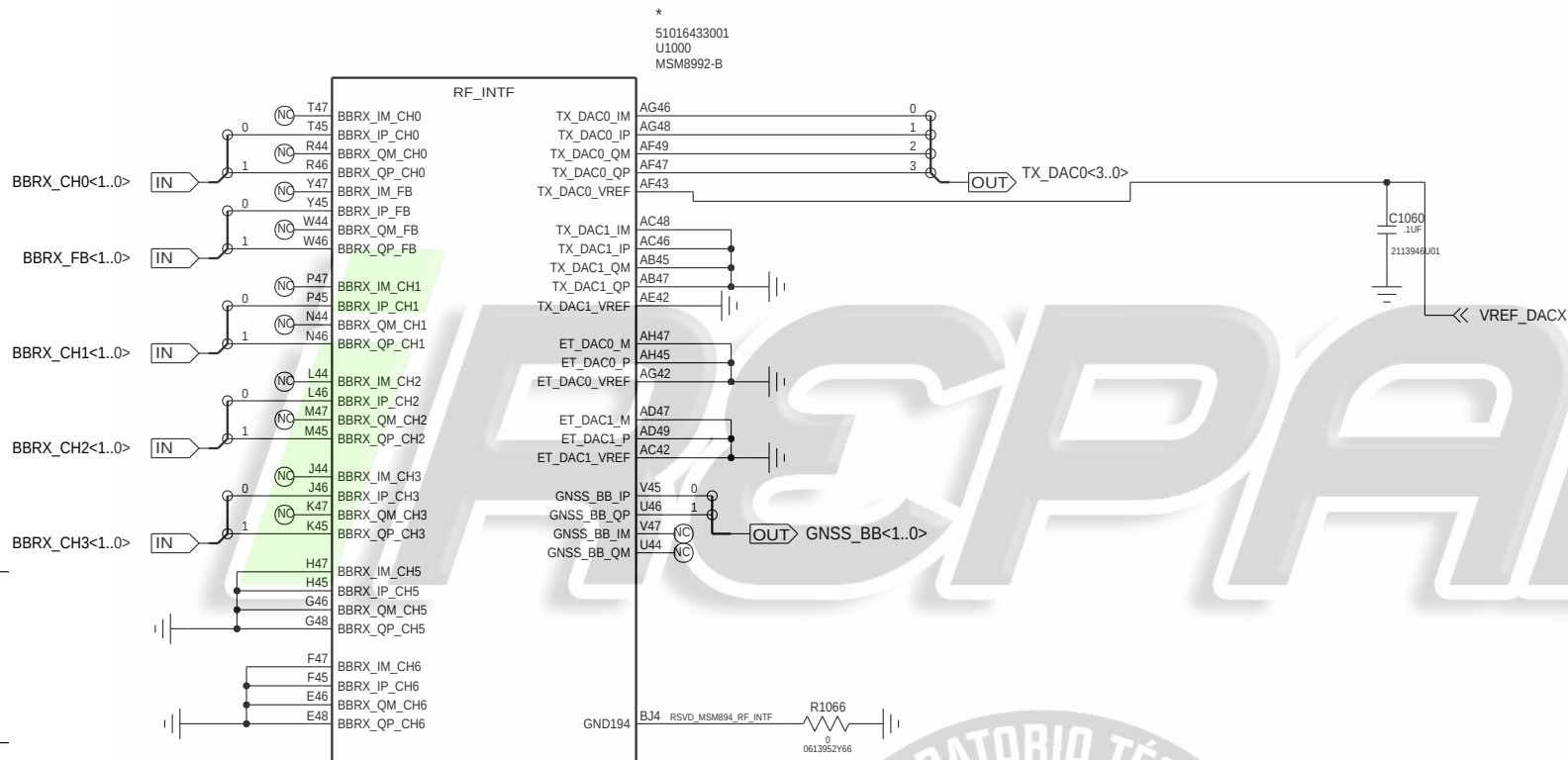
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Tue May 12 10:23:15 2015		ap_pmic	
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		PAGE 44 OF 78	

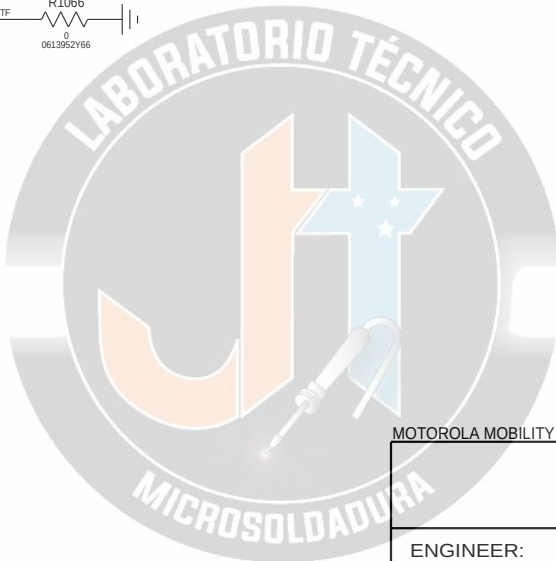
MSM8992: RF INTERFACE

REFDES: 1040-1049

3XCA CHANNELS



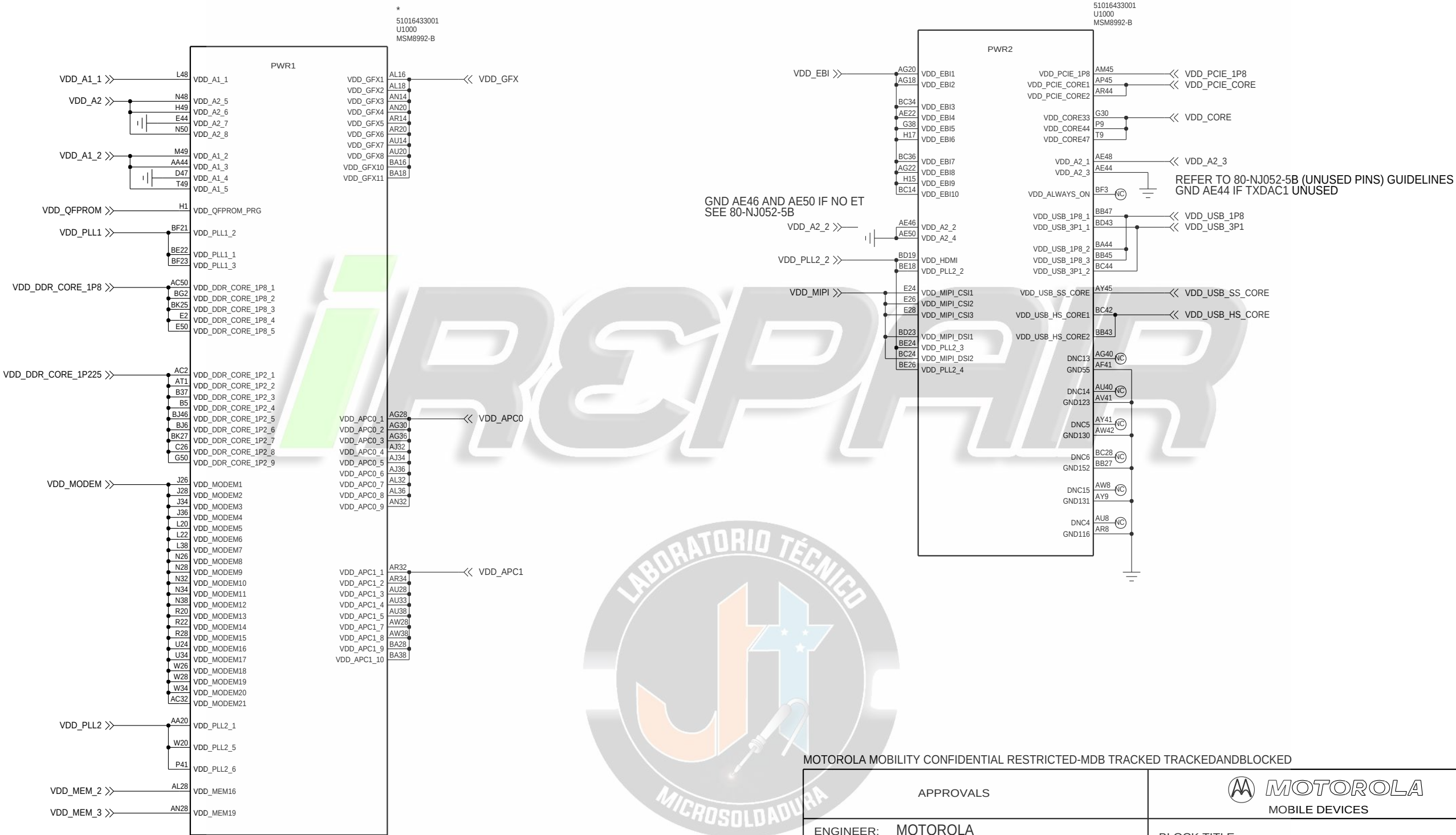
BASED ON 80-NJ051-41 REV. B



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: Tue May 12 10:23:16 2015		ap_pmic	
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		CAD FILE NAME:	
		PAGE 45 OF 78	

MSM8992: POWER 1

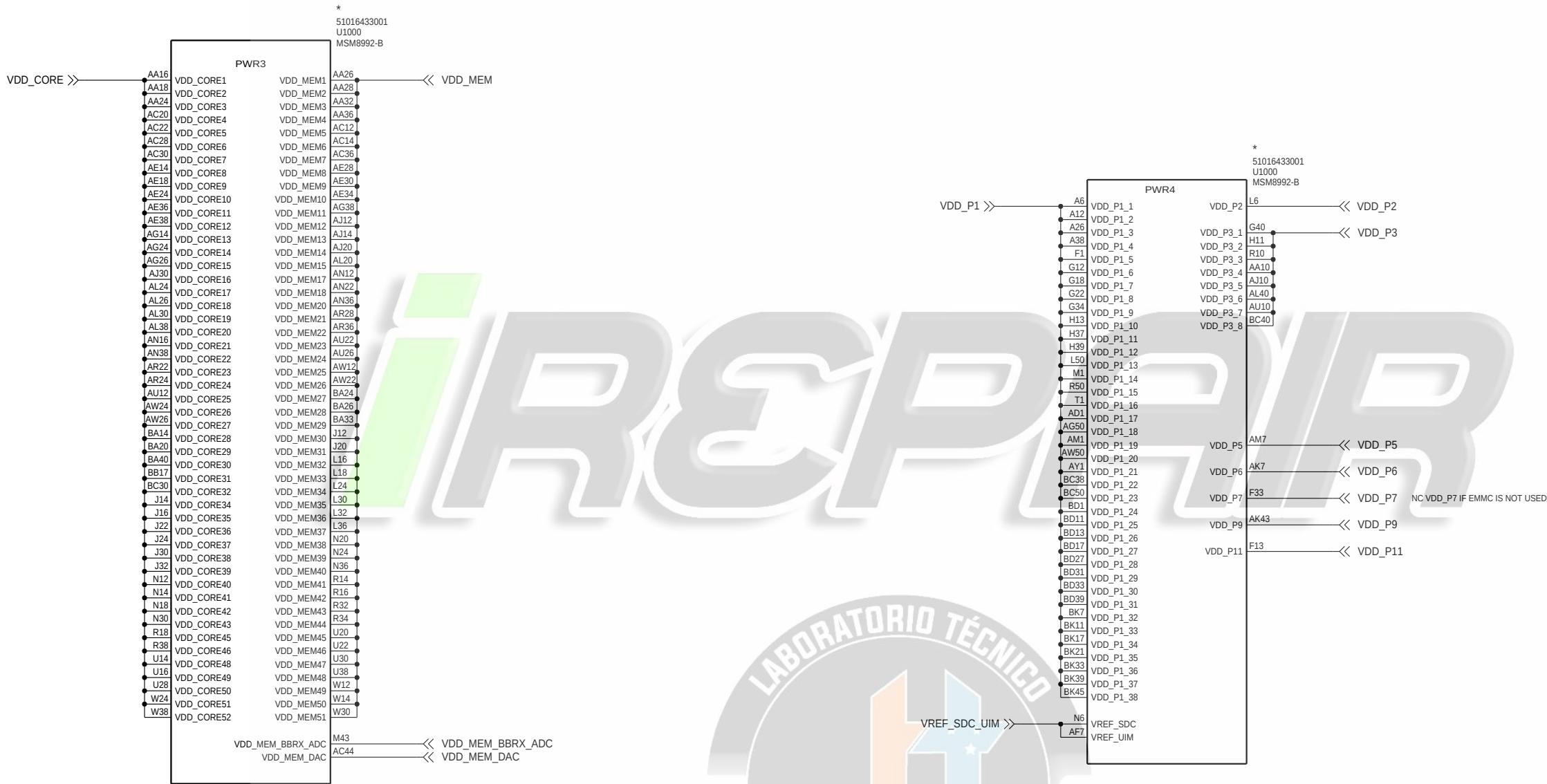


BASED ON 80-NJ052-41 REV. A

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MOTOROLA Tue May 12 10:23:18 2015		ap_pmic	
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		REV 01	
		PAGE 46 OF 78	

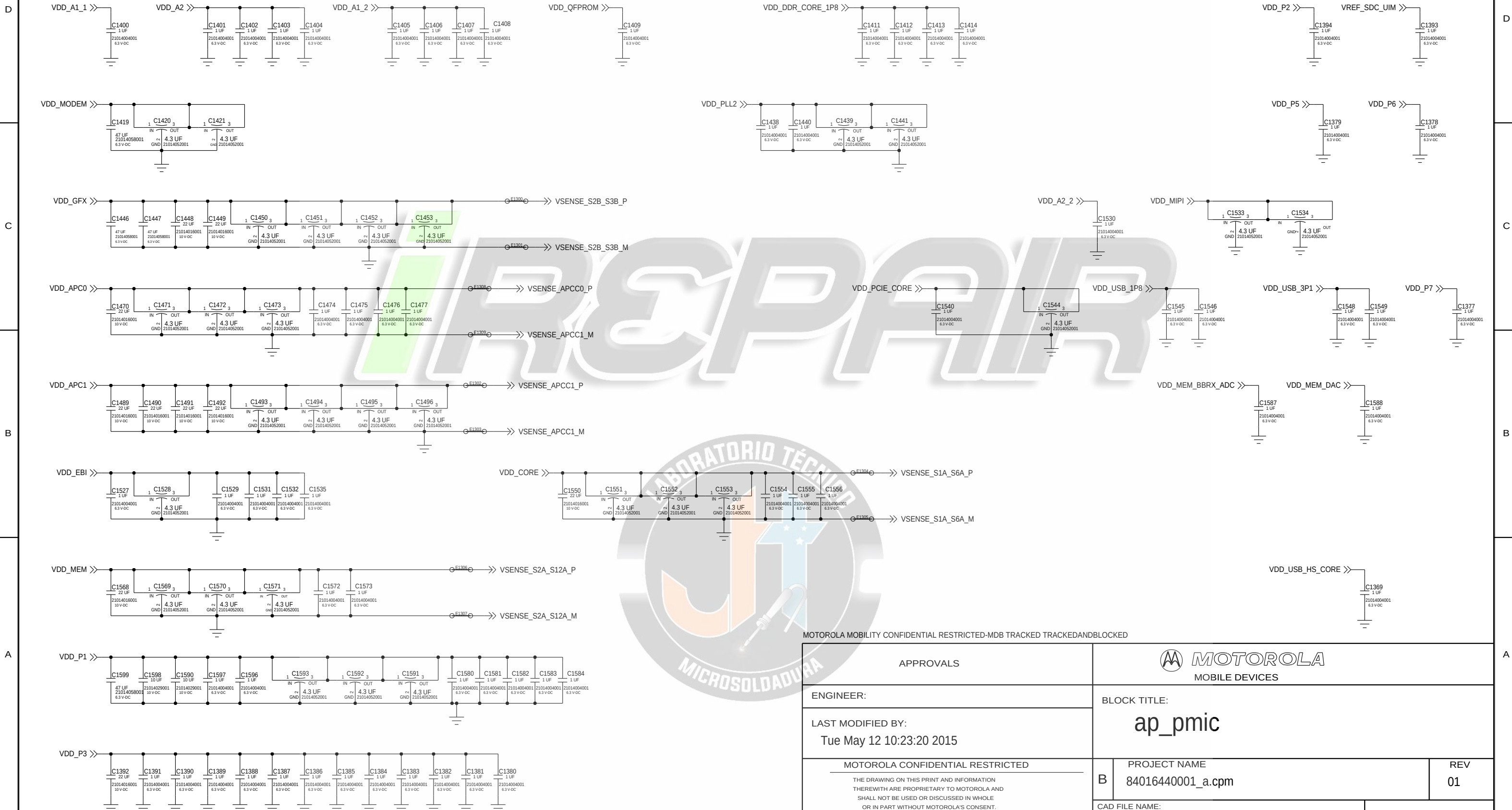
MSM8992: POWER 2



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Dec 04 13:36:32 2014		ap_pmic	
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		CAD FILE NAME:	
		PAGE 47 OF 78	

MSM8992: BYPASS CAPS

REFDES: 1300-1599

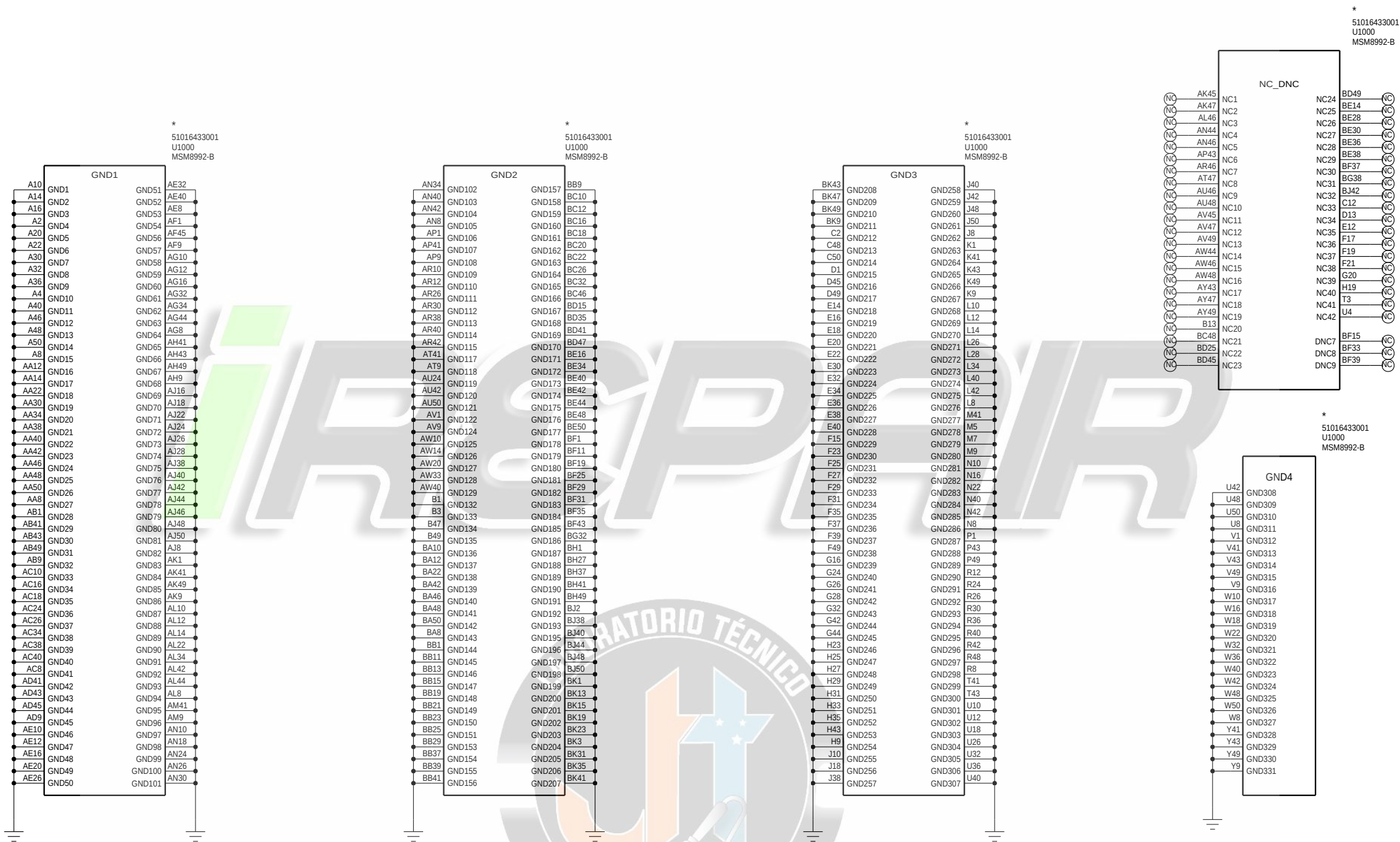


BASED ON 80-NJ051-41 REV. B

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

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Tue May 12 10:23:20 2015			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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CAD FILE NAME:		PAGE 48 OF 78	

MSM8992: GND/NC



BASED ON 80-NJ051-41 REV. A

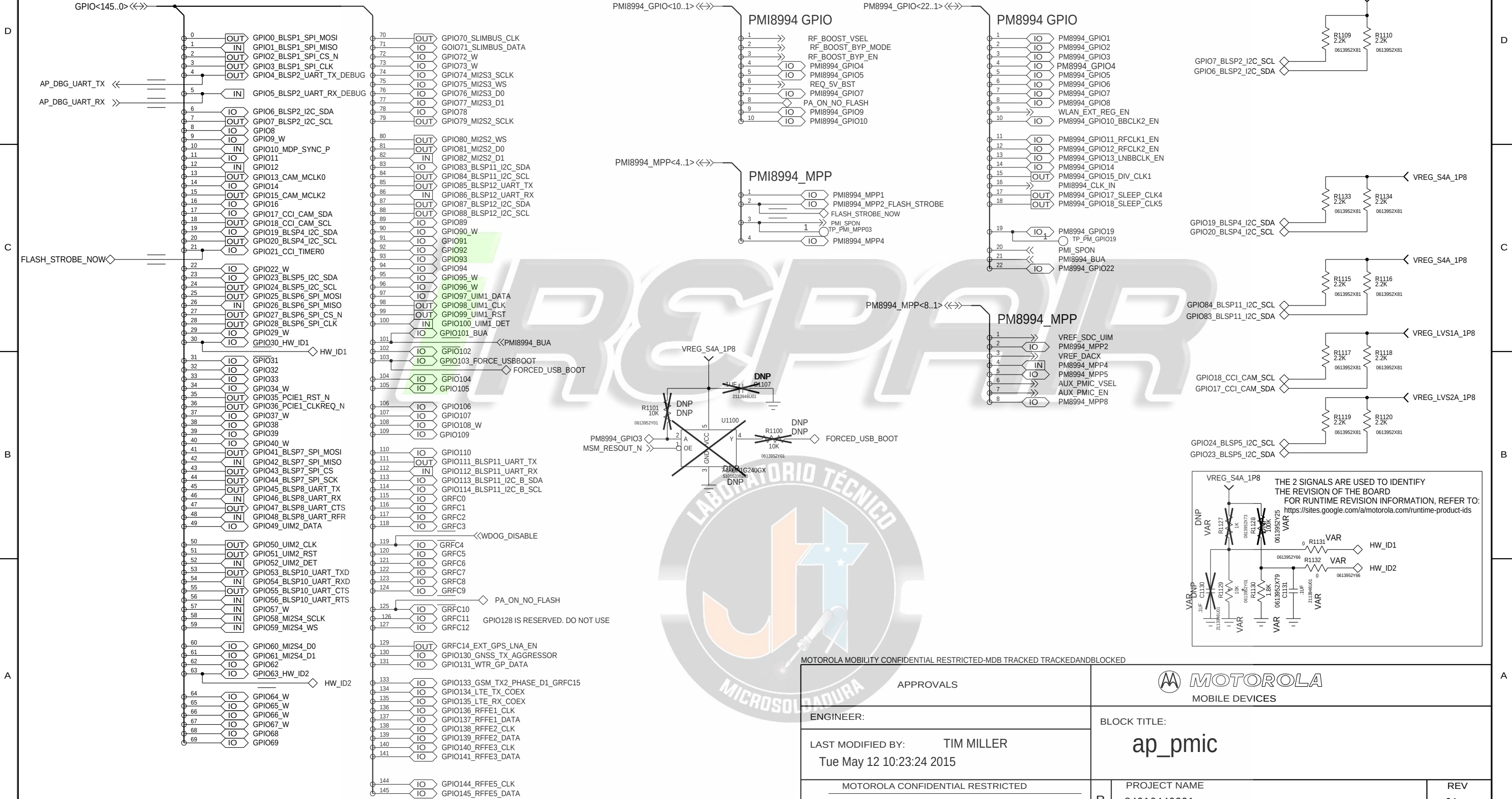
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MSM8992: GPIO MAP

REFDES: 1100-1199

PURPLE=MOT CUSTOM
CYAN=QCOM REFERENCE

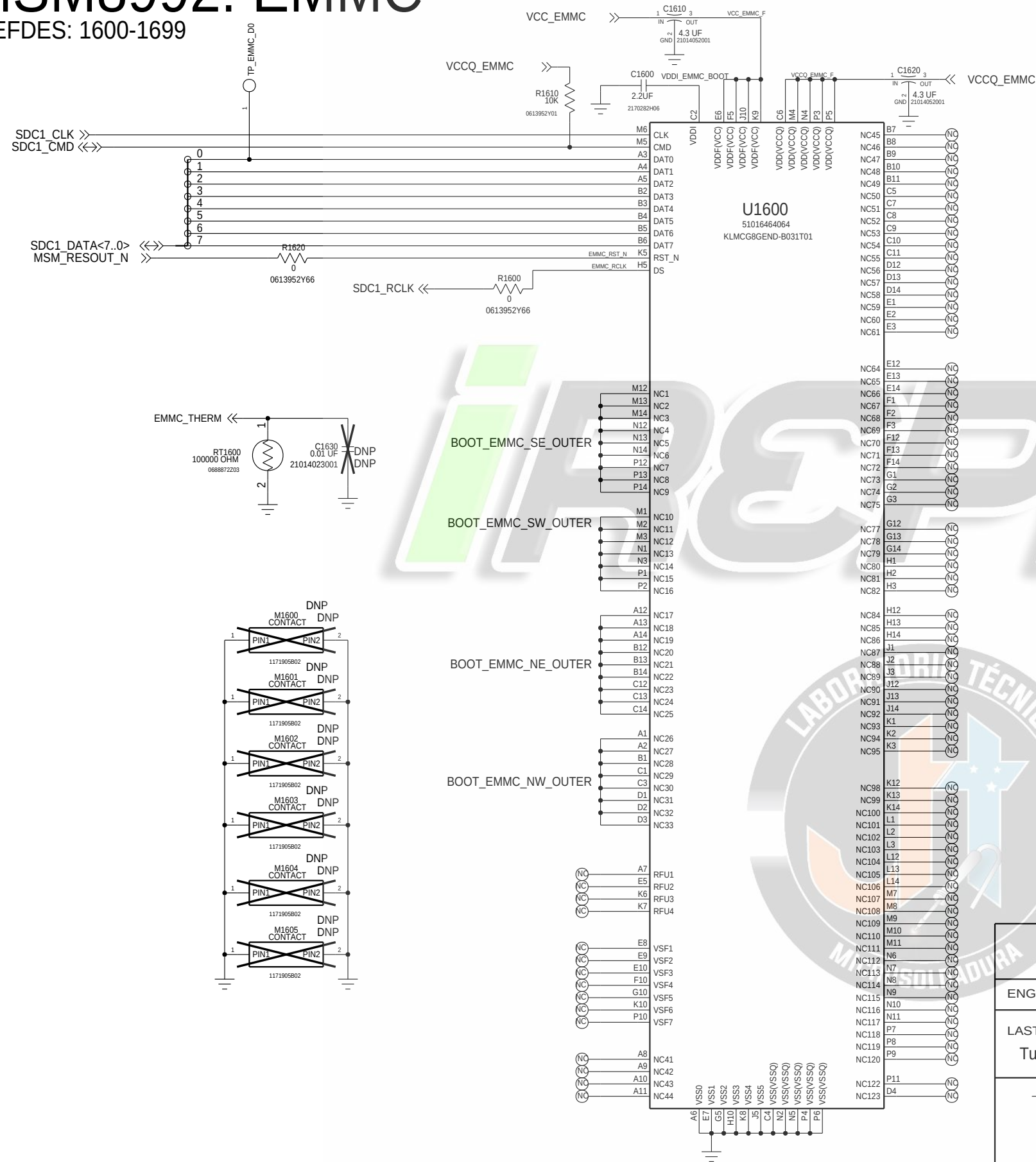


BASED ON 80-NJ051-41 REV. B

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LAST MODIFIED BY: TIM MILLER Tue May 12 10:23:24 2015			
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CAD FILE NAME:		PAGE 50 OF 78	

MSM8992: EMMC

REFDES: 1600-1699



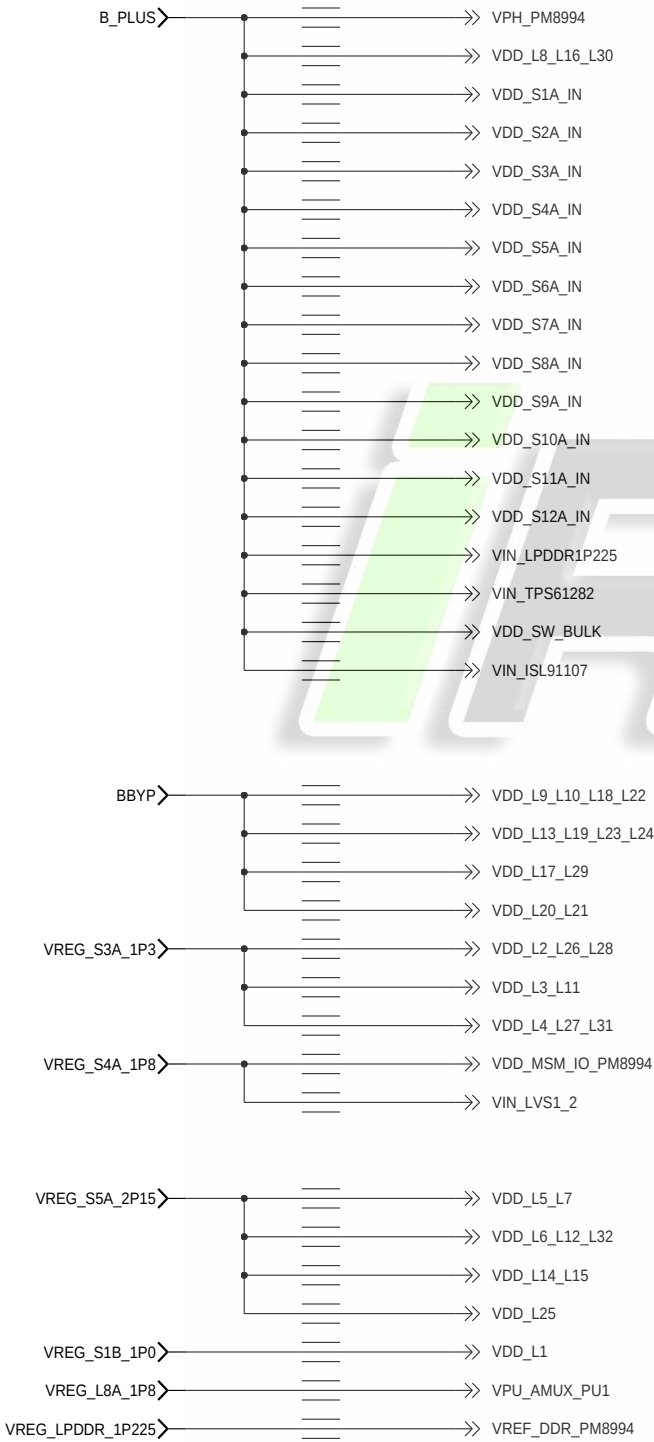
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		B 84016440001_a.cpm	
		CAD FILE NAME:	
		PAGE 51 OF 78	
		REV 01	

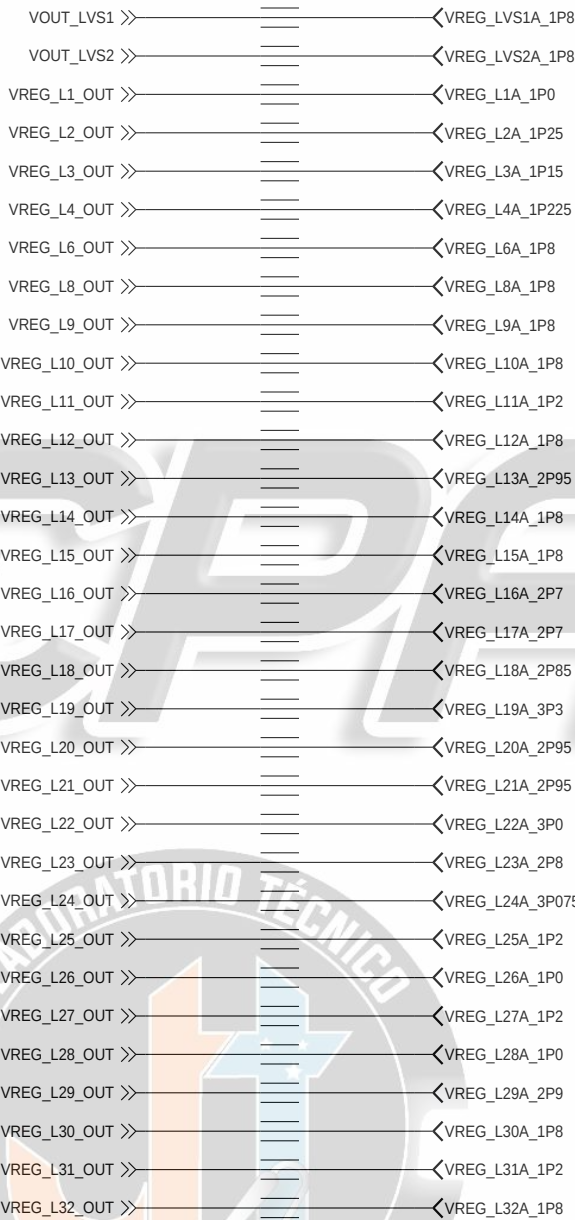
PM8994: POWER

REFDES: 2300-2399

INPUT POWER



OUTPUT POWER



APPROVALS

ENGINEER: NATE CONNELL

LAST MODIFIED BY: NATE CONNELL
Tue May 12 10:23:27 2015

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BLOCK TITLE:

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PROJECT NAME
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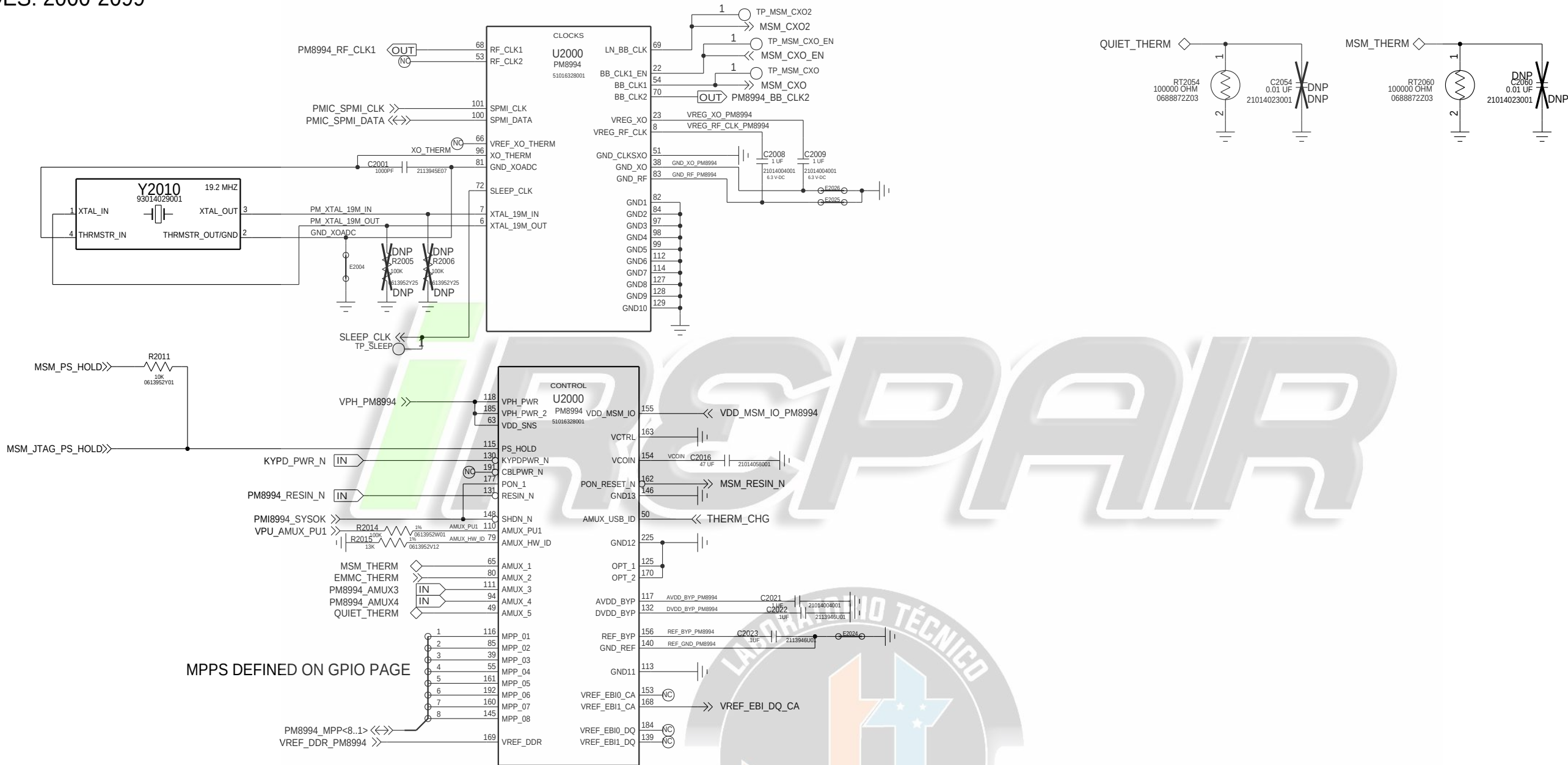
REV
01

CAD FILE NAME:

PAGE 52 OF 78

PM8994: CLK/CTRL/GPIO

REFDES: 2000-2099



FOLLOW CRYSTAL OSCILLATOR PLACEMENT/LAYOUT INSTRUCTIONS IN QCOM DOCUMENT 80-N5423-5 (P. 221 IN REV G)

*19.2MHZ CRYSTAL+THERMISTOR OPTIONS

PACKAGE PART NUMBER
2520 93014012001
2016 93014029001

PM GPIOs DEFINED ON MSM GPIO PAGE

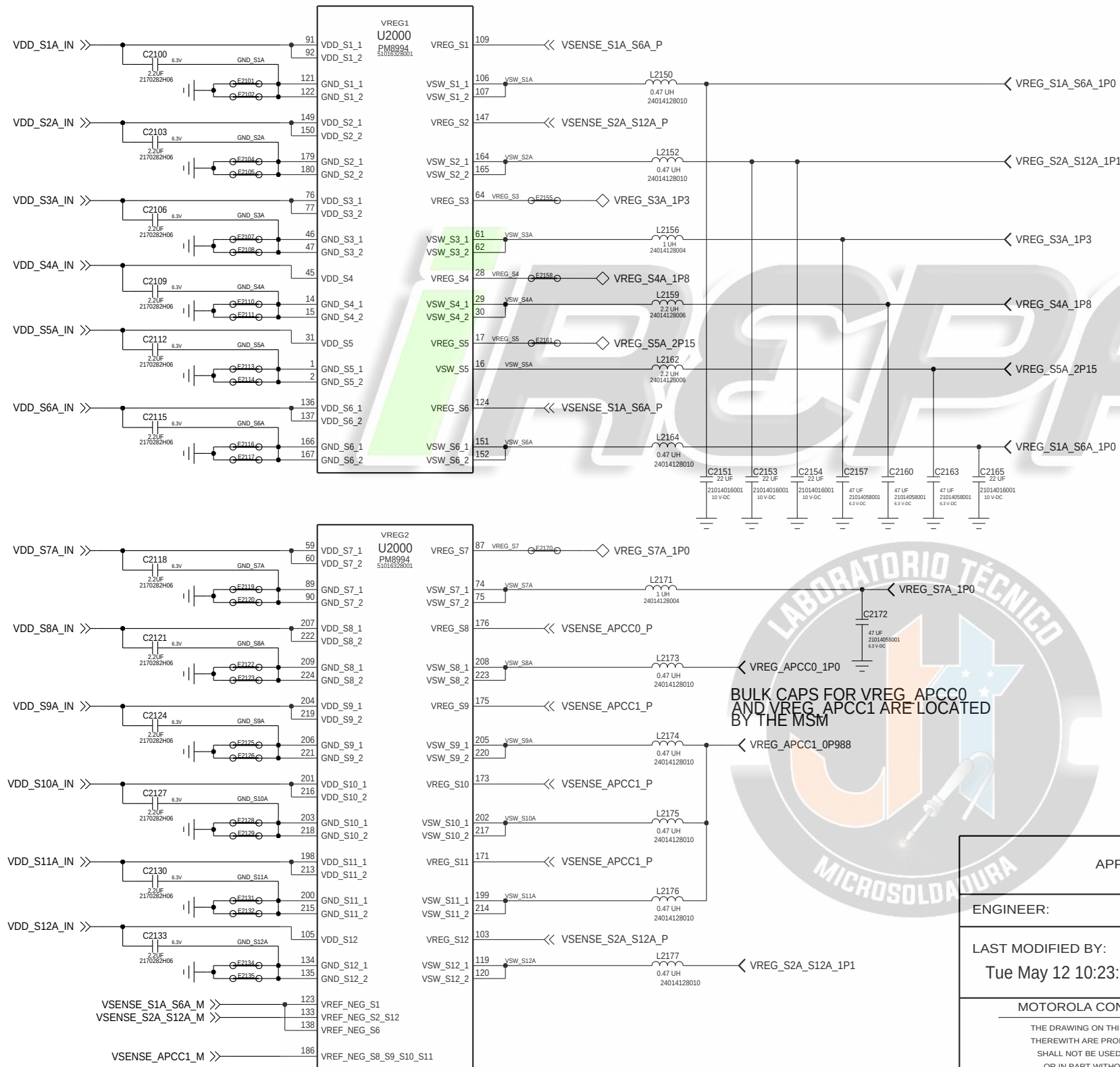
BASED ON 80-NJ051-41 REV. B

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		PAGE 53 OF 78	

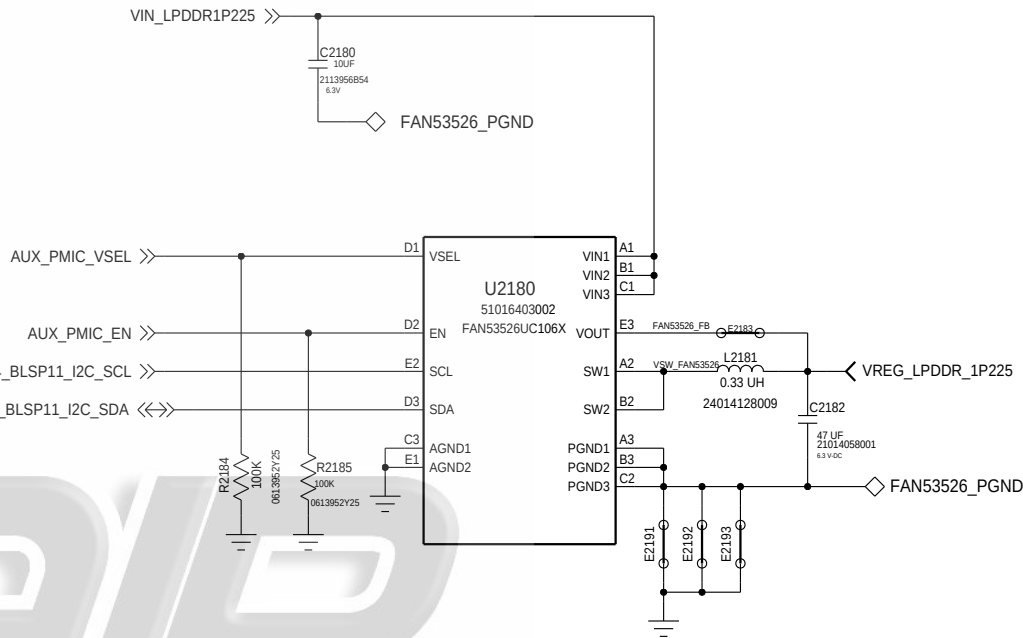
PM8994: SMPS

REFDES: 2100-2199

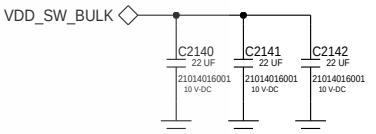


VREG_LPDDR3 EXTERNAL REGULATOR

*** EXT-REGULATOR FOR LPDDR3 I2C ADDR. 0X60



SWITCHER INPUT BULK CAPS



BULK CAPS FOR VREG_APCC0 AND VREG_APCC1 ARE LOCATED BY THE MSM

BASED ON 80-NJ051-41 REV. B

APPROVALS

ENGINEER:

LAST MODIFIED BY:

Tue May 12 10:23:28 2015

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

ap_pmic

PROJECT NAME

84016440001_a.cpm

REV

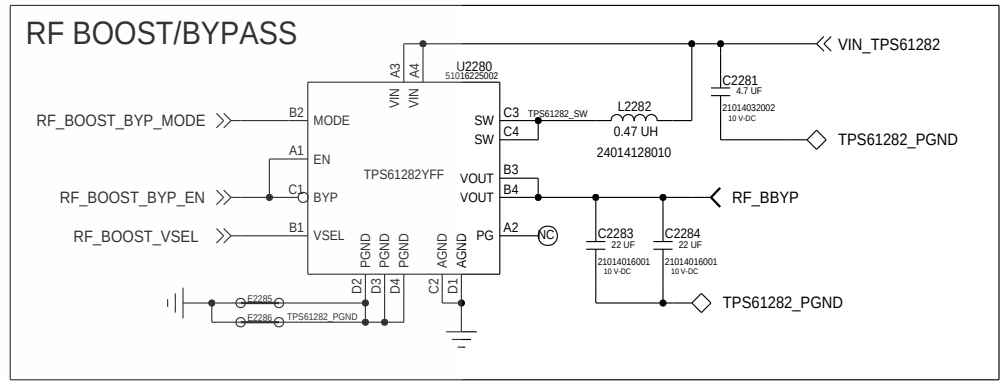
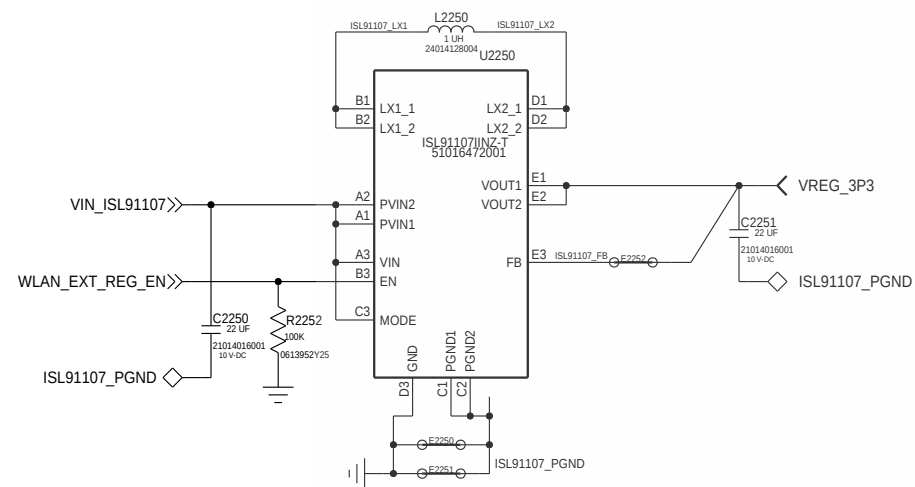
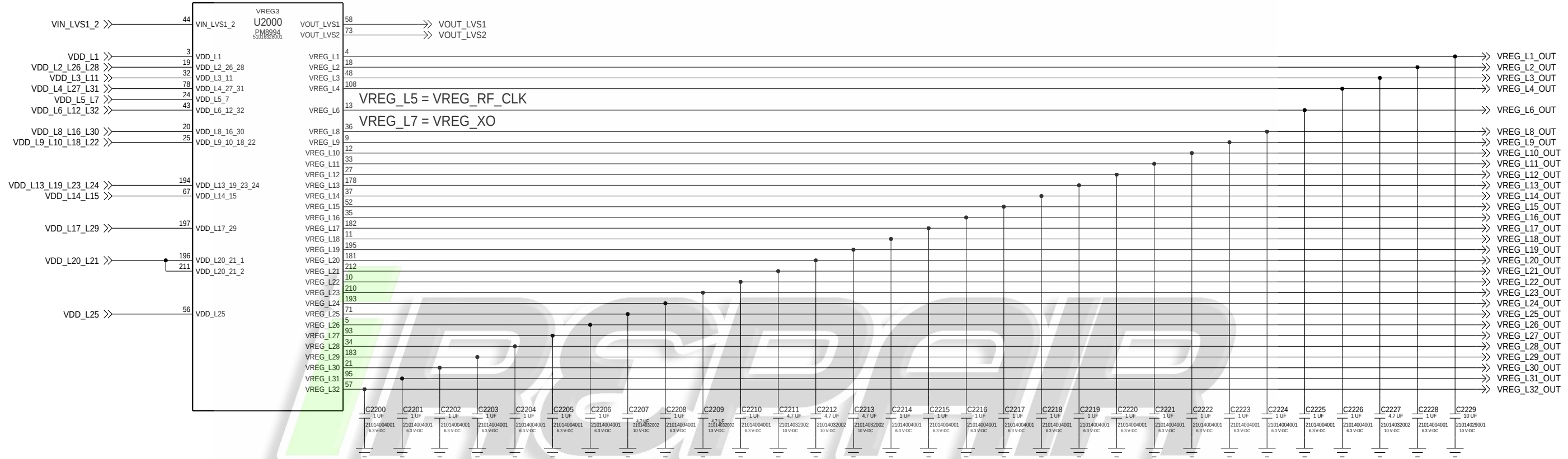
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CAD FILE NAME:

PAGE 54 OF 78

PM8994: VREGS

REFDES: 2200-2299

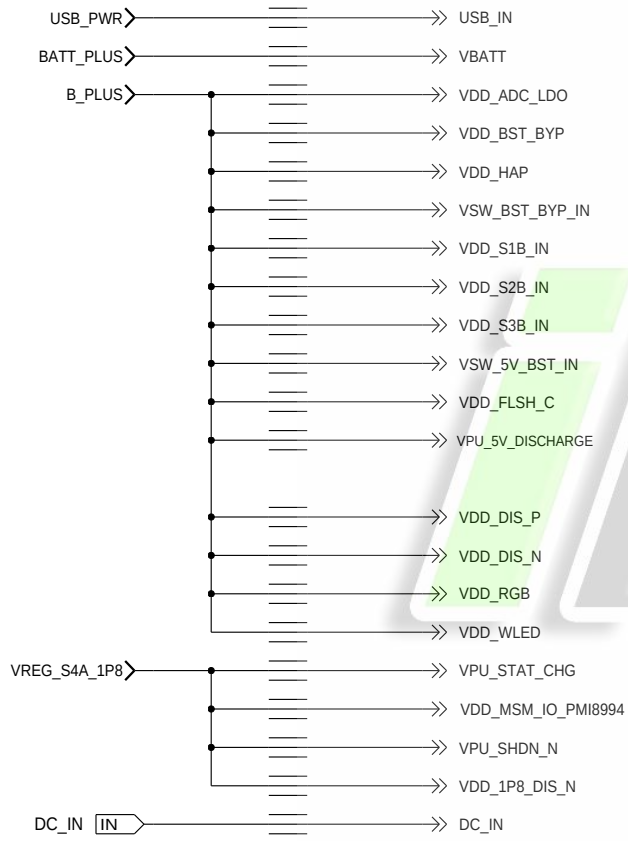


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MOTOROLA CONFIDENTIAL RESTRICTED		REV	
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CAD FILE NAME:		PAGE 55 OF 78	

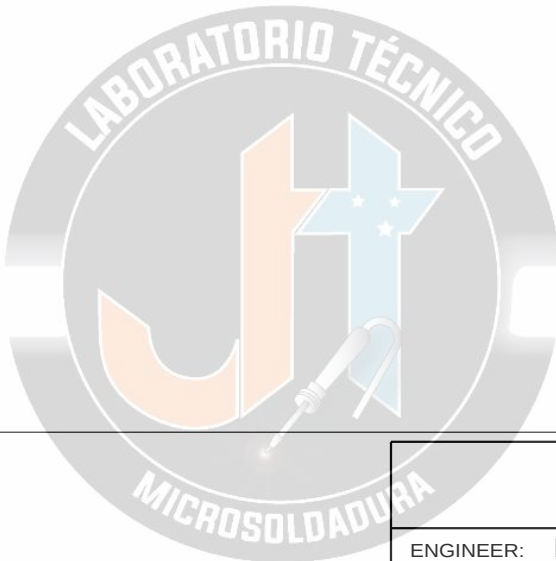
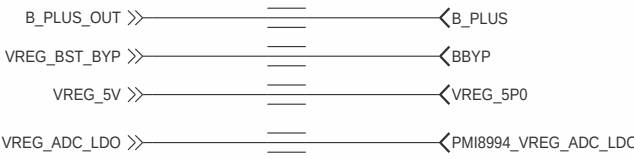
PMI8994: POWER

REFDES: 2400-2499

INPUT POWER



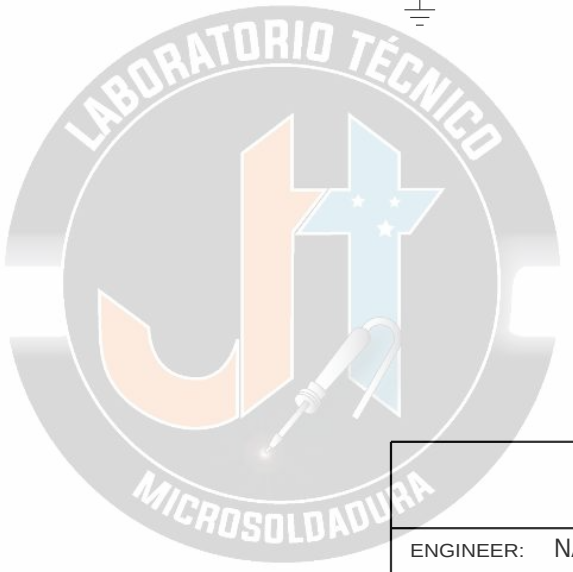
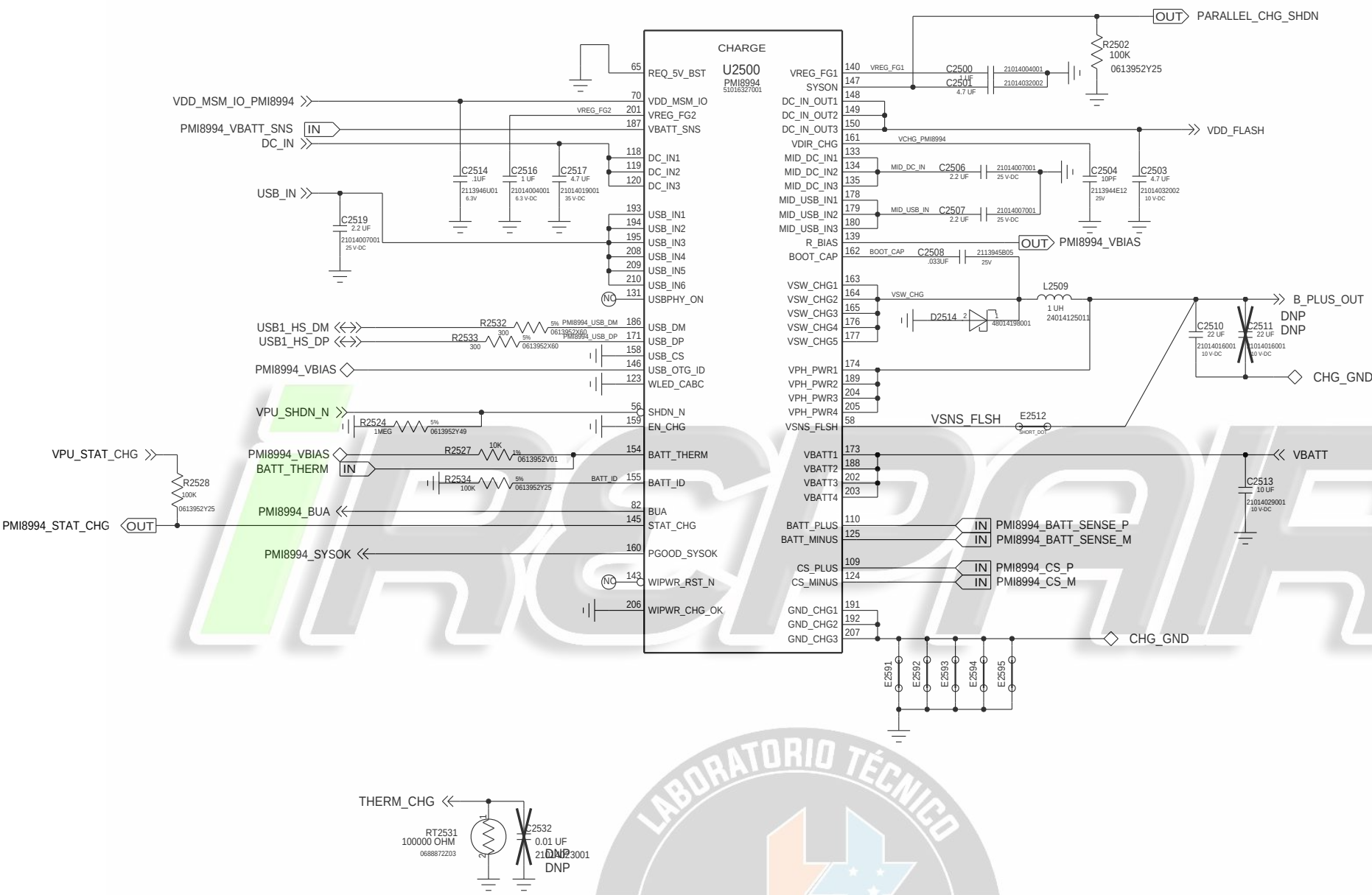
OUTPUT POWER



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ENGINEER: NATE CONNELL		BLOCK TITLE:	
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		CAD FILE NAME:	
		PAGE 56 OF 78	

PMI8994: CHG/USB

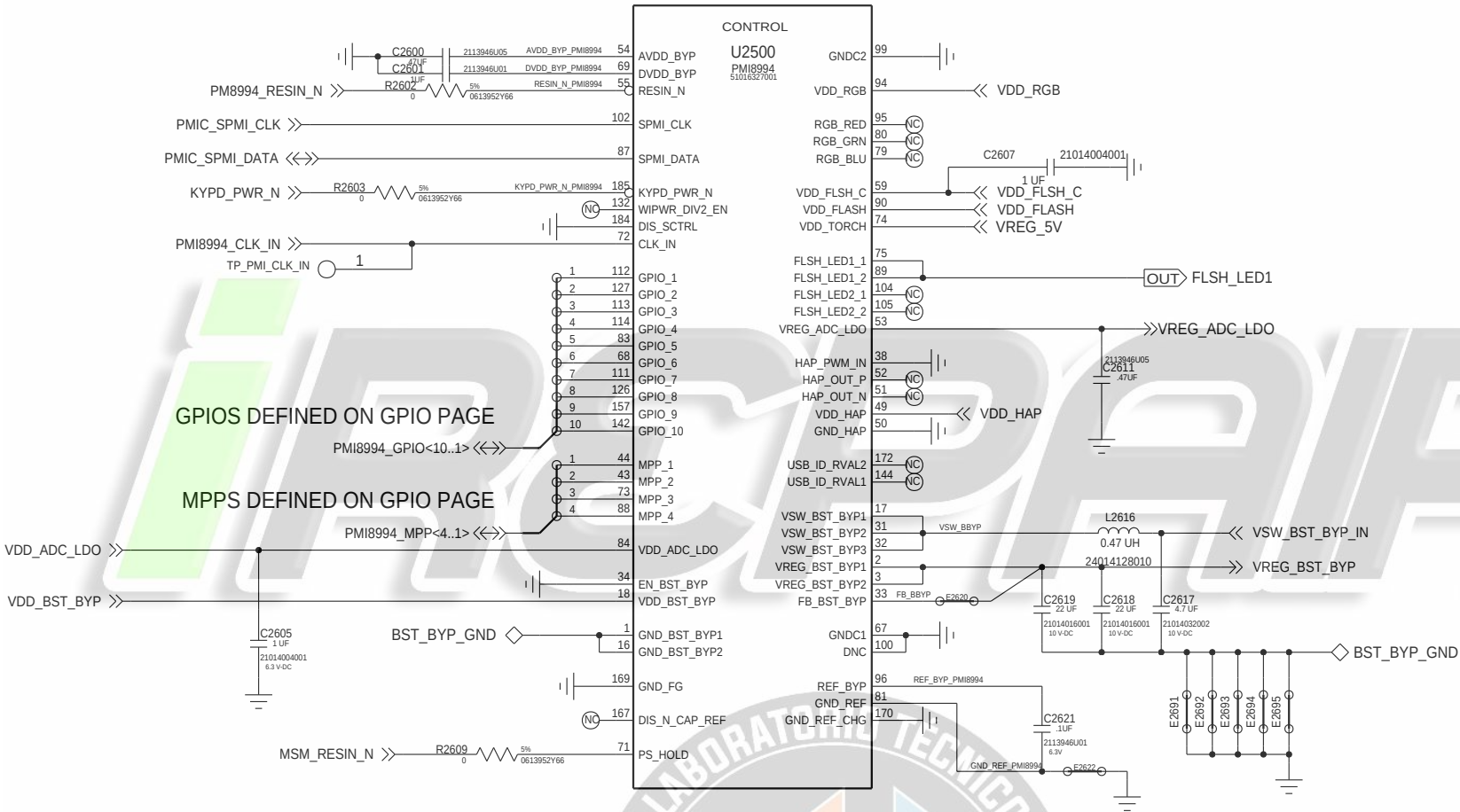
REFDES: 2500-2599



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME:	
		PAGE 57 OF 78	

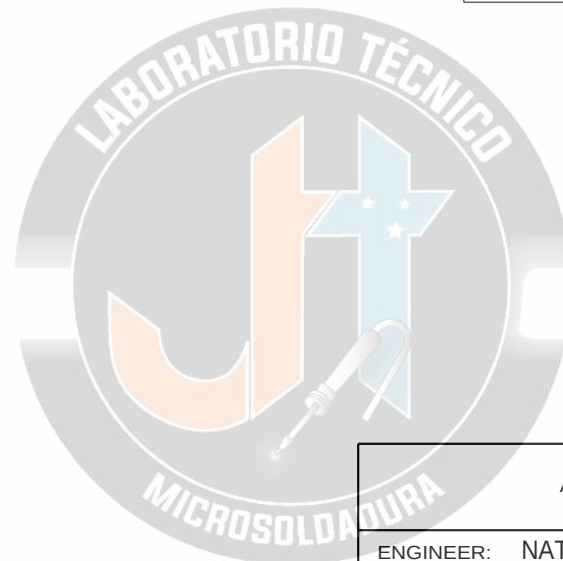
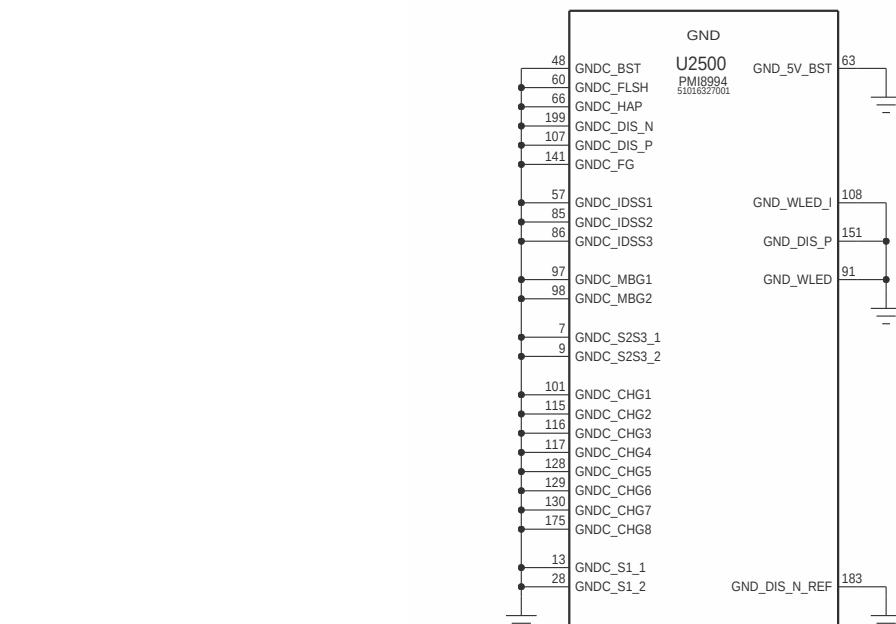
PMI8994: GPIO/LED/RGB

REFDES: 2600-2699



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
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		CAD FILE NAME: PAGE 58 OF 78	

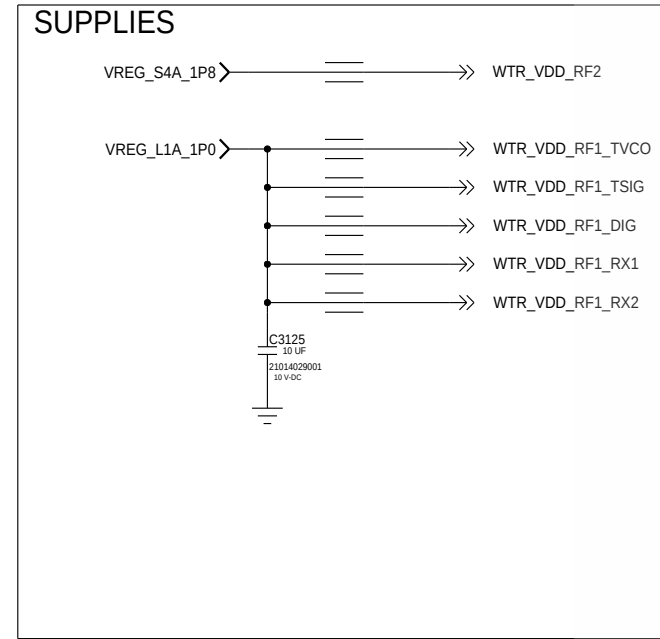
REFDES: 2700-2799



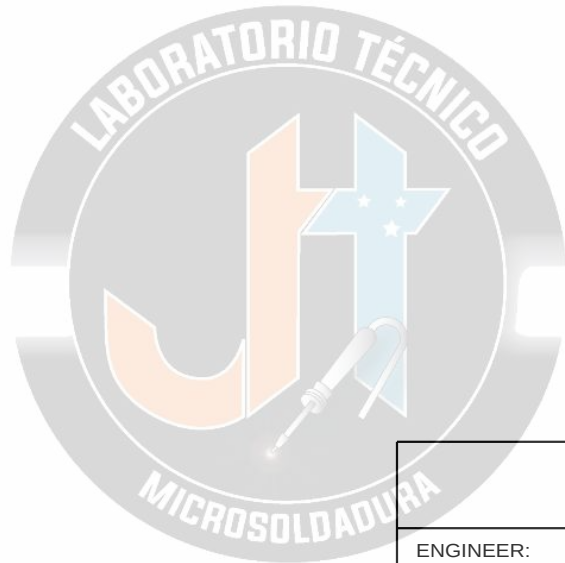
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		CAD FILE NAME:	
		PAGE 59 OF 78	

WTR3925: POWER

REF 3120-3199



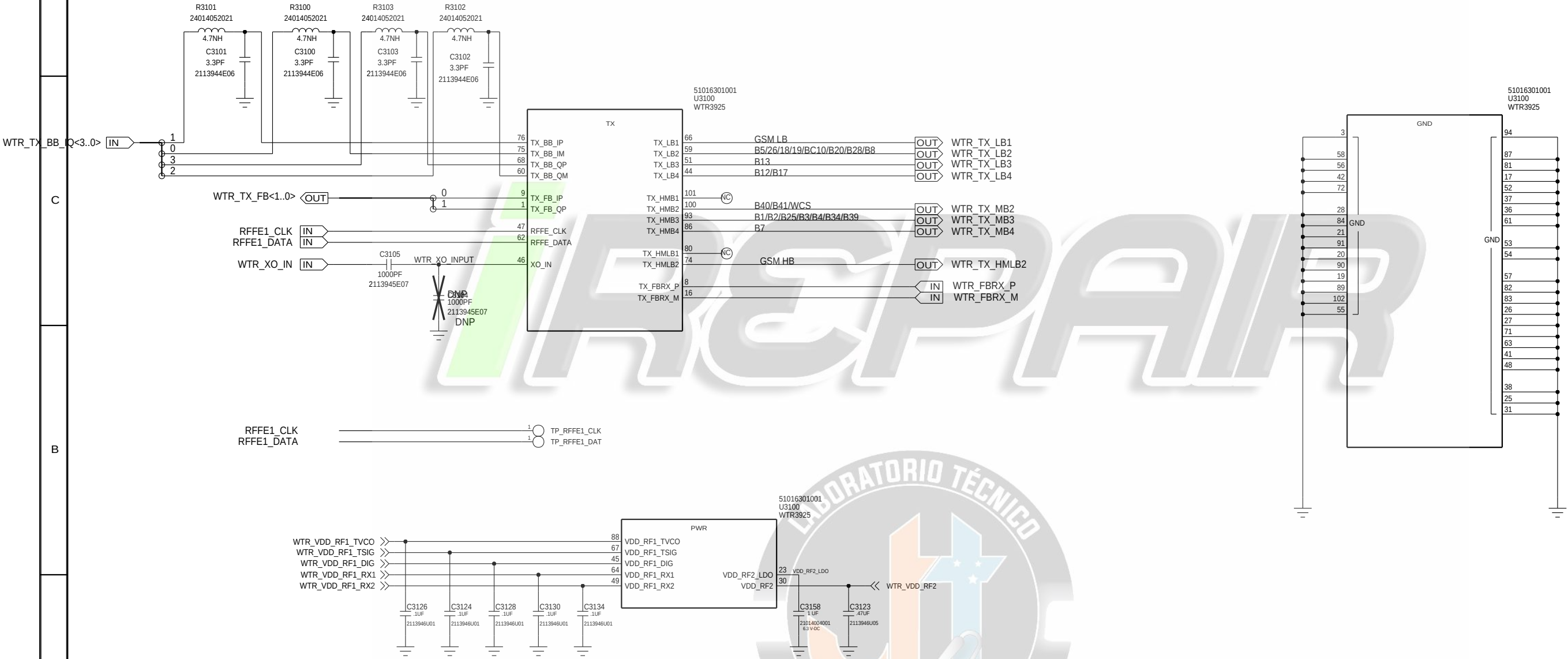
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APPROVALS		MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME: PAGE 60 OF 78	

WTR3925: TX/PWR/GROUND

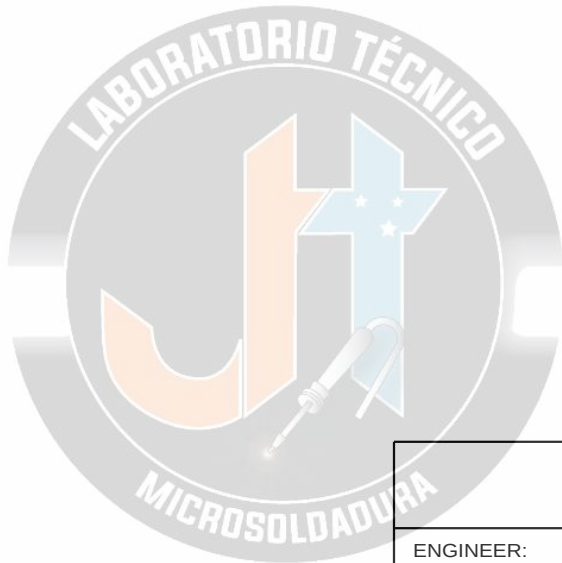
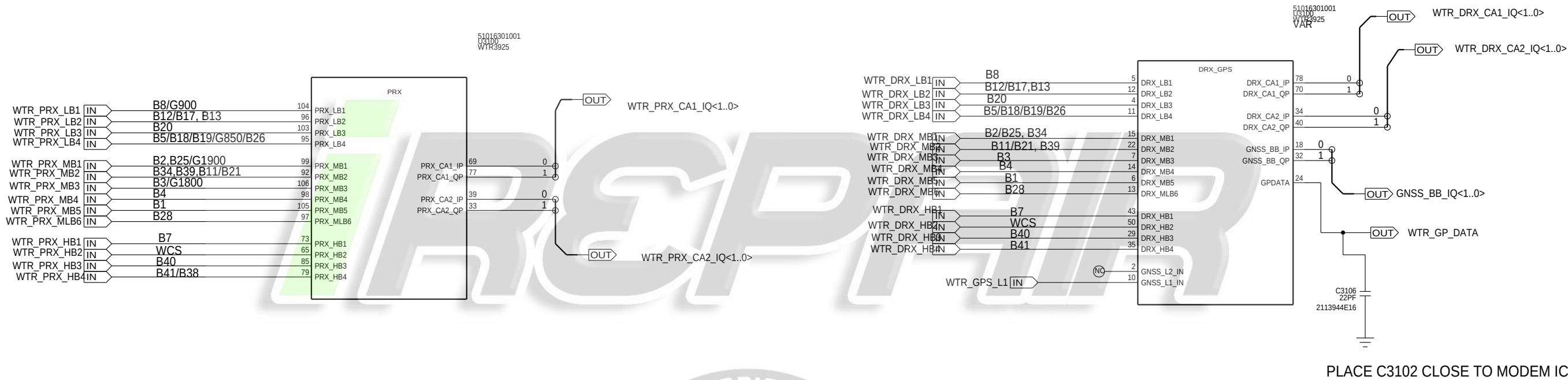
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		REV 01	
		CAD FILE NAME:	
		PAGE 61 OF 78	

WTR3925: PRX/DRX/GPS

REF 3110-3119



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ENGINEER:		BLOCK TITLE:	
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		CAD FILE NAME:	PAGE 62 OF 78

87654321

SLICE: I/F SLICE DELETED

D

SL_VBUS<<>>IO>SL_VBUS

D

SL_I2C_SDA <<>>IO>SL_I2C_SDA

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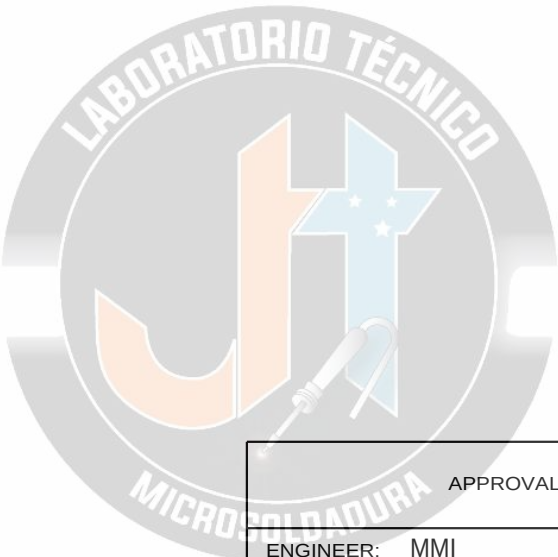
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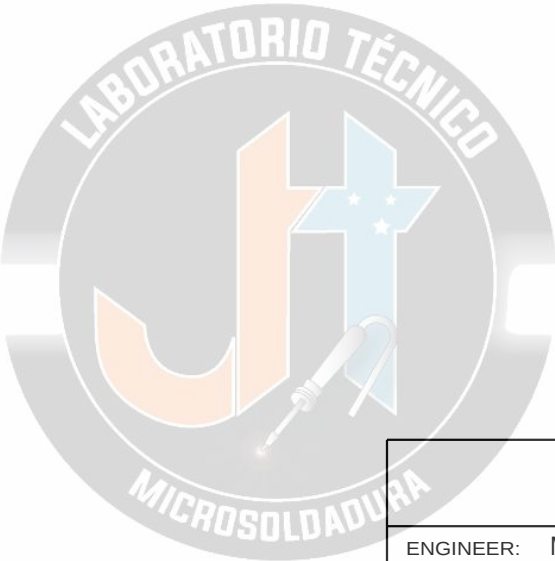
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		CAD FILE NAME: PAGE 63 OF 78	

87654321

RFFE: BLOCK DIAGRAM

REFDES: 7000-8999

iREPAIR

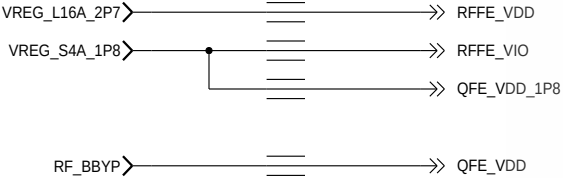


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LAST MODIFIED BY: MMI Tue Sep 30 10:58:34 2014		rffe_3ca_dsda_combo	
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		REV 01	
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		PAGE 64 OF 78	

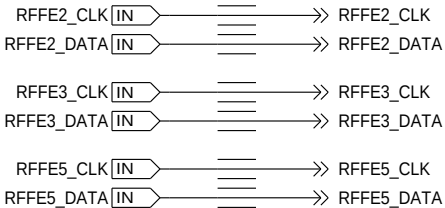
RFFE: PORTS

REFDES: 7000-8999

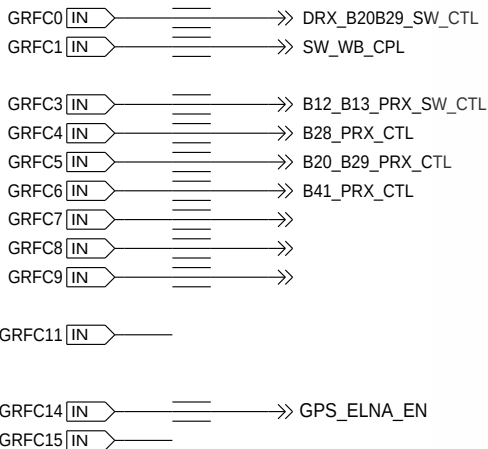
POWER



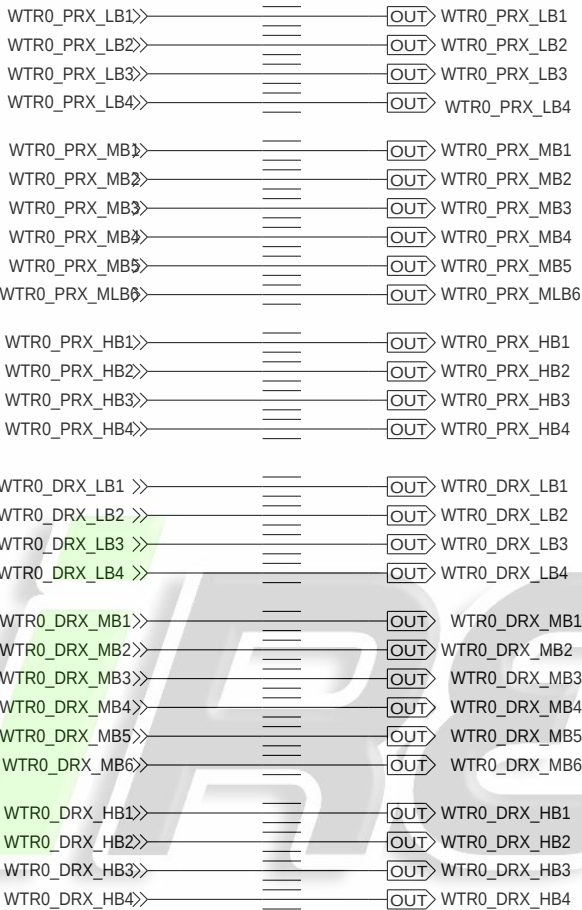
INTERFACES - RFFE MIPI



INTERFACES - RFFE GPIO/GRFC



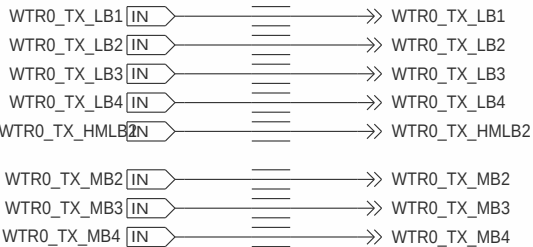
OUTPUTS - RF RECEIVER - WTR0



OUTPUTS - RF RECEIVER - 3DL

OUTPUTS - RF RECEIVER - DSDA

INPUTS - RF DRIVER - WTR0



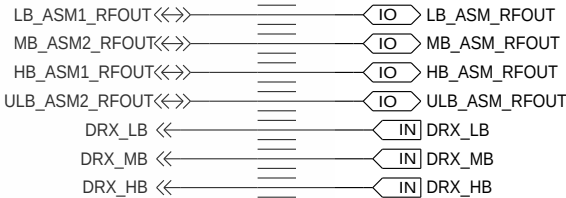
INPUTS - RF DRIVER - 3DL

INPUTS - RF DRIVER - DSDA

INPUTS /OUTPUTS - DETECTORS



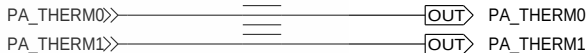
OUTPUTS - RF ANTENNA CONNECTORS



GPS



THERMISTORS



APPROVALS

ENGINEER: MMI

LAST MODIFIED BY: MMI
Tue May 12 10:22:56 2015

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

rffe_3ca_dsda_combo

PROJECT NAME
B 84016440001_a.cpm

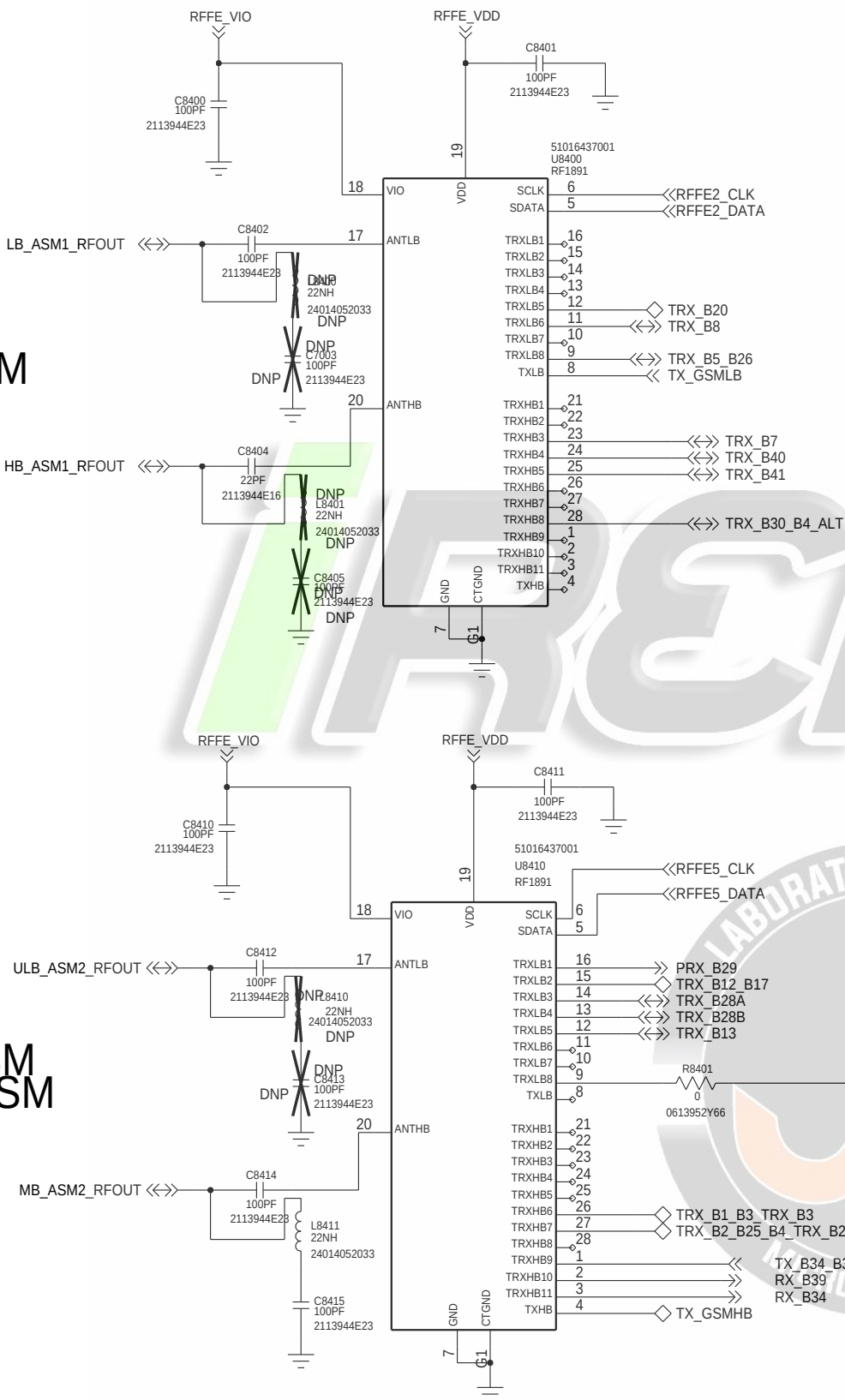
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REV
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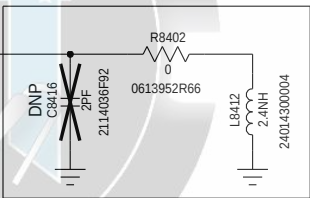
RFFE: ASM
REF 8400-8499

ASM1
CA: LB+HB ASM
DSDA: DATA ASM

ASM2
CA: ULB+MB ASM
DSDA: VOICE ASM

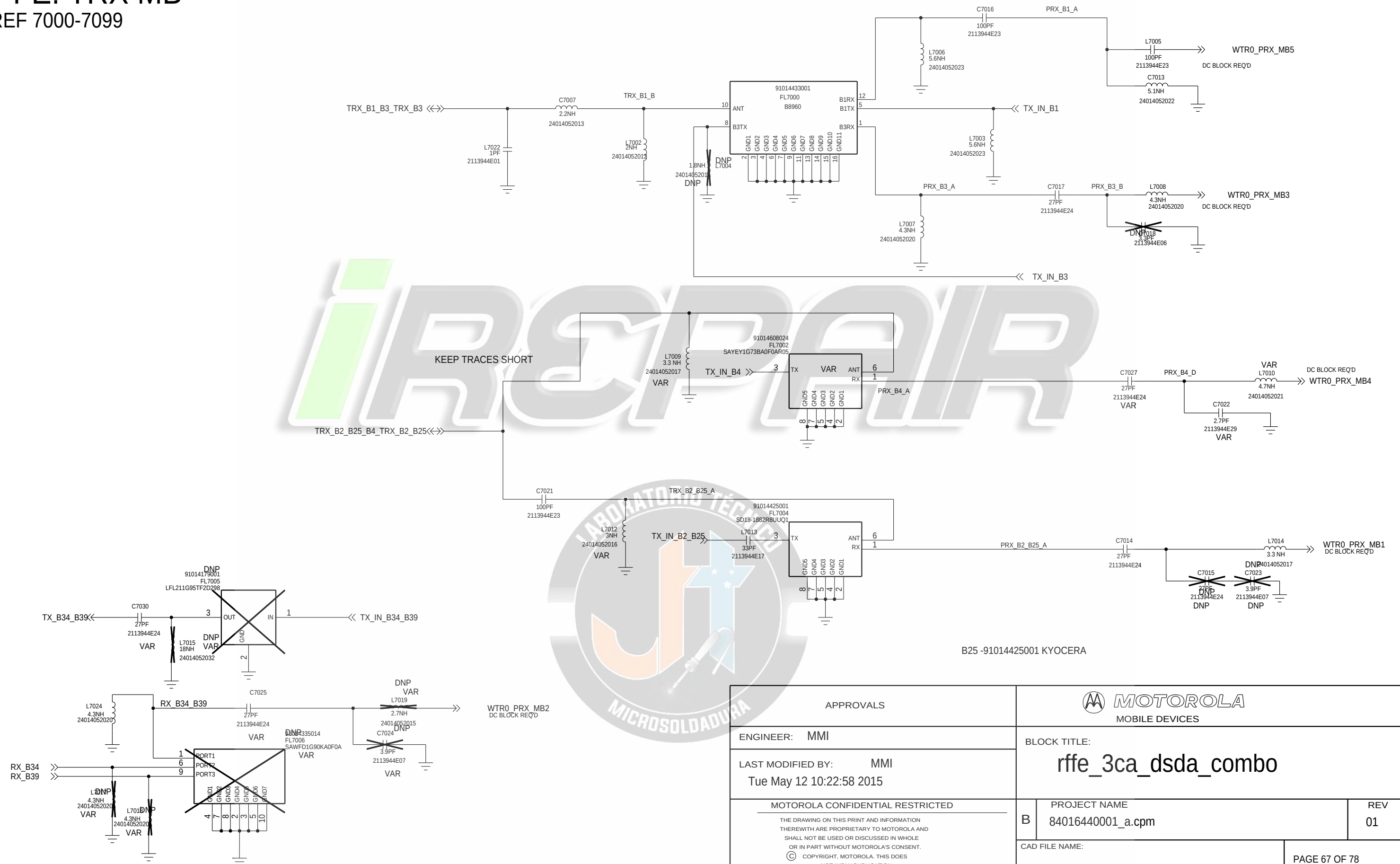


PLACE PI OUTSIDE OF SHIELD CAN



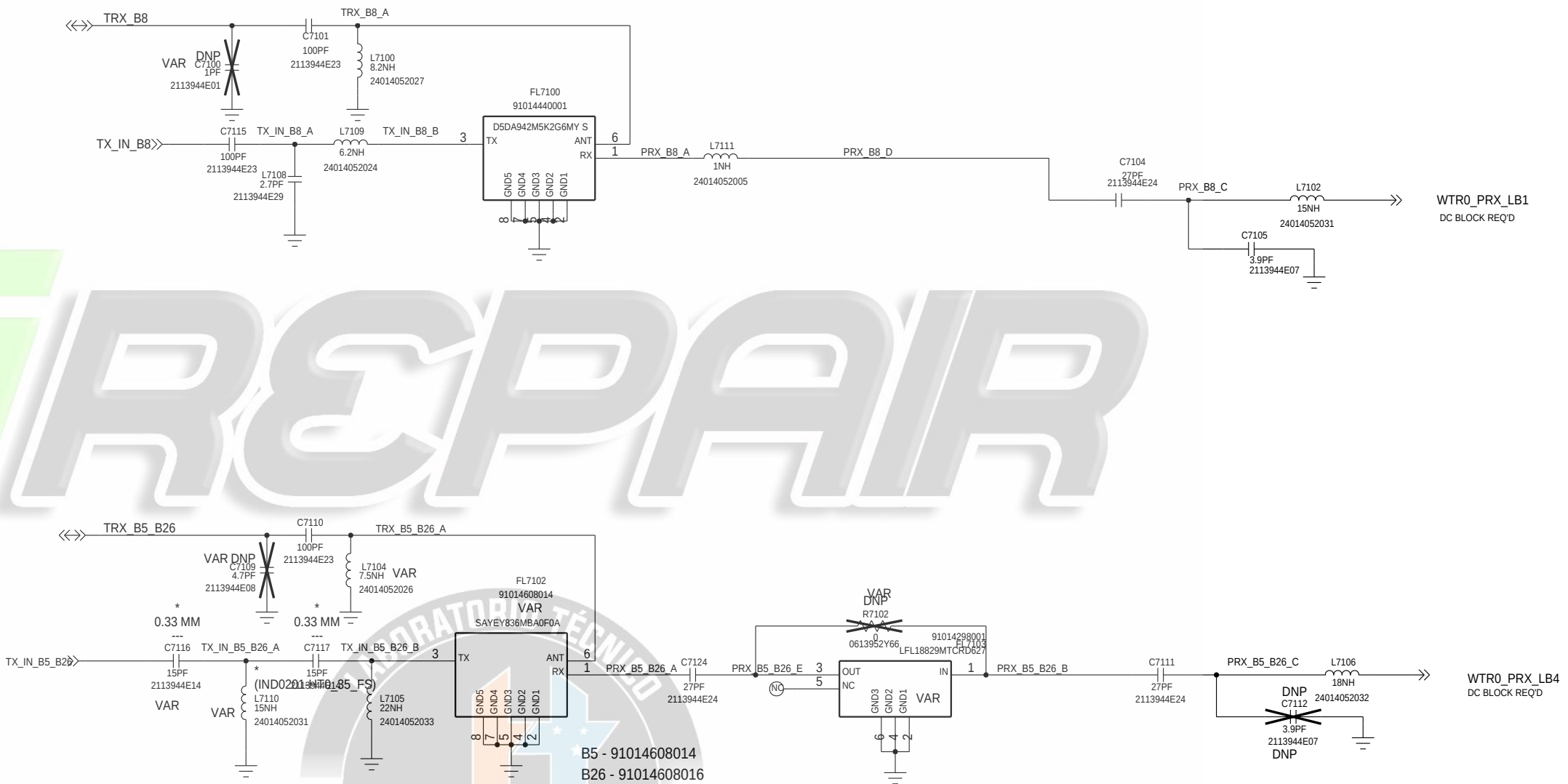
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RFFE: TRX MB
REF 7000-7099



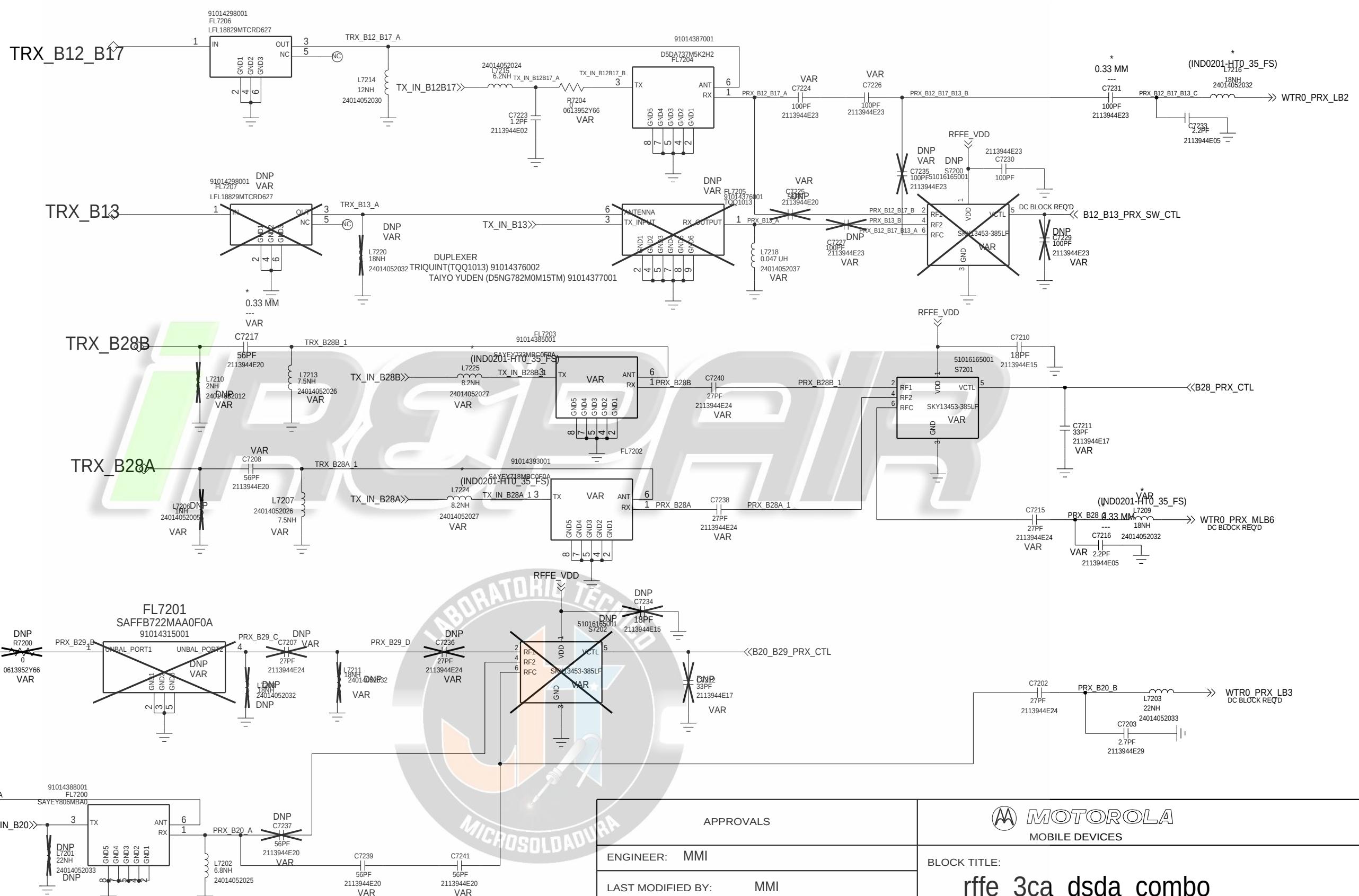
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LAST MODIFIED BY: MMI Tue May 12 10:22:58 2015			
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		CAD FILE NAME:	PAGE 67 OF 78

RFFE: TRX LB
REF 7100-7199



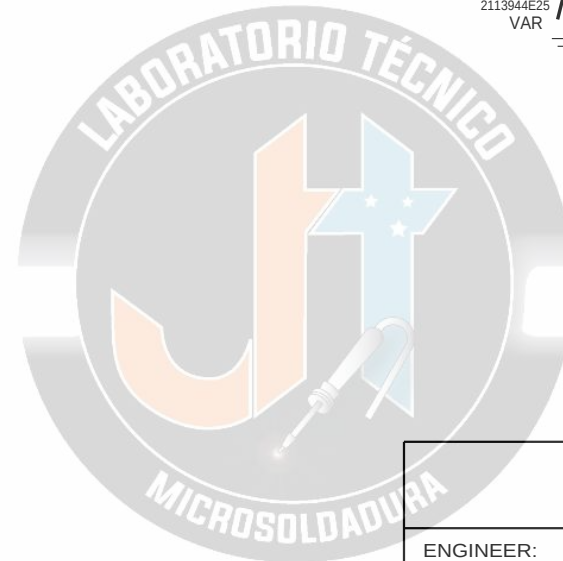
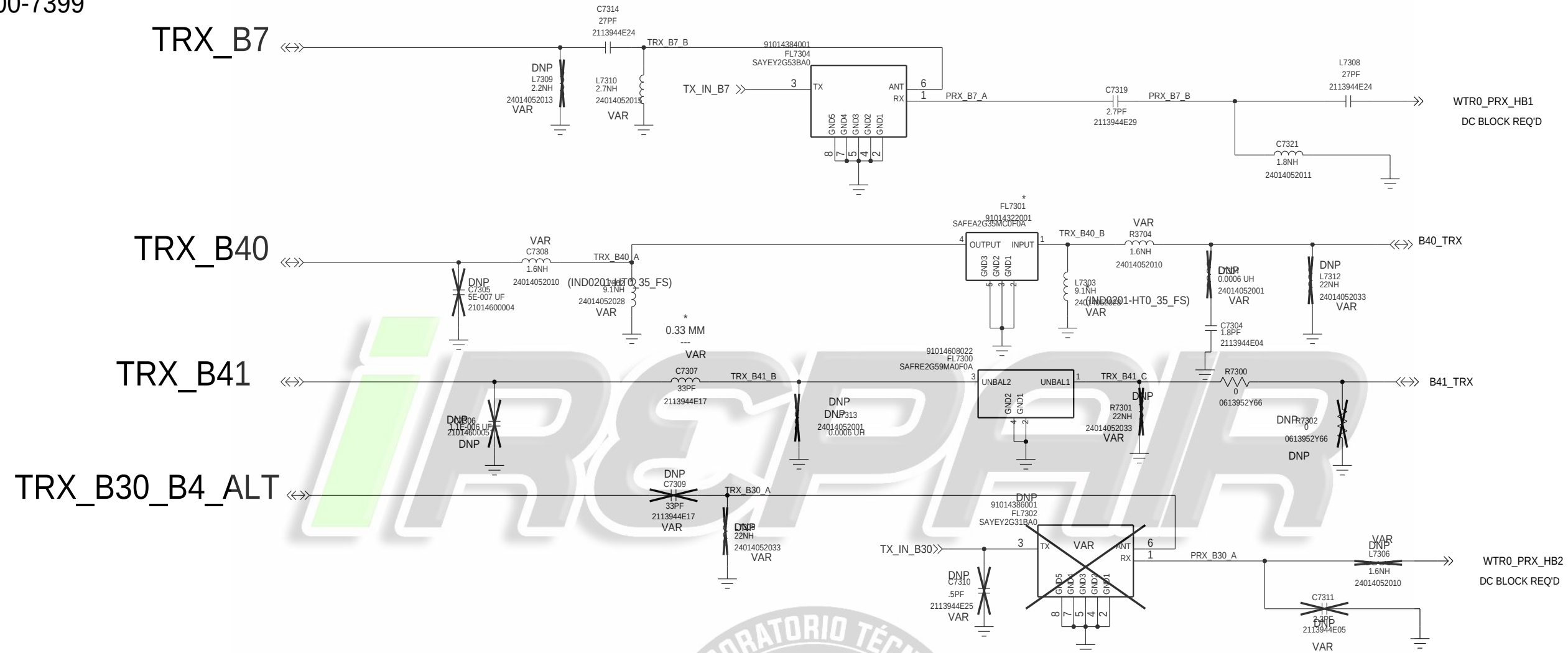
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: rffe_3ca_dsda_combo	
LAST MODIFIED BY: Tue May 12 10:22:59 2015			
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		CAD FILE NAME: 89016410001_a.cpm	
		PAGE 68 OF 78	

RFFE: TRX ULB
REF 7200-7299



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: rffe_3ca_dsda_combo	
LAST MODIFIED BY: MMI Tue May 12 10:23:00 2015			
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		CAD FILE NAME: PAGE 69 OF 78	

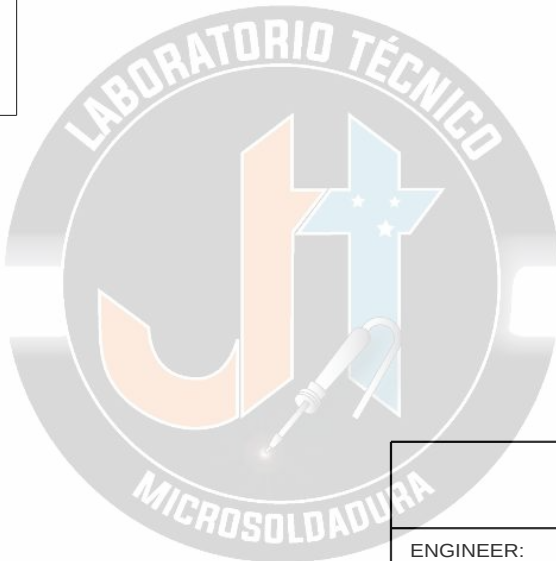
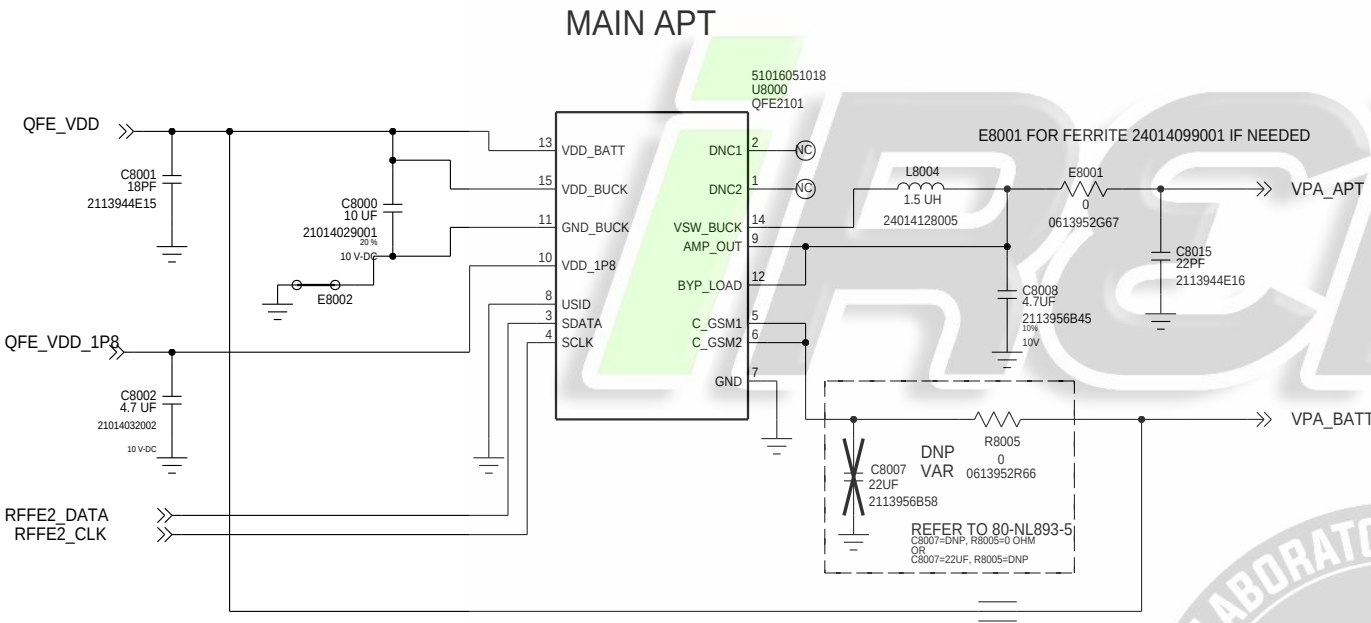
RFFE: TRX HB
REF 7300-7399



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: rffe_3ca_dsda_combo		
LAST MODIFIED BY: MMI Tue May 12 10:23:00 2015			
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		CAD FILE NAME:	PAGE 70 OF 78

RFFE: APT

REF 8000-8099



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Tue May 12 10:23:01 2015		bfeck3_cafedsda_combo	
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		CAD FILE NAME:	
		PAGE 71 OF 78	

RFFE: TX MMB PA

REF 7400-7499

WTR0_TX_MB3

D

WTR0_TX_LB2

C

WTR0_TX_LB4

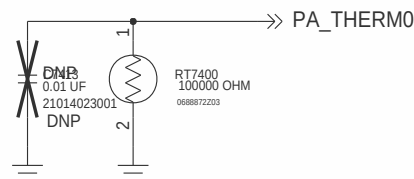
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A

WTR0_TX_HMLB2

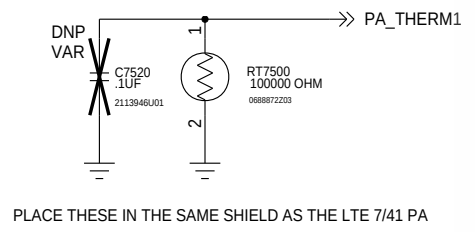
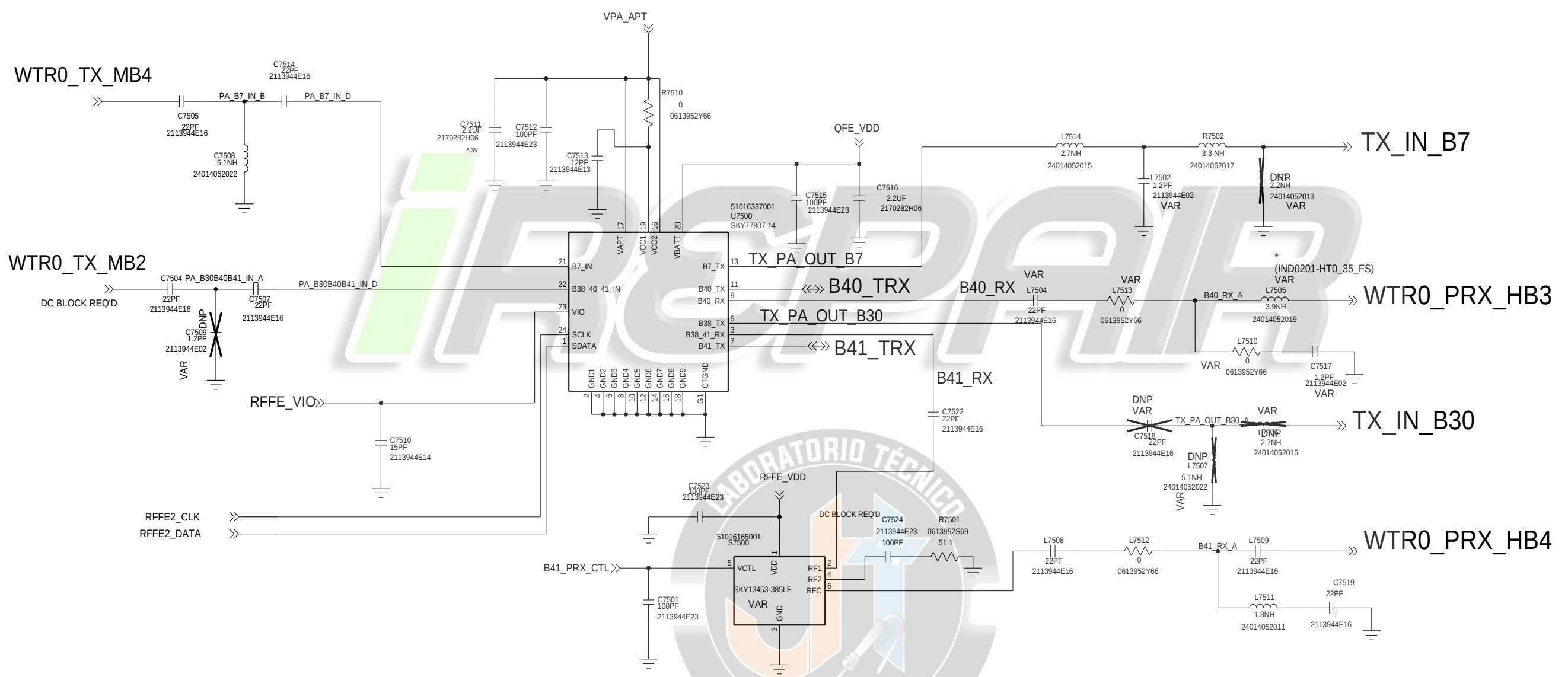
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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: rffe_3ca_dsda_combo	
LAST MODIFIED BY: MMI Tue May 12 10:23:02 2015		PROJECT NAME 8001a10001_a.cpm	
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		PAGE 72 OF 78	

RFFE: TX B7/B38/B41

REF 7500-7599

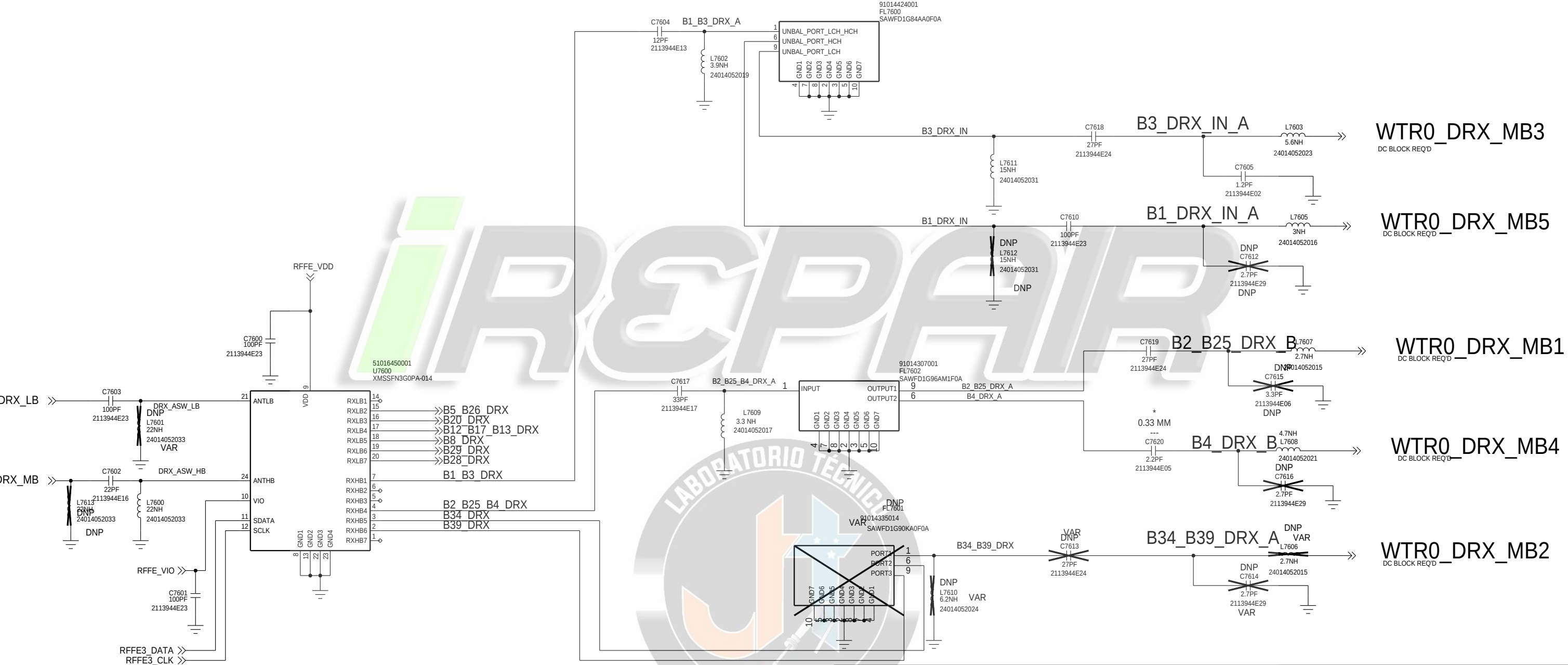


MUTE SWITCH FOR B7 HIGH CHANNEL DENSENSE

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: rffe_3ca_dsda_combo	
LAST MODIFIED BY: Tue May 12 10:23:02 2015		PROJECT NAME 89016410001_a.cpm	
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		PAGE 73 OF 78	

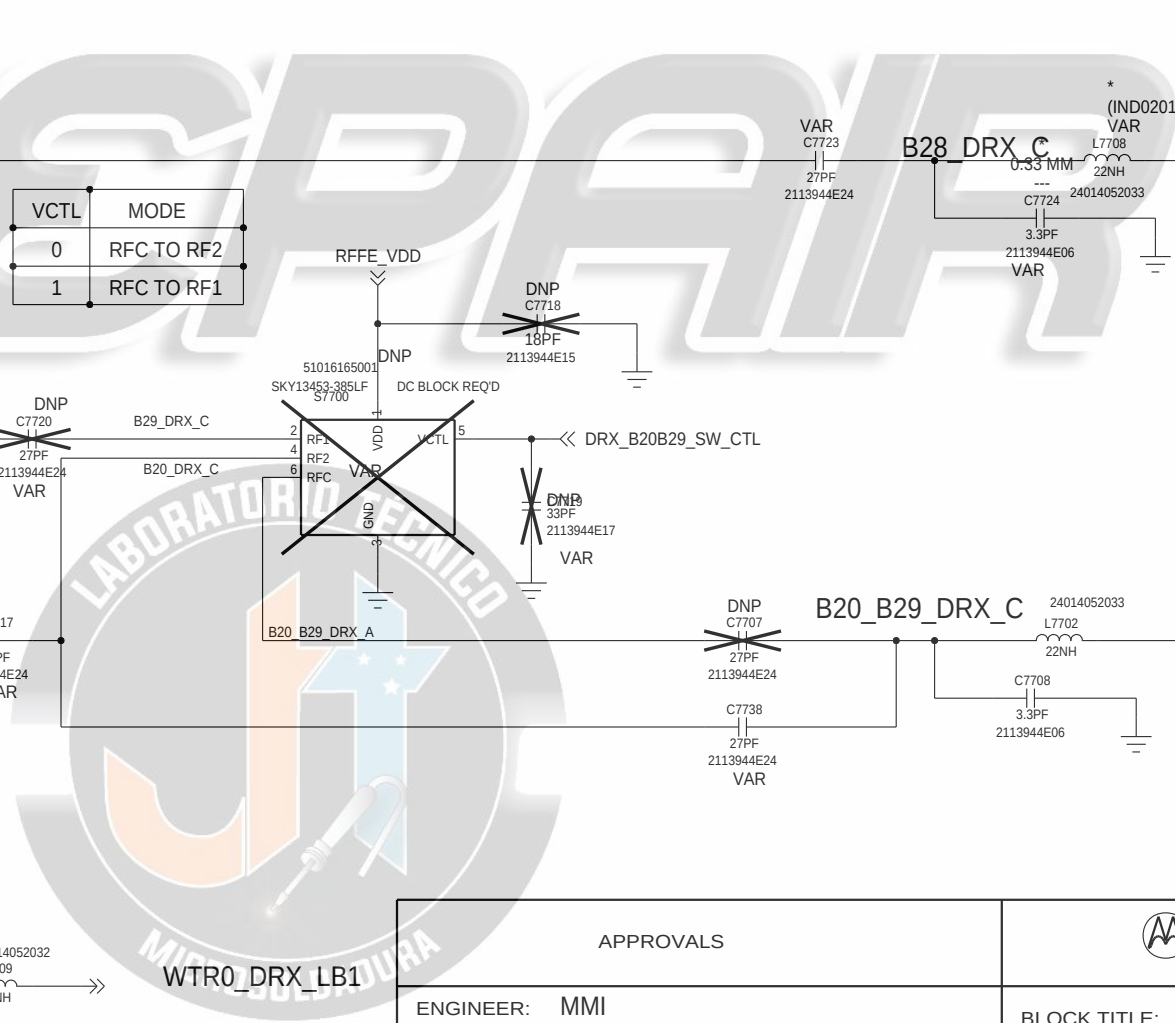
RFFE: DRX ANTENNA SWITCH

REF 7600-7699



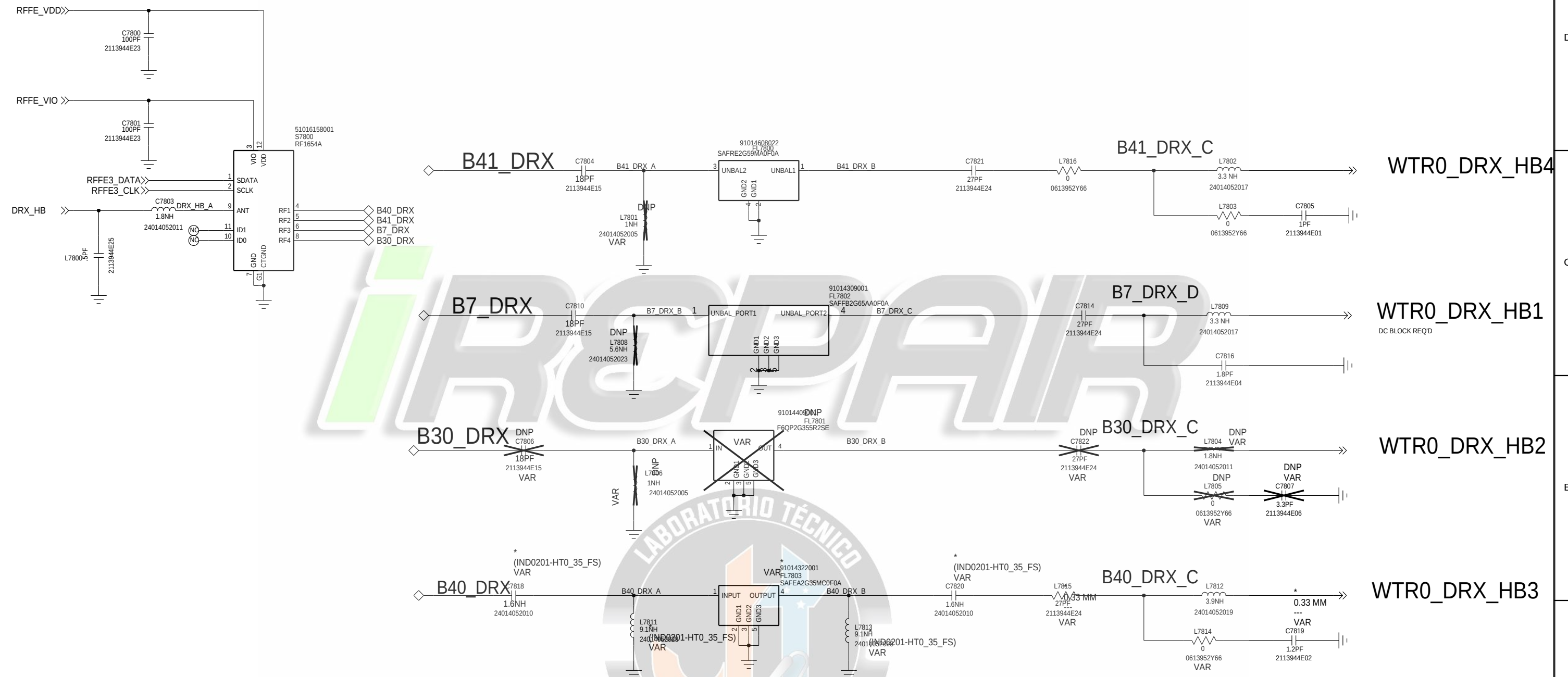
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Tue May 12 10:23:03 2015		rffe_3ca_dsda_combo	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	REV
		B 8016450001_a.cpm	01
		CAD FILE NAME:	
		PAGE 74 OF 78	

REF : 7700-7799



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: rffe_3ca_dsda_combo		
LAST MODIFIED BY: MMI Tue May 12 10:23:04 2015			
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		CAD FILE NAME:	PAGE 75 OF 78

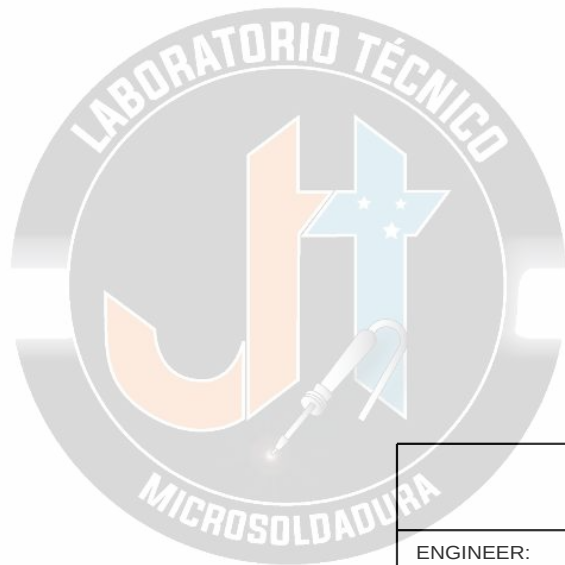
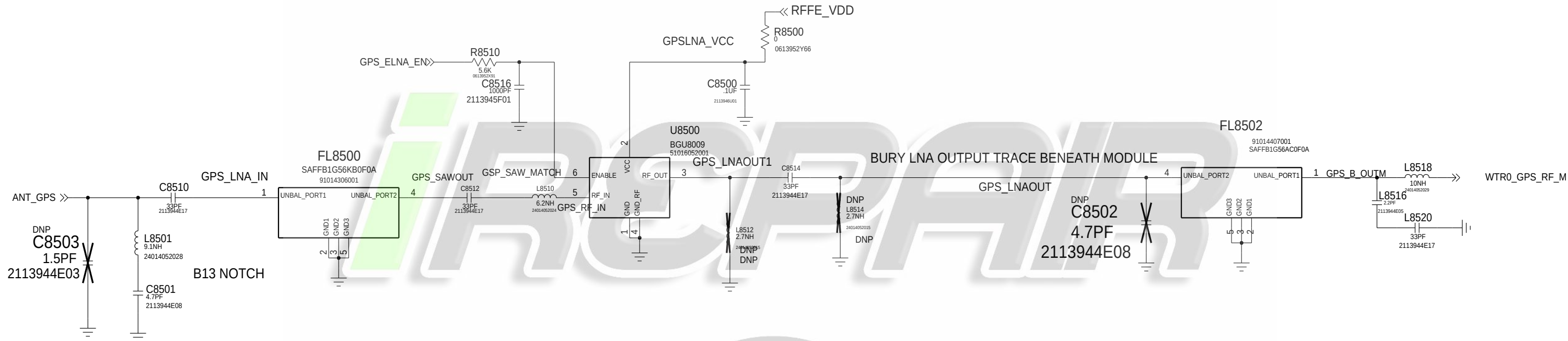
RFFE: DRX HB
REF 7800-7899



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: rffe_3ca_dsda_combo		
LAST MODIFIED BY: MMI Tue May 12 10:23:04 2015			
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		CAD FILE NAME:	PAGE 76 OF 78

RFFE: GPS

REF : 8500-8549



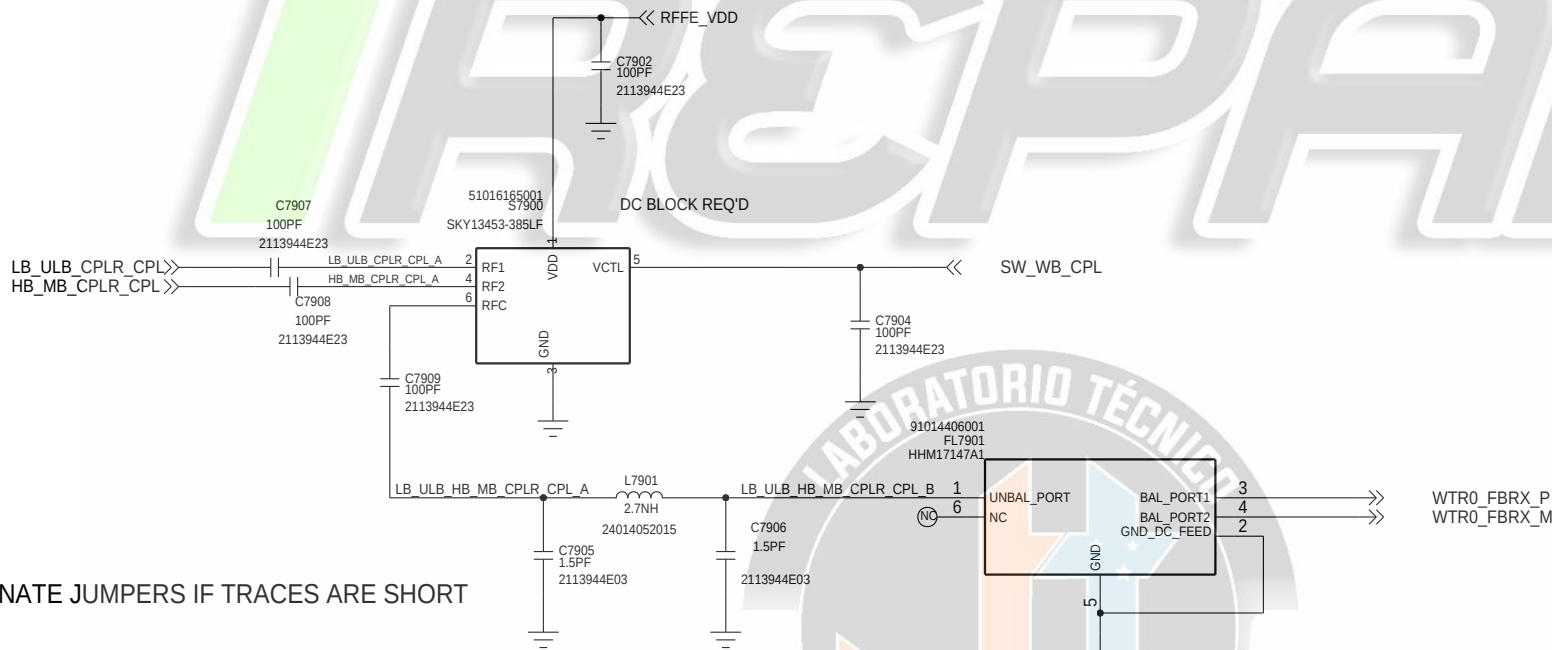
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Tue May 12 10:23:05 2015		rffe_3ca_dsda_combo	
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		CAD FILE NAME:	
		REV 01	
		PAGE 77 OF 78	

RFFE: PDET

REF : 7900-7949

COULD ELIMINATE JUMPERS IF TRACES ARE SHORT

DEVICE LOCATION - TBD



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Tue May 12 10:23:06 2015		rffe_3ca_dsda_combo	
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		CAD FILE NAME:	
		PAGE 78 OF 78	